

Structures of General Relativity and Gauge Theory

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Institution: AstroParticles and Cosmology
laboratory/Stockholm University

Degree: Master's project - 30 hp

Programme: Master's degree in Theoretical Physics - 120 hp

Period: Spring 2025

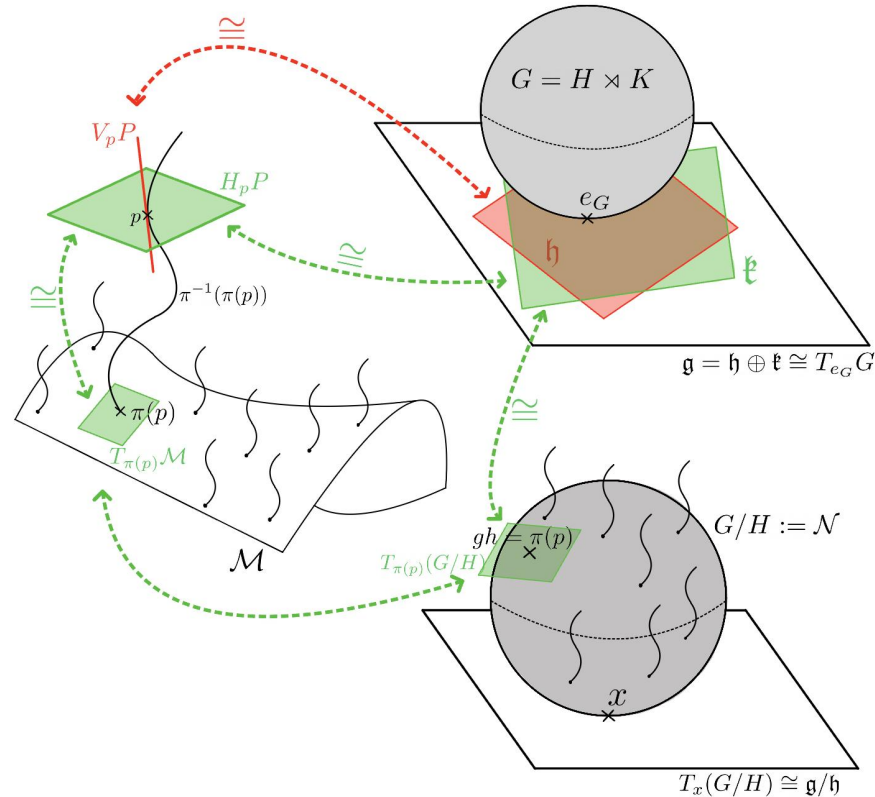
Supervisor: Eric Huguet

Co-supervisor: Fawad Hassan

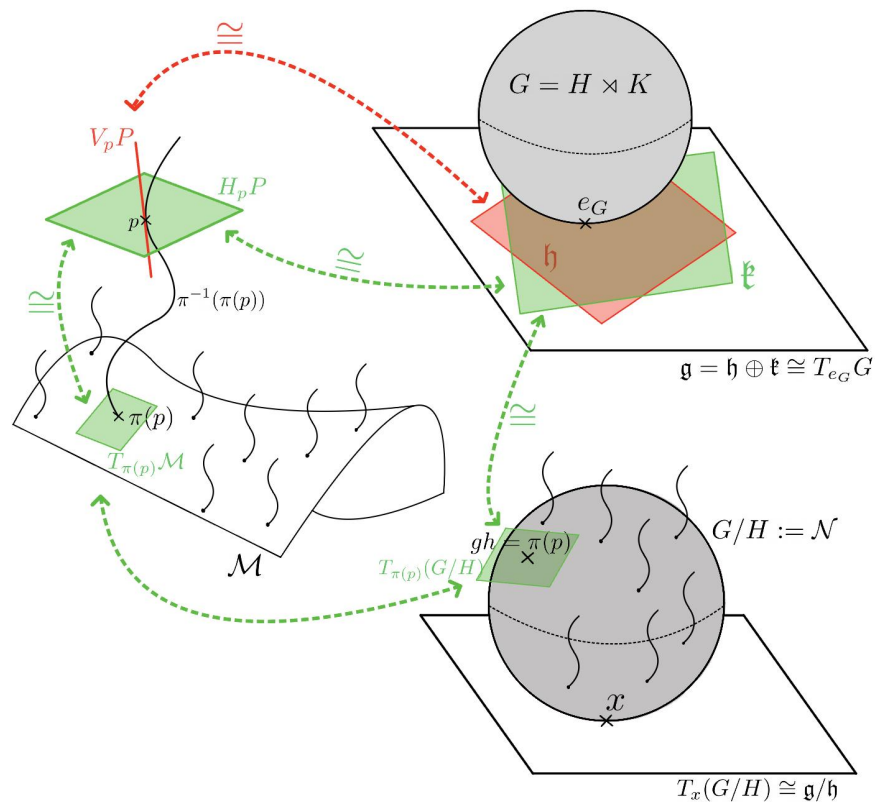
Examiner: Eddy Ardonne - Vladimir Krasnov



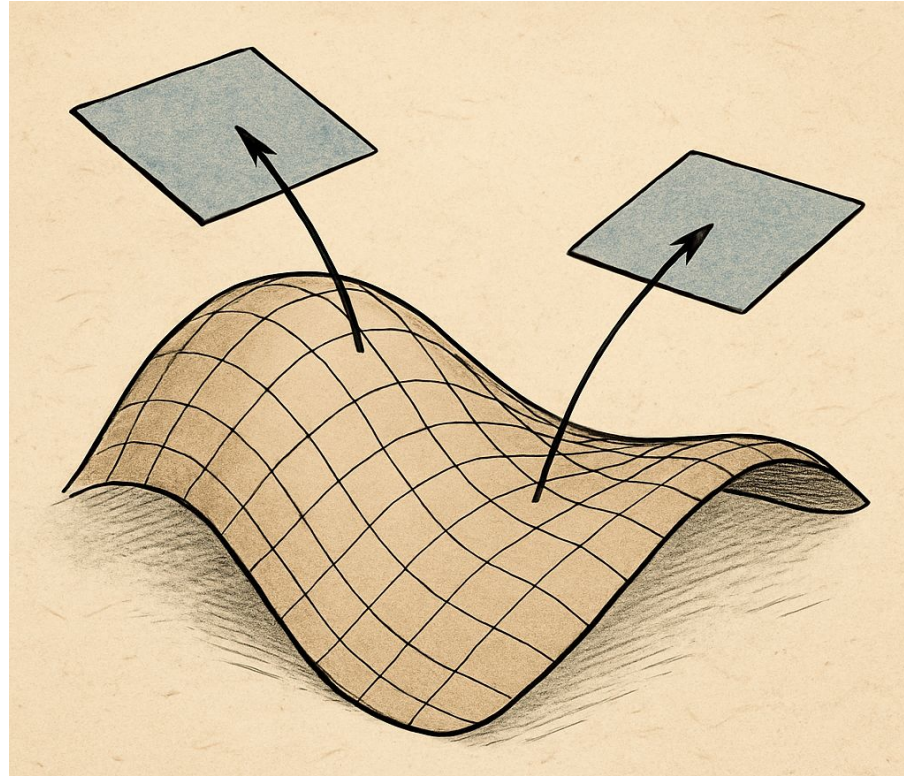
General Relativity as a Lorentz gauge theory: gauge field from the Levi-Civita connection?



Why is the Levi-Civita connection not a gauge field?



Differential geometry and Cartan formalism



Differential geometry and Cartan formalism

Differential forms

Differential geometry and Cartan formalism

Differential forms:

- Scalar fields

Differential geometry and Cartan formalism

Differential forms:

- Scalar fields: 0-forms

Differential geometry and Cartan formalism

Differential forms:

- Scalar fields: 0-forms ϕ

Differential geometry and Cartan formalism

Differential forms:

- Scalar fields: 0-forms ϕ
- Derivative of scalar fields

Differential geometry and Cartan formalism

Differential forms:

- Scalar fields: 0-forms ϕ
- Derivative of scalar fields: 1-forms

Differential geometry and Cartan formalism

Differential forms:

- Scalar fields: 0-forms ϕ
- Derivative of scalar fields: 1-forms $d\phi = \partial_\mu \phi dx^\mu$

Differential geometry and Cartan formalism

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Differential geometry and Cartan formalism

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- Vector fields: 1-forms (connections)

Differential geometry and Cartan formalism

Differential forms:

- Scalar fields: 0-forms ϕ
- Derivative of scalar fields: 1-forms $d\phi = \partial_\mu \phi dx^\mu$
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Differential geometry and Cartan formalism

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- Noether currents

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- Noether currents: 3-forms

Differential geometry and Cartan formalism

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- Lagrangian

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- Lagrangian: top-degree (4)-form

Differential geometry and Cartan formalism

Differential forms

Differential geometry and Cartan formalism

Differential forms:

- Wedge product

Differential geometry and Cartan formalism

Differential forms:

- Wedge product $\wedge$

Differential geometry and Cartan formalism

Differential forms:

- Wedge product $\wedge$
- Hodge operator

Differential geometry and Cartan formalism

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$$\eta^{\mu\nu} \partial_\mu \phi^* \partial_\nu \phi$$

Differential geometry and Cartan formalism

Differential forms:

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$$\eta^{\mu\nu} \partial_\mu \phi^* \partial_\nu \phi \rightarrow d\phi^* \wedge \star d\phi$$

Differential geometry and Cartan formalism

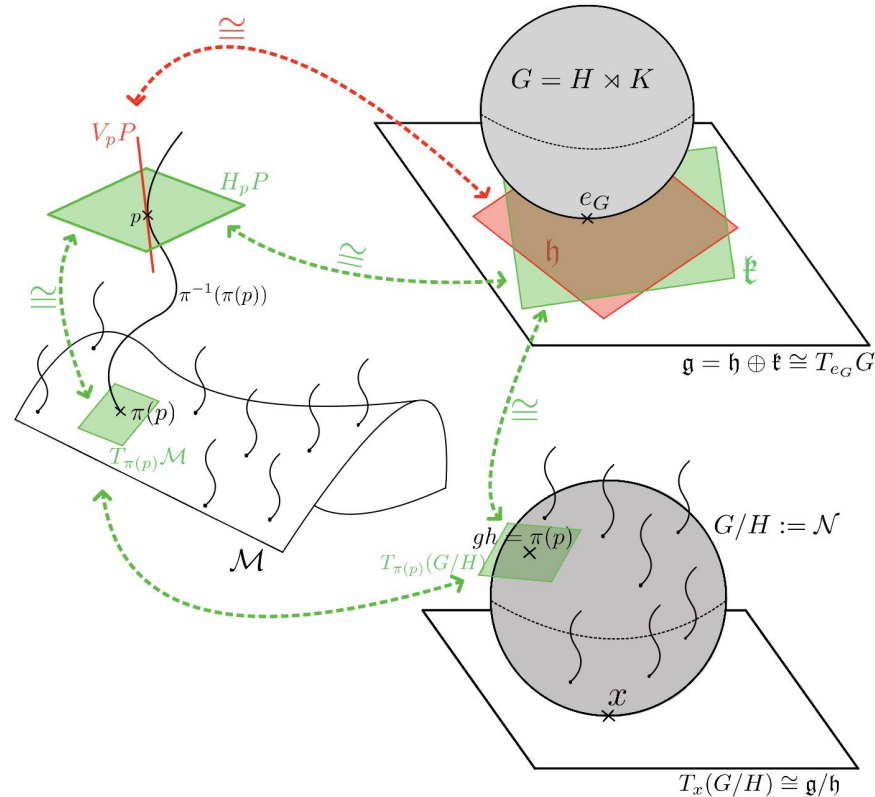
Differential forms:

- Wedge product $\wedge$
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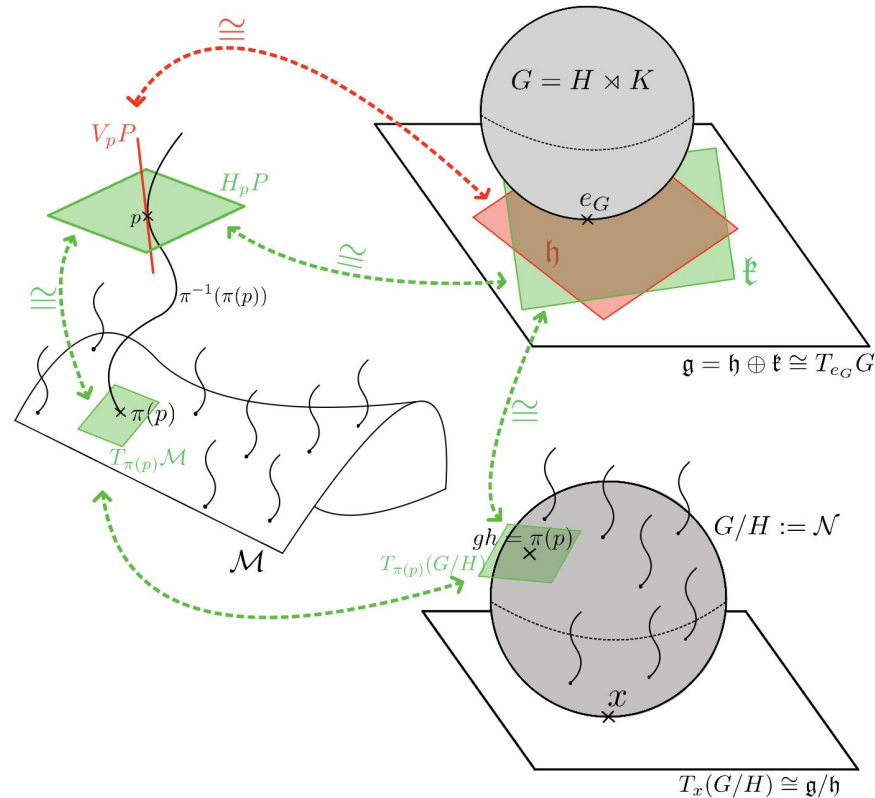
$$\eta^{\mu\nu} \partial_\mu \phi^* \partial_\nu \phi \rightarrow d\phi^* \wedge \star d\phi$$

$$S[\phi] = \int_{\mathcal{M}} d\phi^* \wedge \star d\phi$$

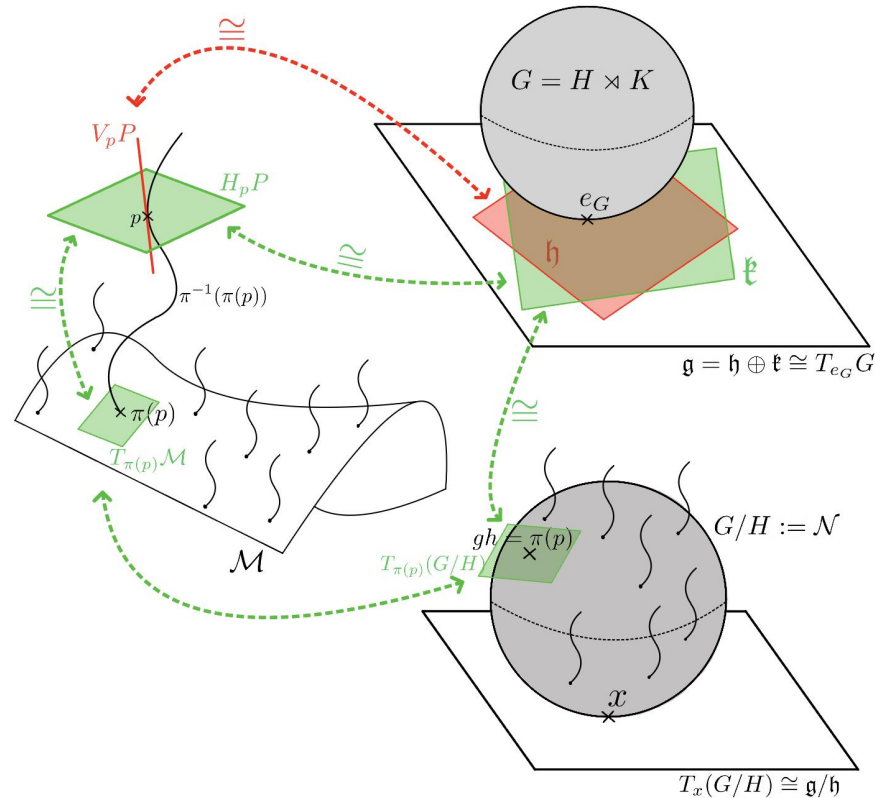
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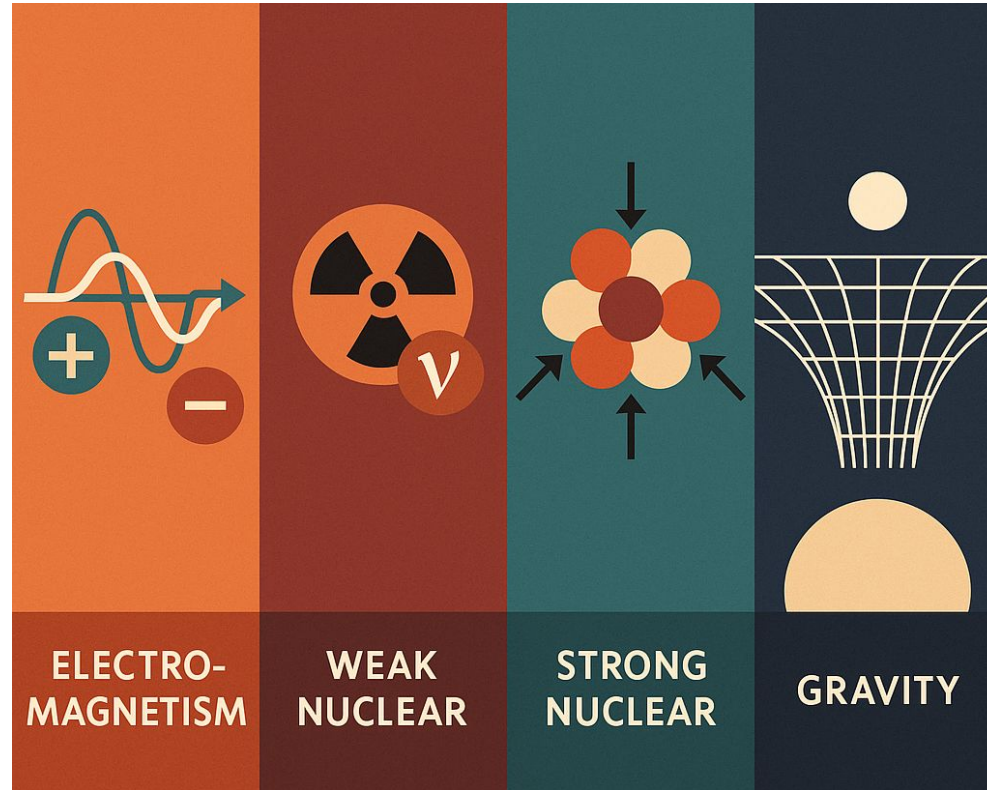
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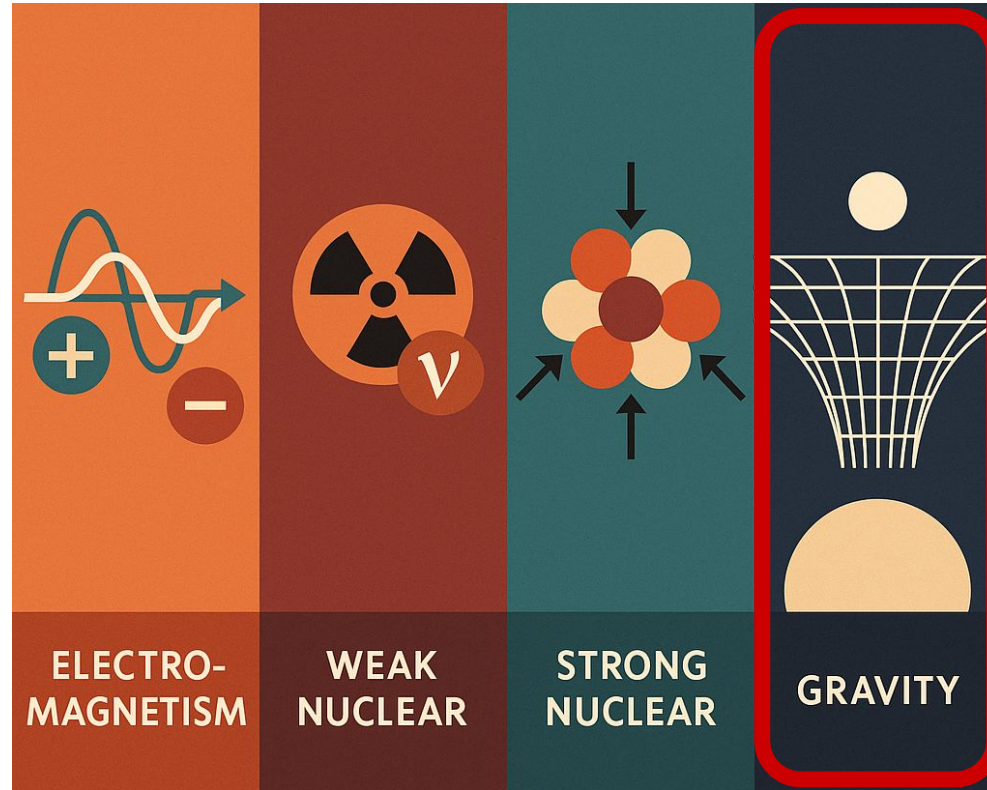
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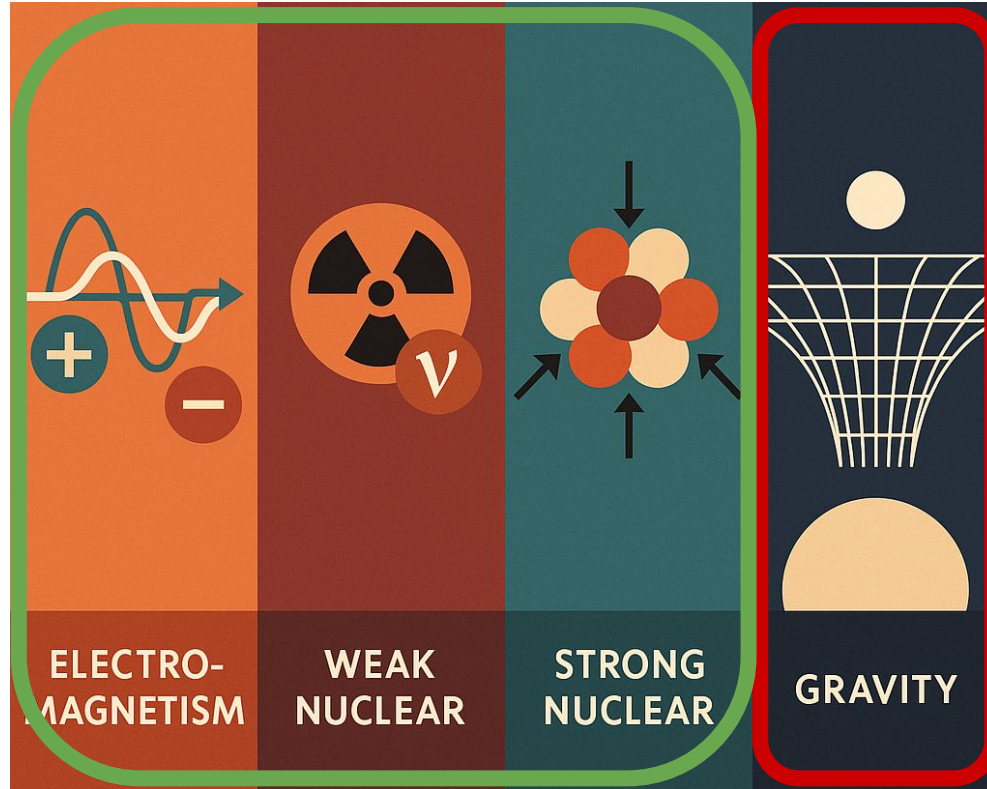
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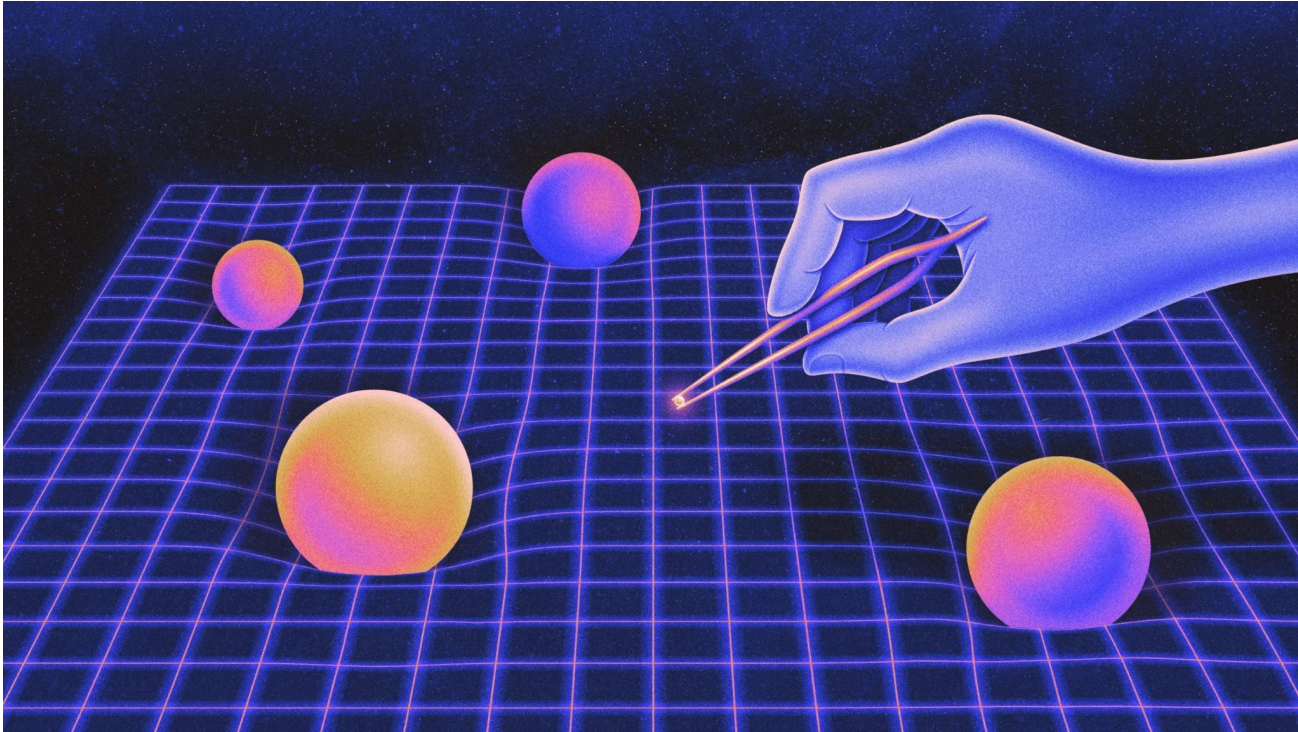
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General Relativity



General Relativity

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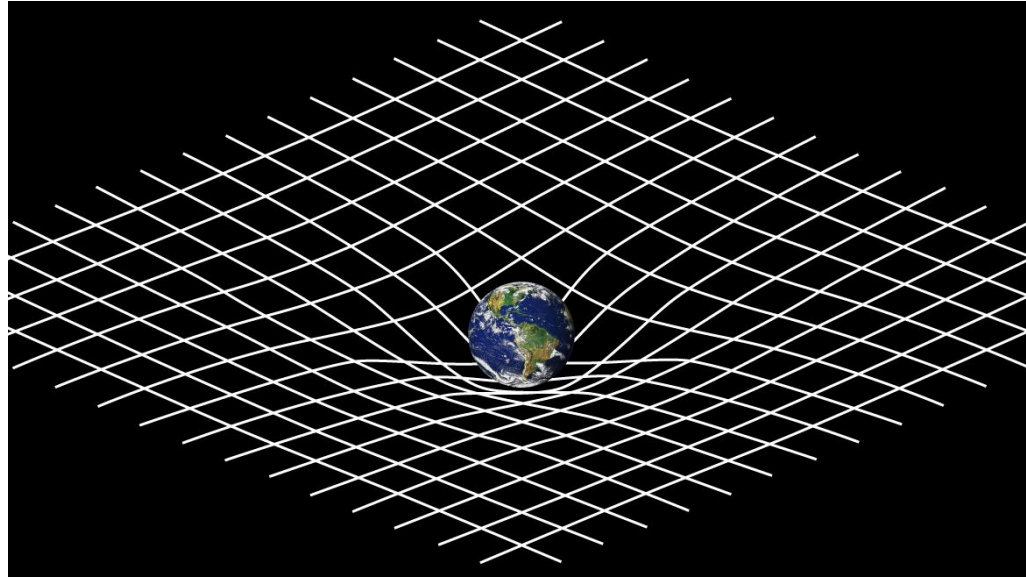
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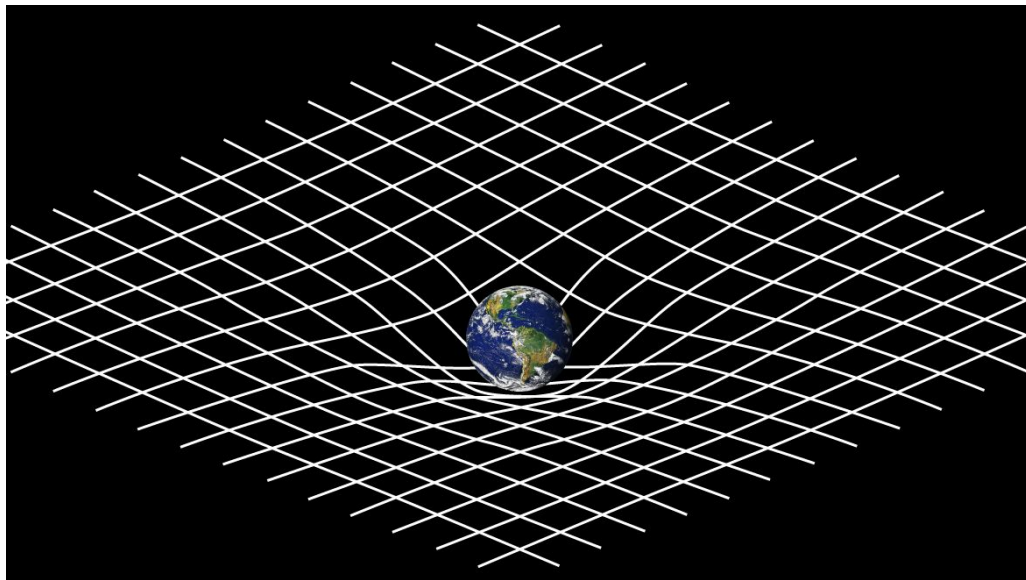
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General Relativity



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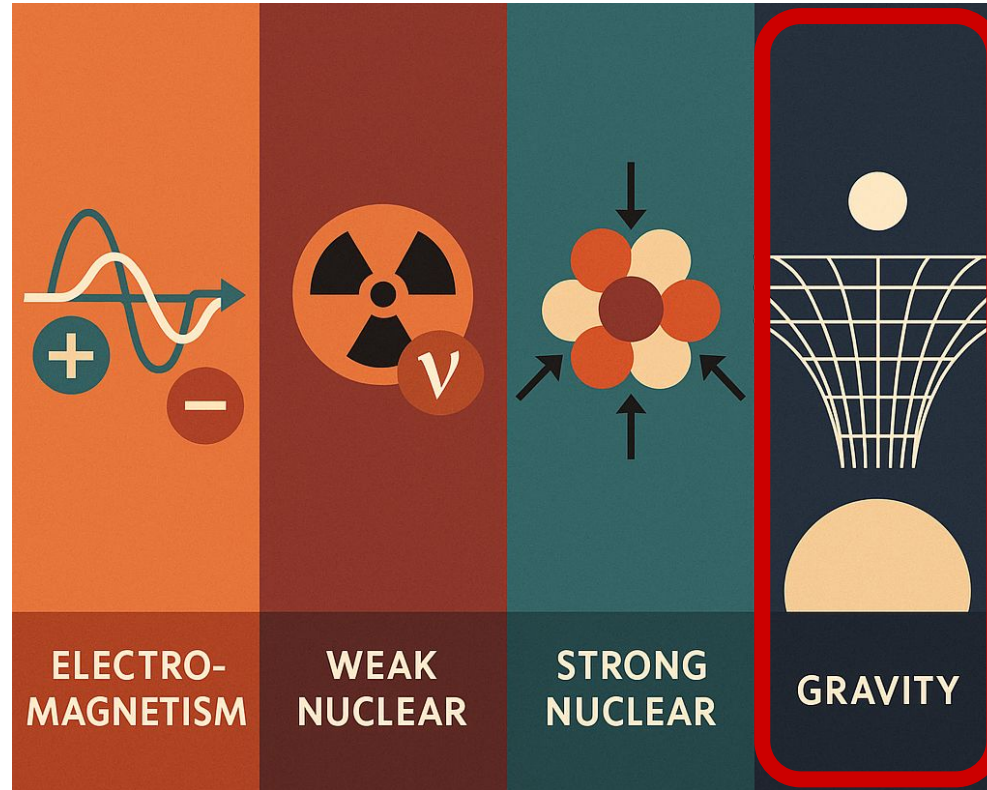
General Relativity



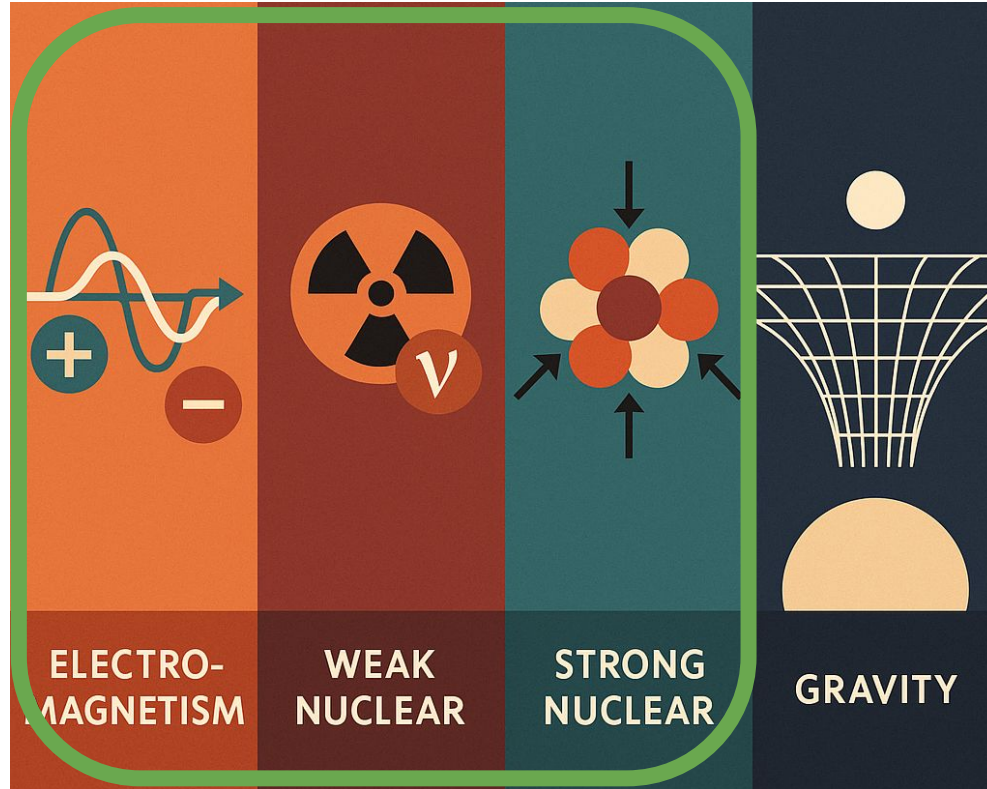
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$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$$

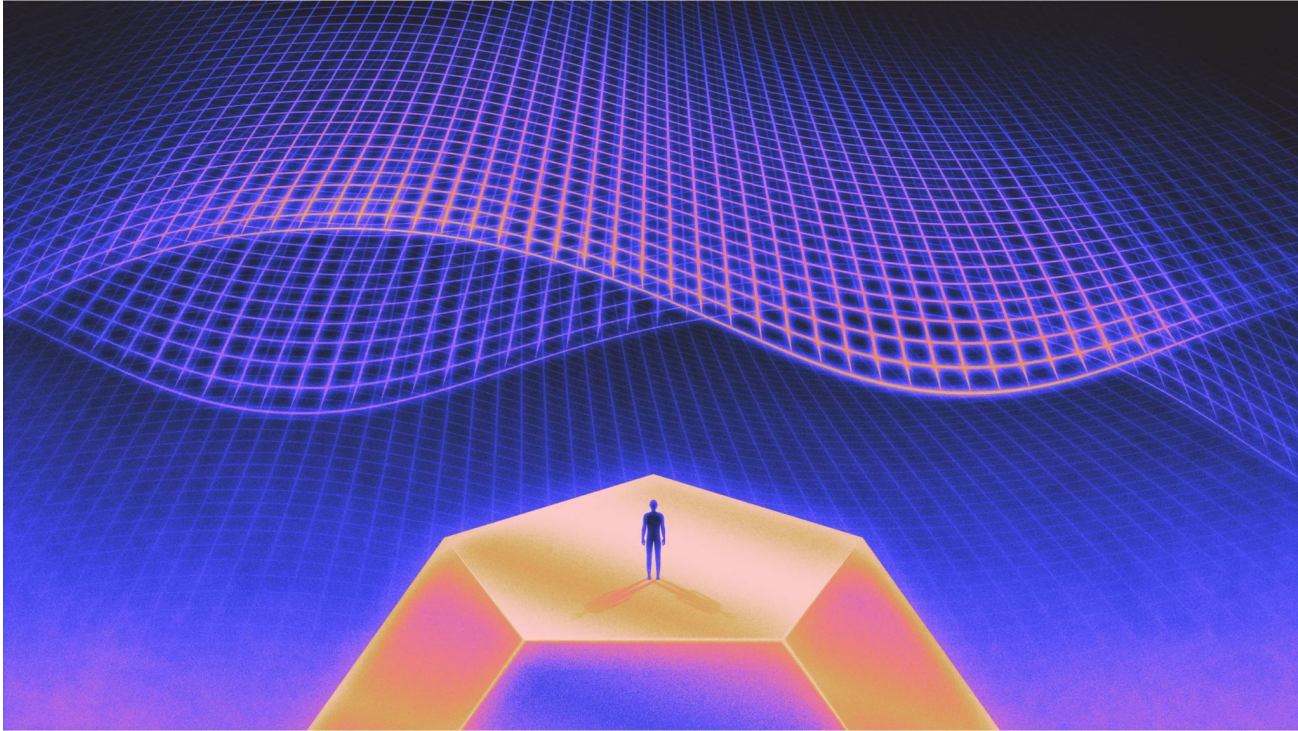
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Gauge theories



Gauge theories

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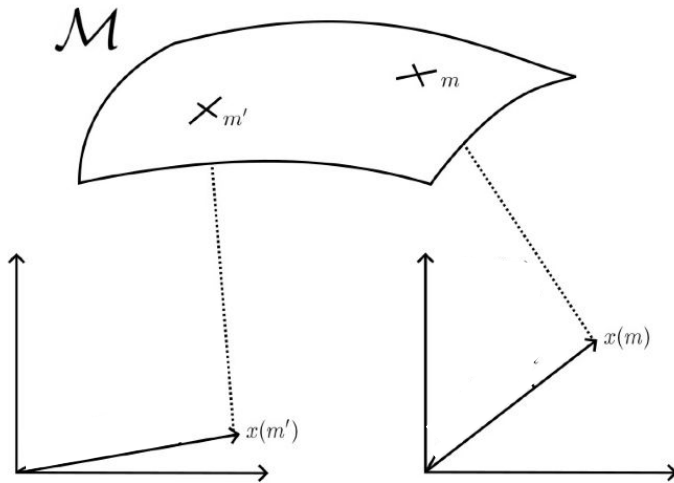
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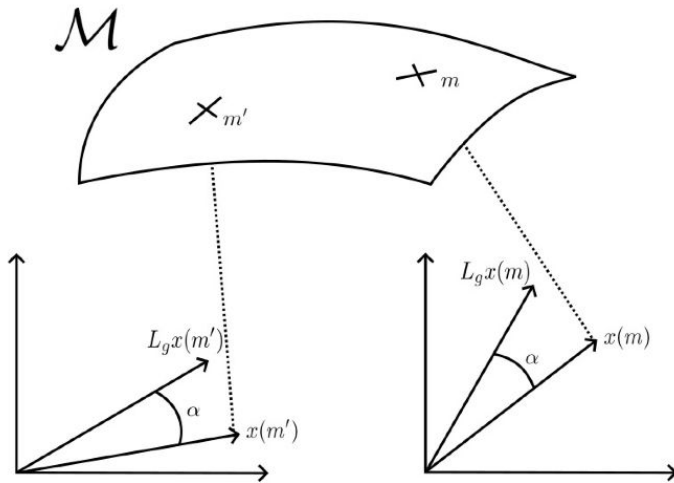
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Gauge theories

Gauge theories

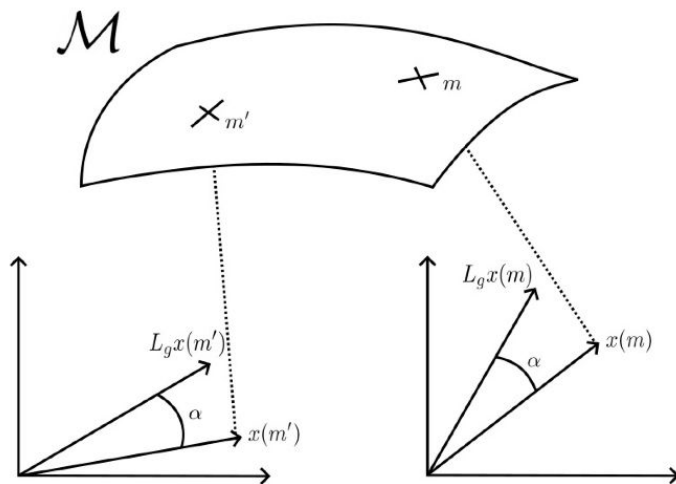


Gauge theories

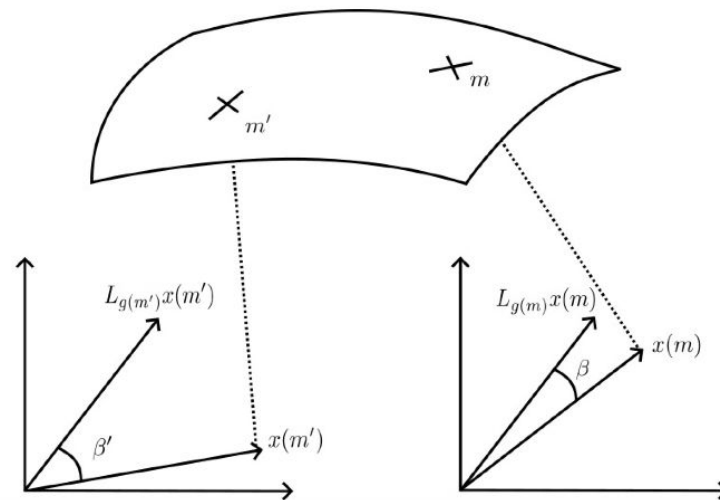


Global

Gauge theories



Global



Local

Gauge theories

$$S = \int d^4x \, \eta^{\mu\nu} \partial_\mu \phi^* \partial_\nu \phi$$

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$$S = \int d^4x \, \eta^{\mu\nu} \partial_\mu \phi^* \partial_\nu \phi$$

$$\left\{ \begin{array}{l} \phi \rightarrow e^{i\alpha} \phi \\ \phi^* \rightarrow e^{-i\alpha} \phi^* \end{array} \right.$$

Global

Gauge theories

$$S = \int d^4x \, \eta^{\mu\nu} \partial_\mu \phi^* \partial_\nu \phi$$

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Global

Local

Gauge theories

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Global

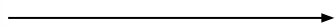
Local

Gauge theories

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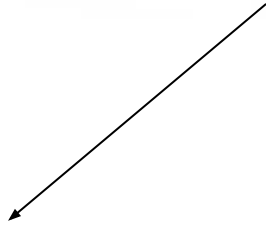
$$\partial_\mu$$

Gauge theories

$$\partial_\mu \rightarrow D_\mu$$

Gauge theories

$$\partial_\mu \rightarrow D_\mu$$

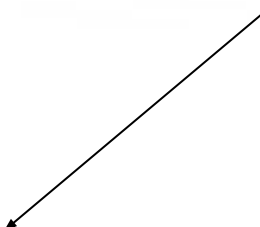


Covariant derivative

Gauge theories

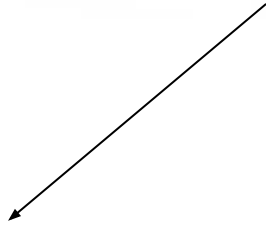
$$\partial_\mu \rightarrow D_\mu = \partial_\mu$$

Covariant derivative



Gauge theories

$$\partial_\mu \rightarrow D_\mu = \partial_\mu + iqA_\mu$$



Covariant derivative

Gauge theories

$$\partial_\mu \rightarrow D_\mu = \partial_\mu + iqA_\mu$$

Covariant derivative

Vector field

Gauge theories

$$\partial_\mu \rightarrow D_\mu = \partial_\mu + iqA_\mu$$

Covariant derivative

Vector field

$$A_\mu \rightarrow A_\mu - \partial_\mu \alpha$$

Gauge theories

$$\partial_\mu \rightarrow D_\mu = \partial_\mu + iqA_\mu$$

Covariant derivative

Gauge
transformation

Vector field

$$A_\mu \rightarrow A_\mu - \partial_\mu \alpha$$

Gauge theories

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Covariant derivative

Gauge field

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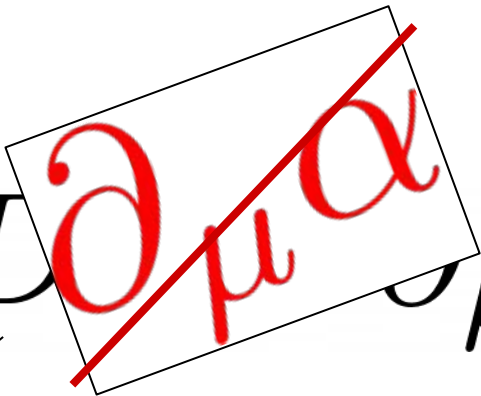
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Gauge theories

$$\partial_\mu \rightarrow D_\mu = \partial_\mu + iqA_\mu$$



Gauge field

Gauge theories

$$\partial_\mu \rightarrow D_\mu = \partial_\mu + iqA_\mu$$

The diagram illustrates the transition from the partial derivative ∂_μ to the covariant derivative D_μ in gauge theories. The equation $\partial_\mu \rightarrow D_\mu = \partial_\mu + iqA_\mu$ is shown. An arrow points from the term A_μ in the equation down to the text "Gauge field". A horizontal arrow points from the text "Gauge field" to the left, with the word "Quantization" written above it.

Gauge theories

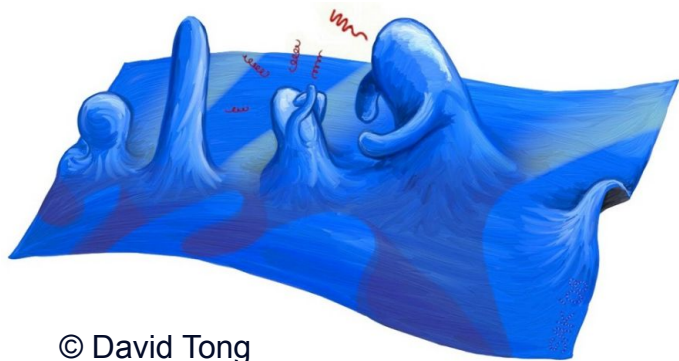
$$\partial_\mu \rightarrow D_\mu = \partial_\mu + iqA_\mu$$

Interaction
quantum field

Quantization ← Gauge field

Gauge theories

$$\partial_\mu \rightarrow D_\mu = \partial_\mu + iqA_\mu$$

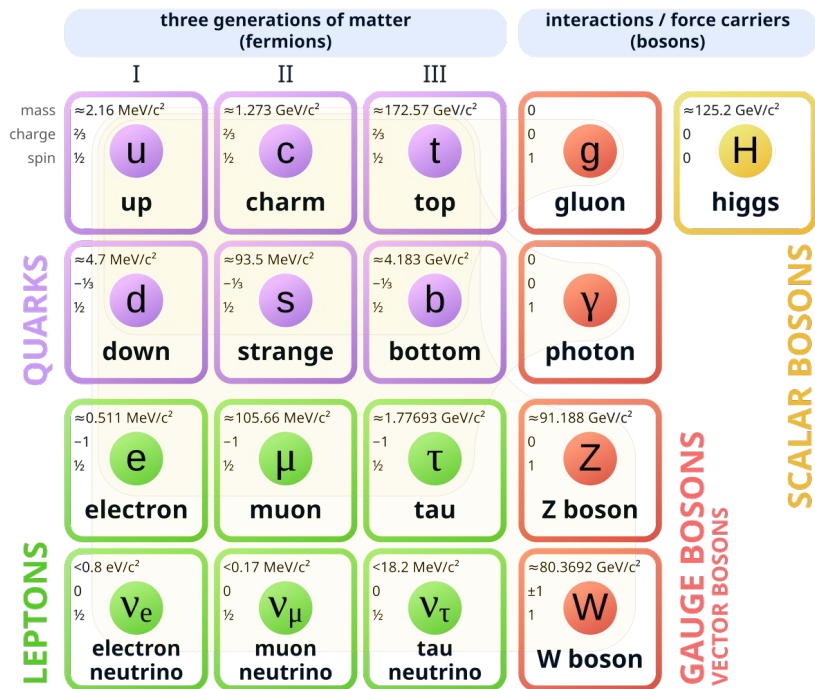


Interaction
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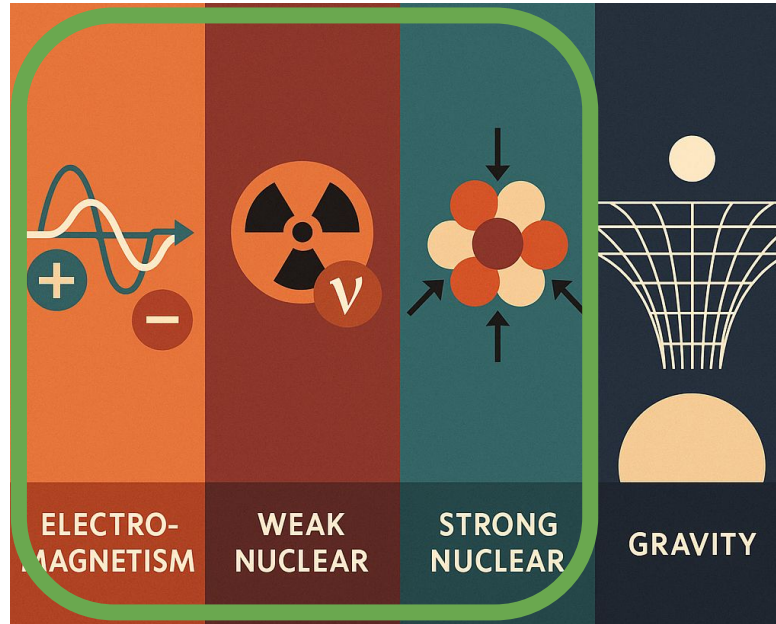
Quantization ← Gauge field

Gauge theories

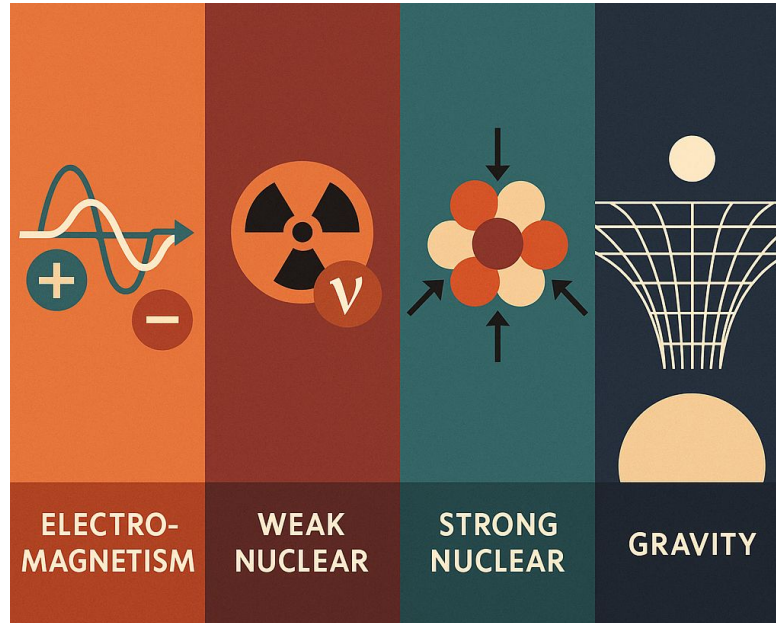
Standard Model of Elementary Particles



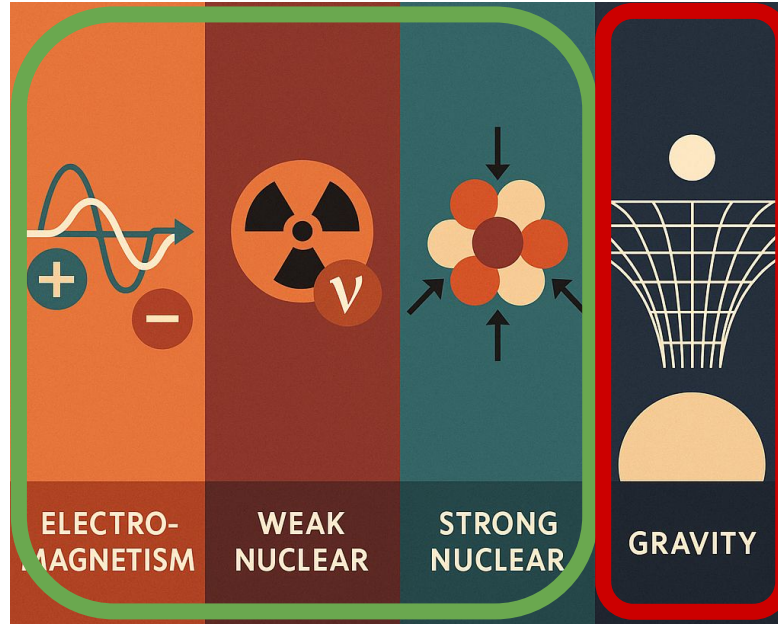
Gauge theories



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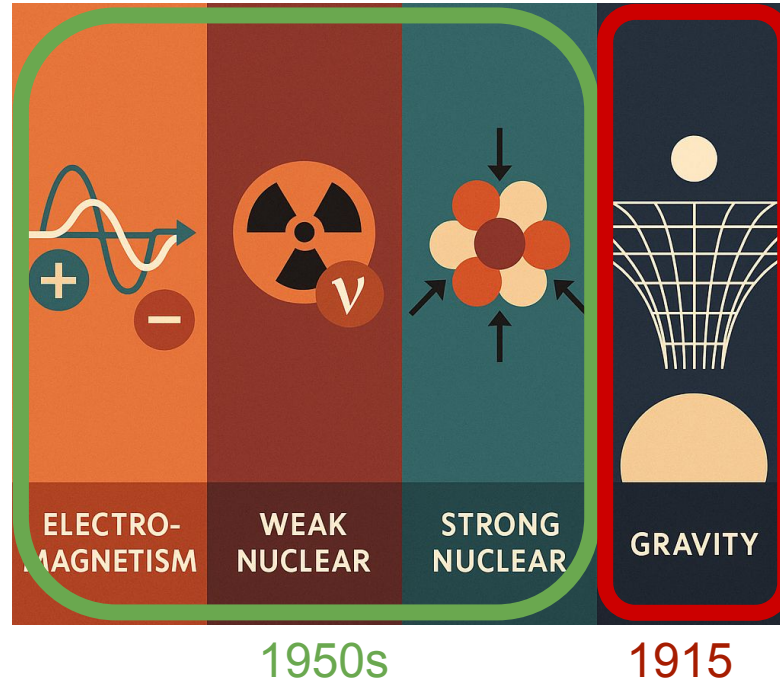


Gauge theories

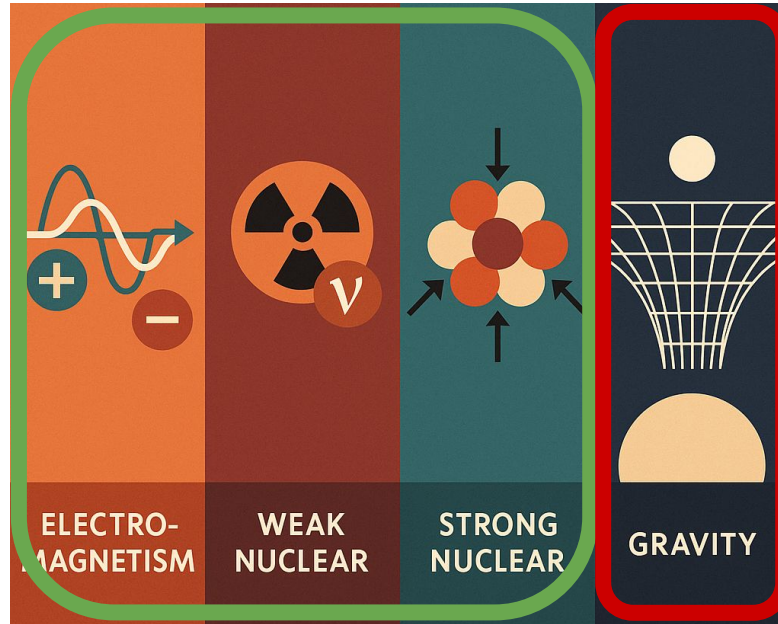


1915

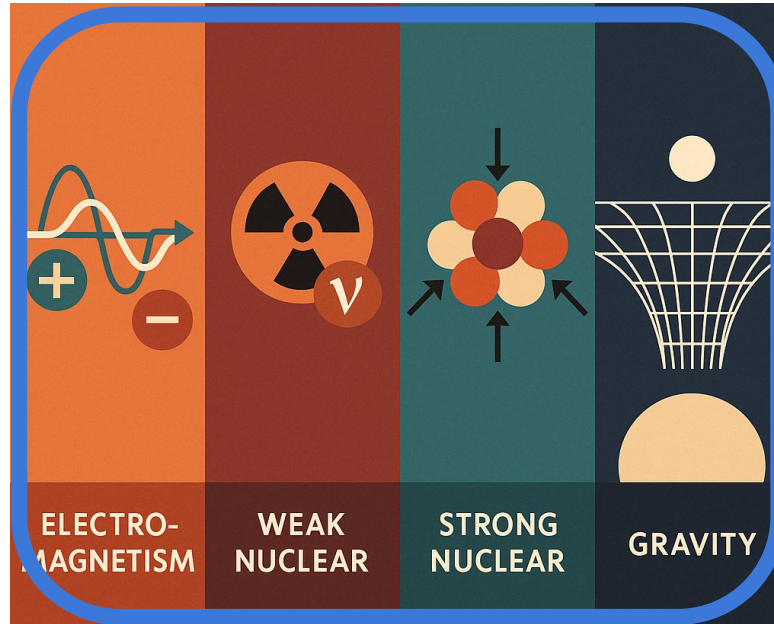
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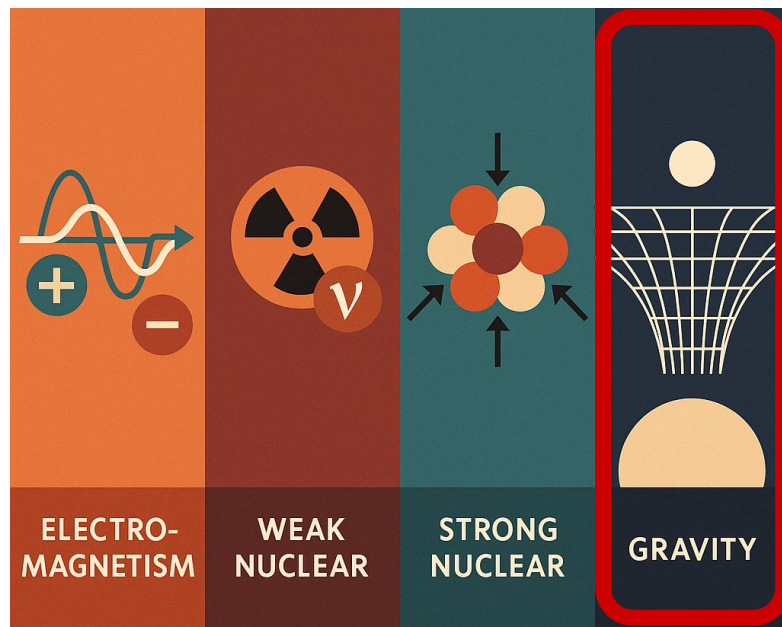
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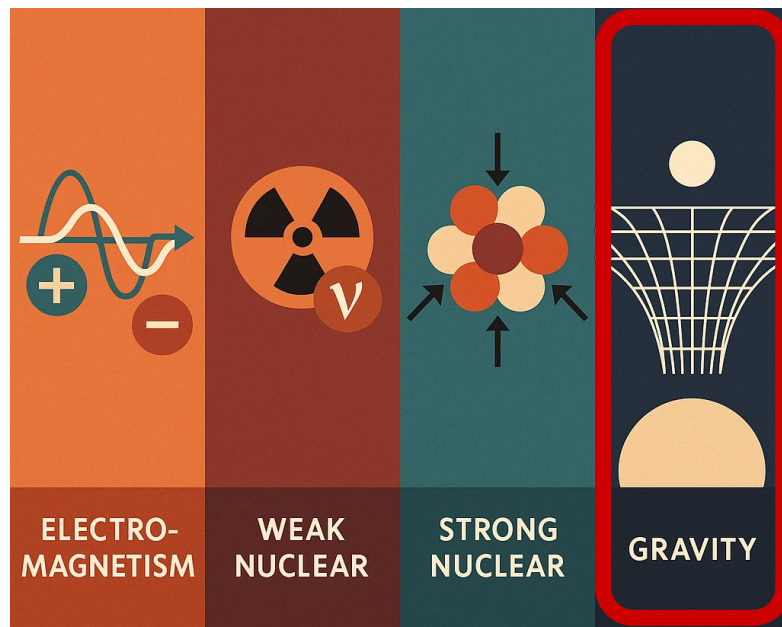
Unified gauge theory



Gauge theory of gravity

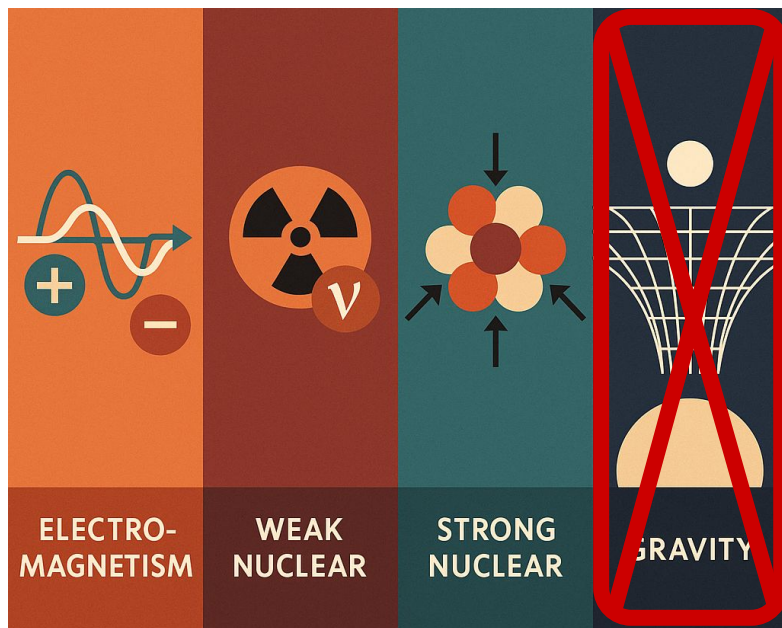


Gauge theory of gravity



Milutin Blagojevic and Friedrich W. Hehl. Gauge Theories of Gravitation. 2022.
arXiv:1210.3775 [gr-qc]. url: <https://arxiv.org/abs/1210.3775>.

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M. Peskin and D. Schroeder. *An Introduction to Quantum Field Theory*. Reading, USA: Addison-Wesley (1995) 842 p. Westview Press, 1995.

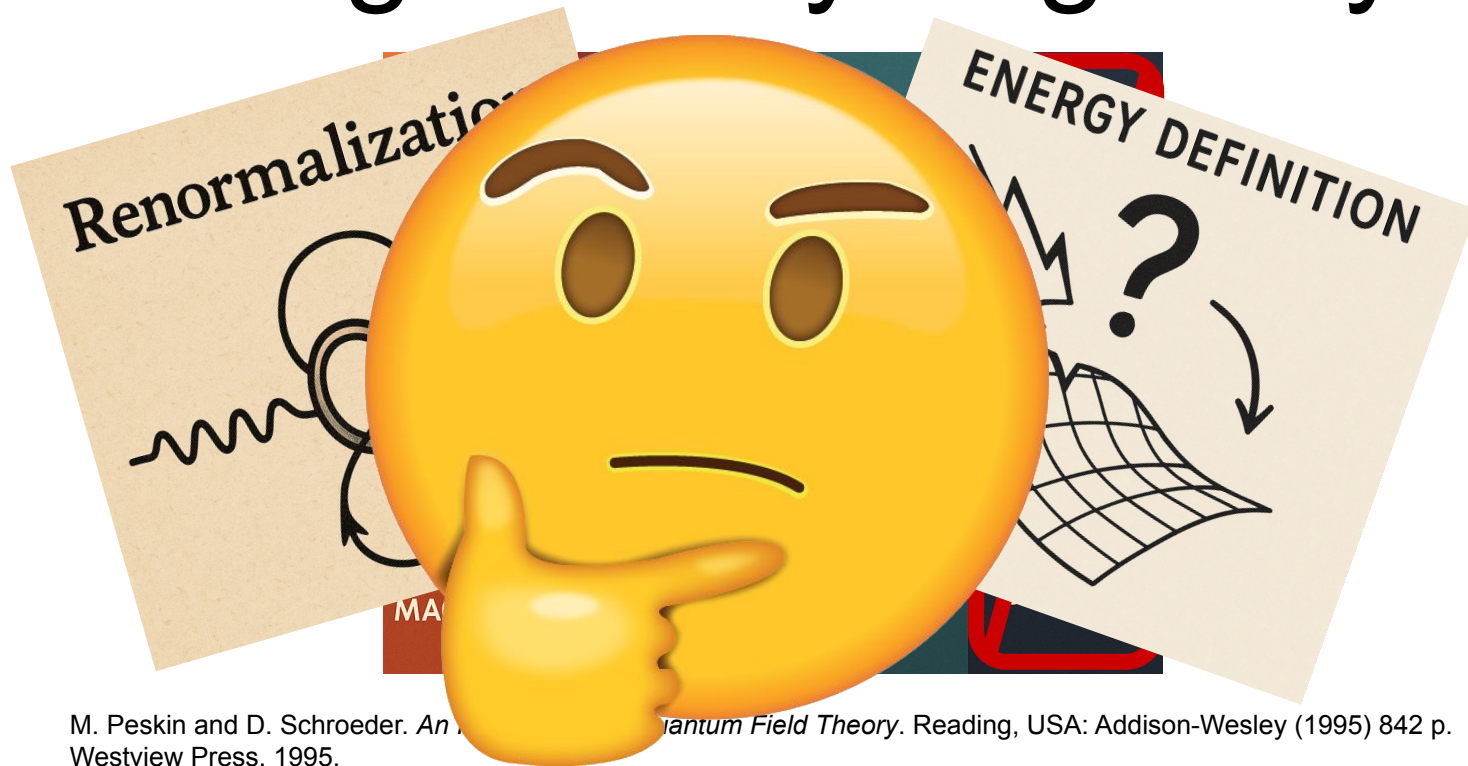
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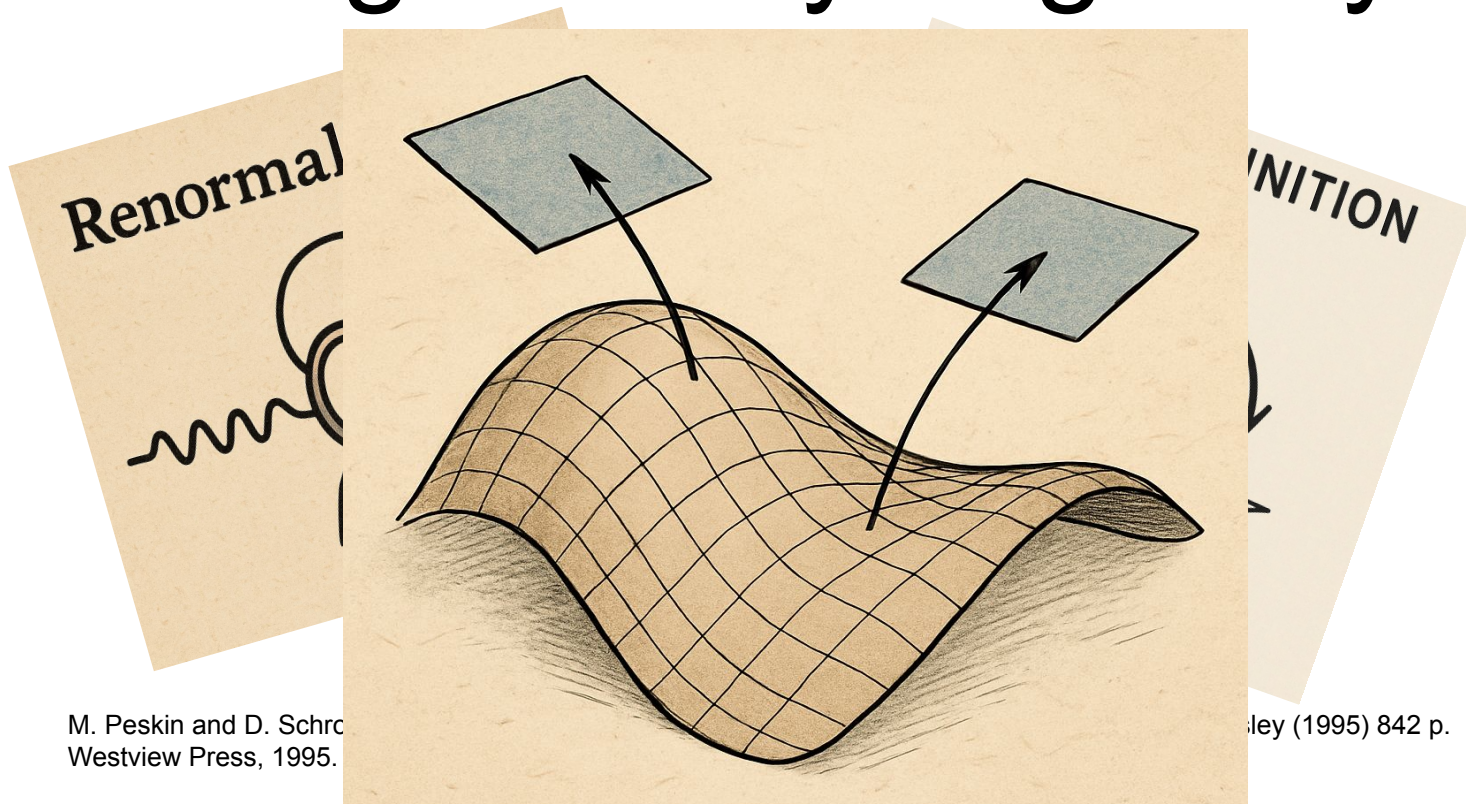
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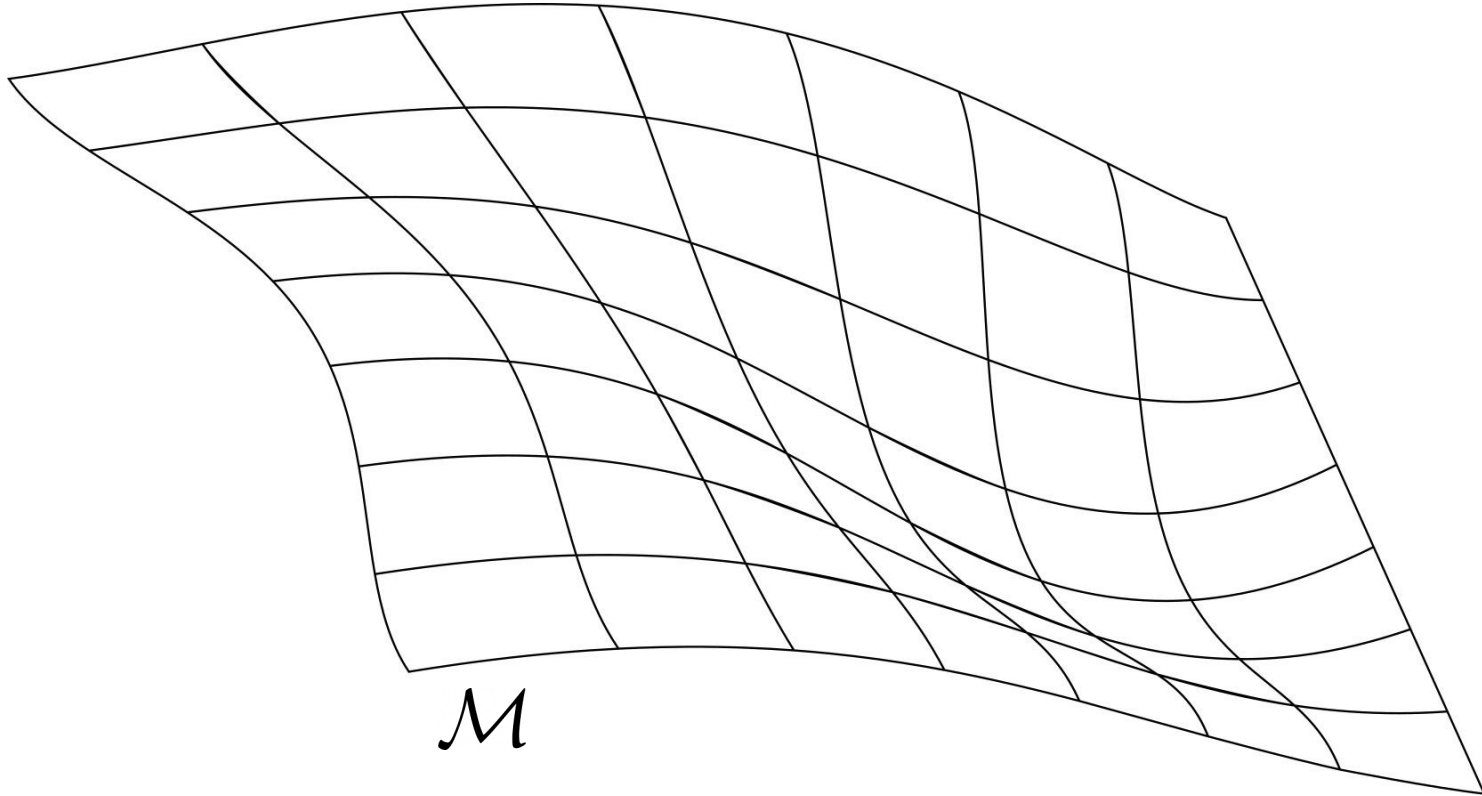
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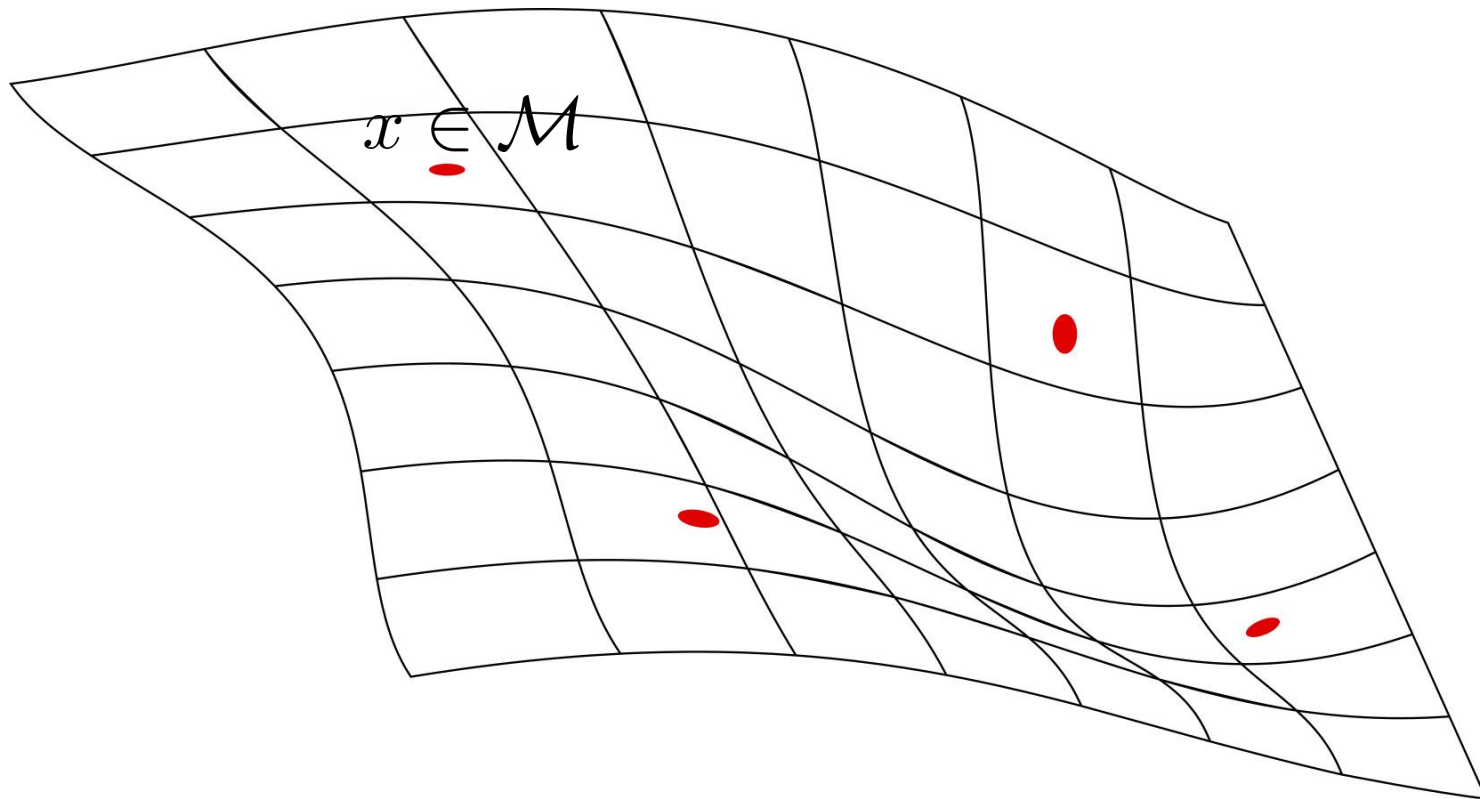
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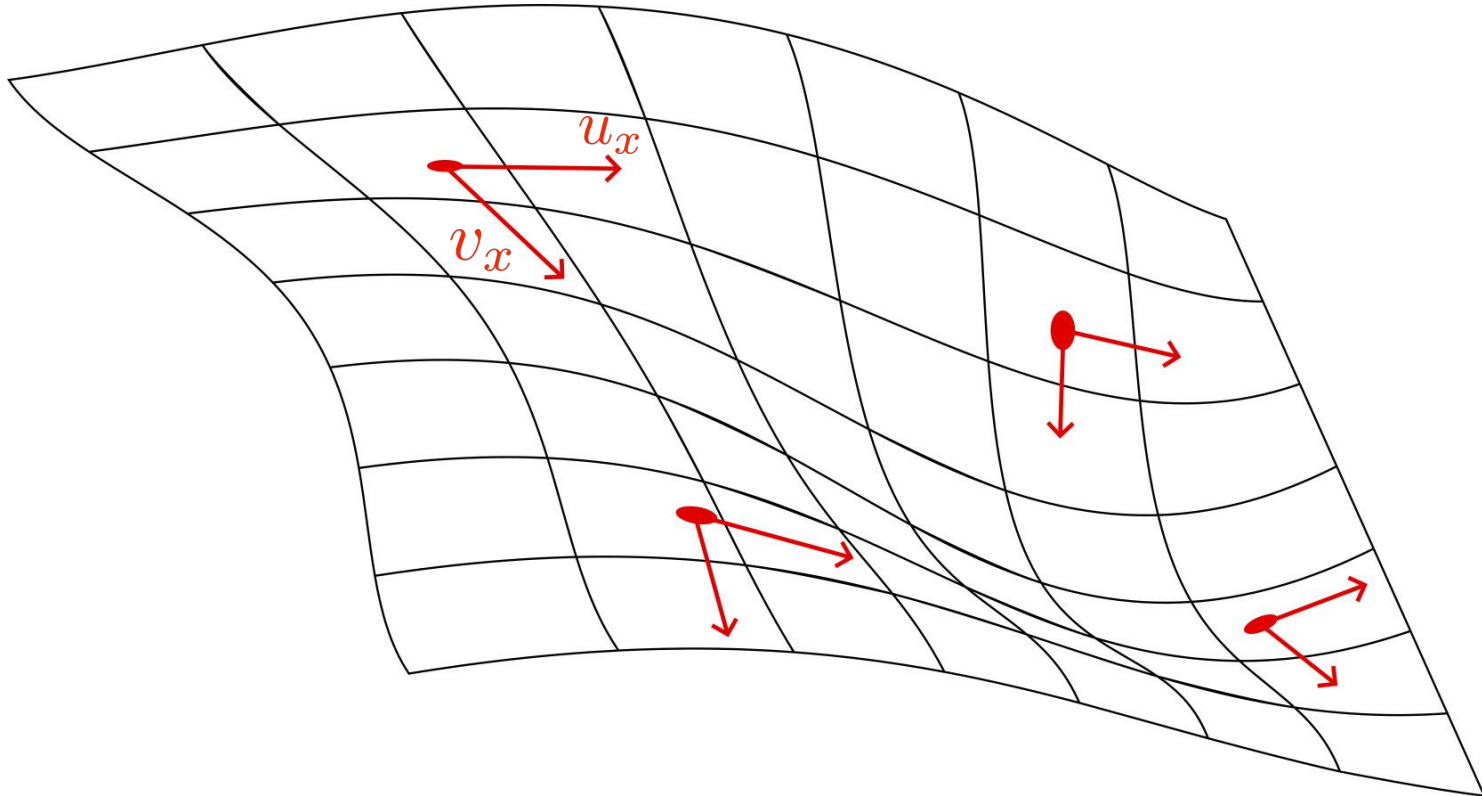
Geometry of General Relativity



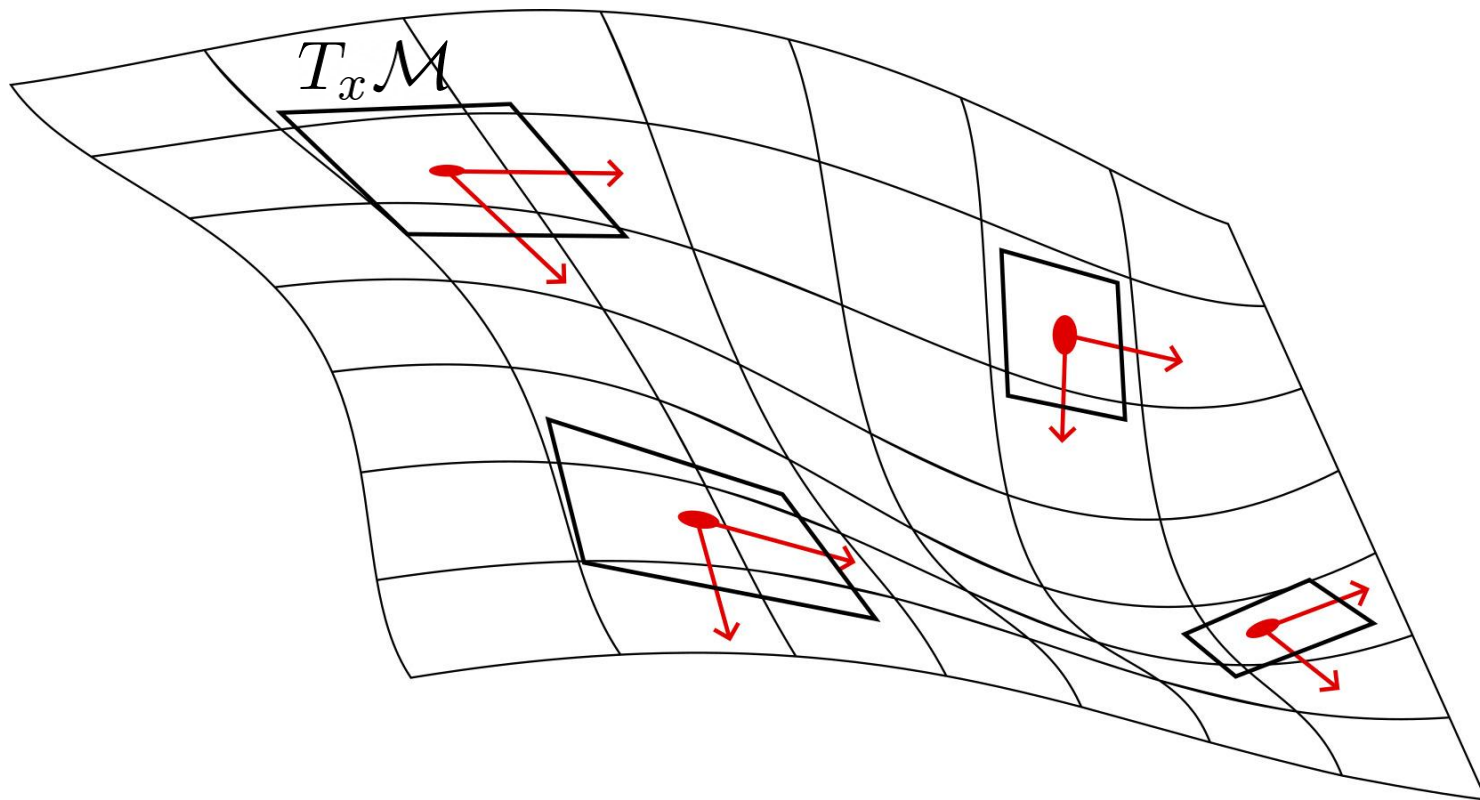
Geometry of General Relativity



Geometry of General Relativity

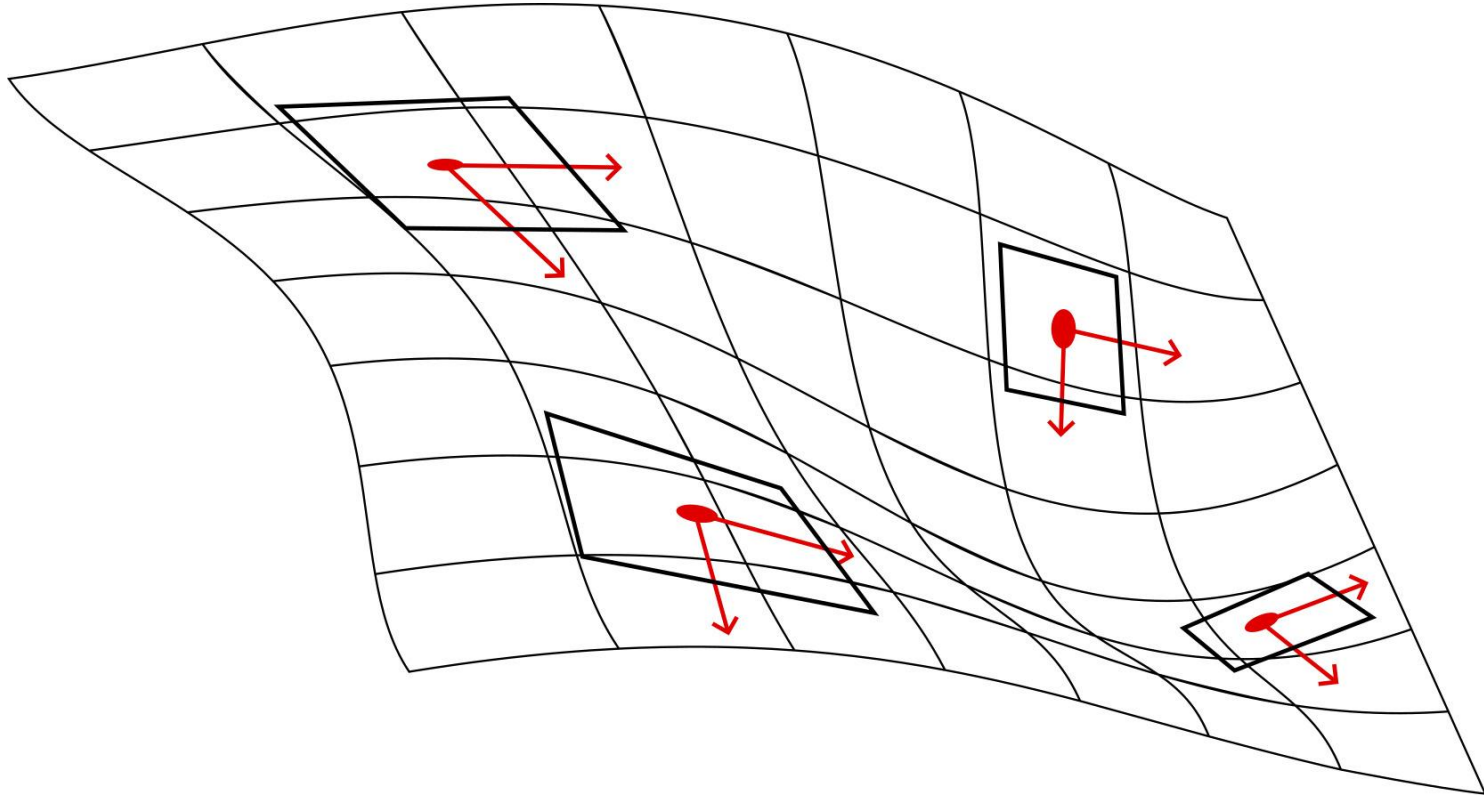


Geometry of General Relativity

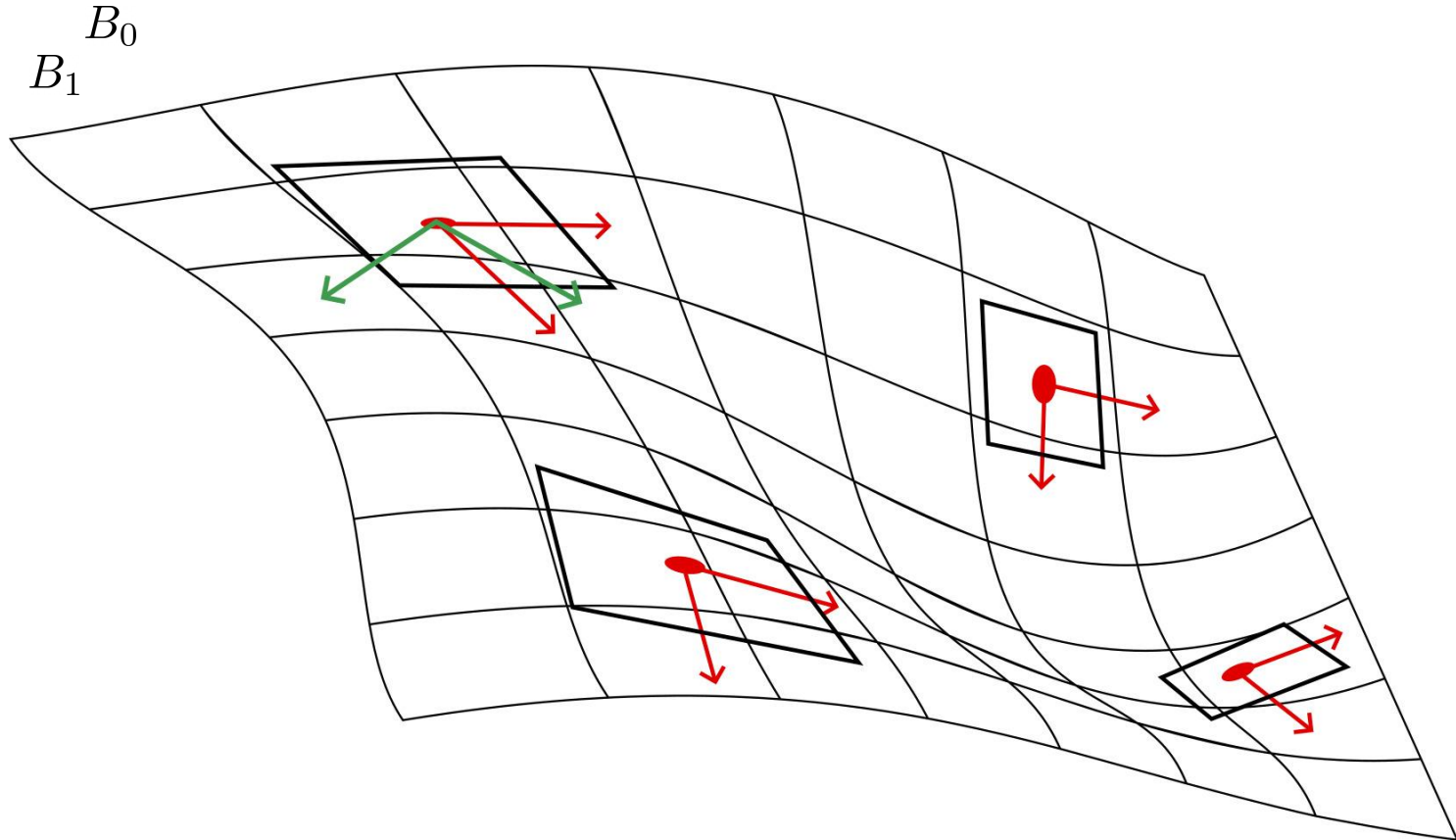


Geometry of General Relativity

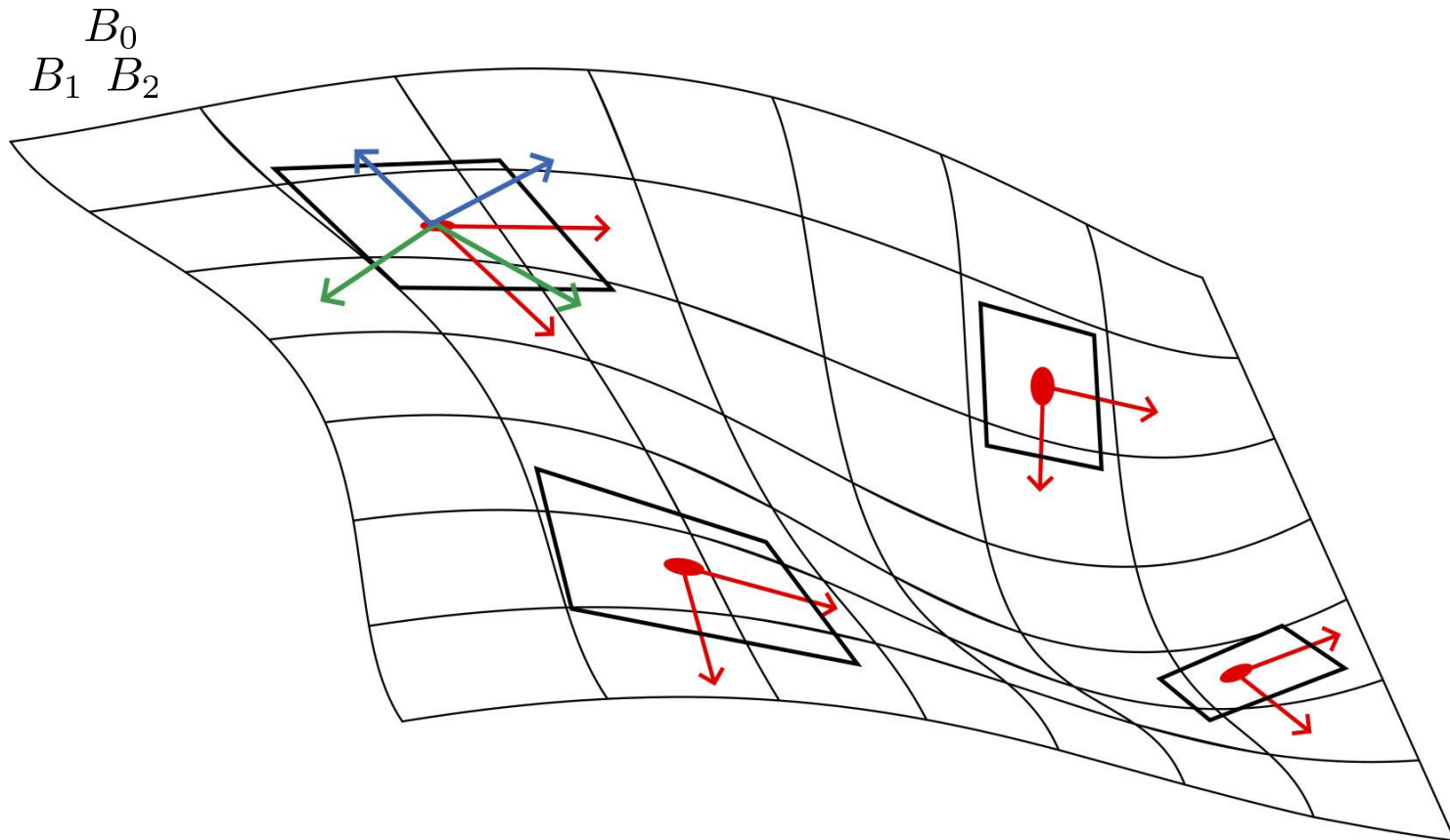
B_0



Geometry of General Relativity

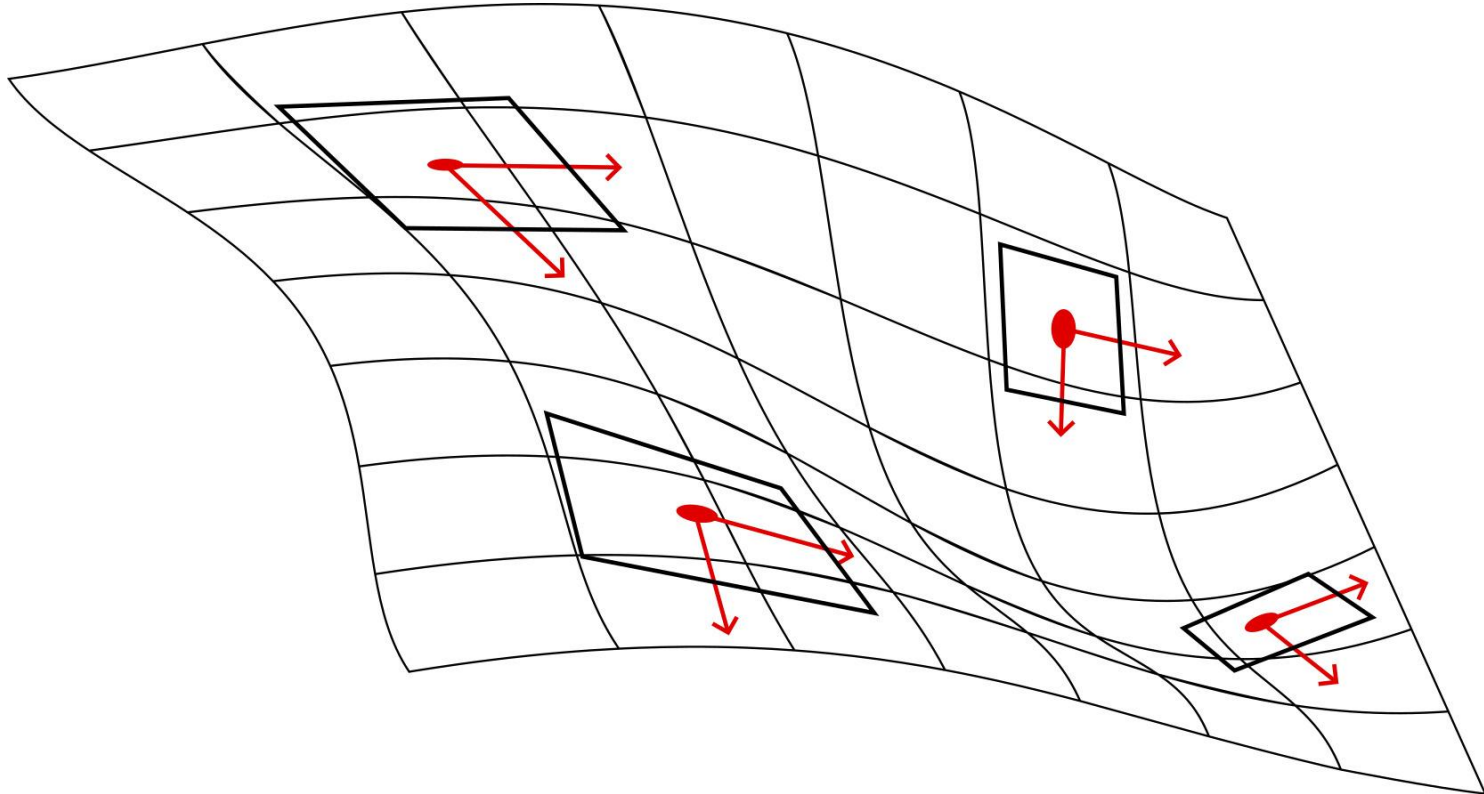


Geometry of General Relativity



Geometry of General Relativity

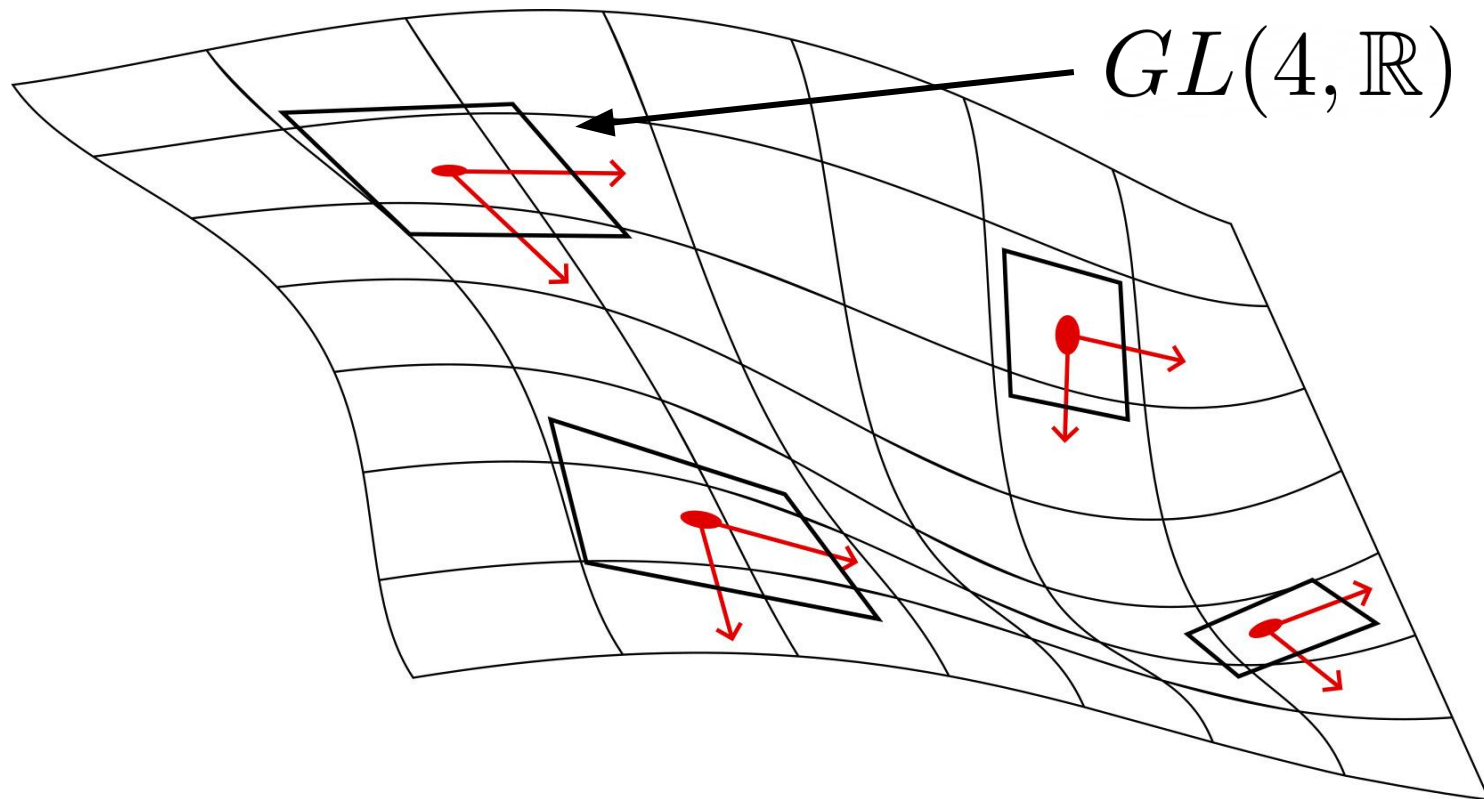
B_0



$$B_0 \rightarrow B_1$$

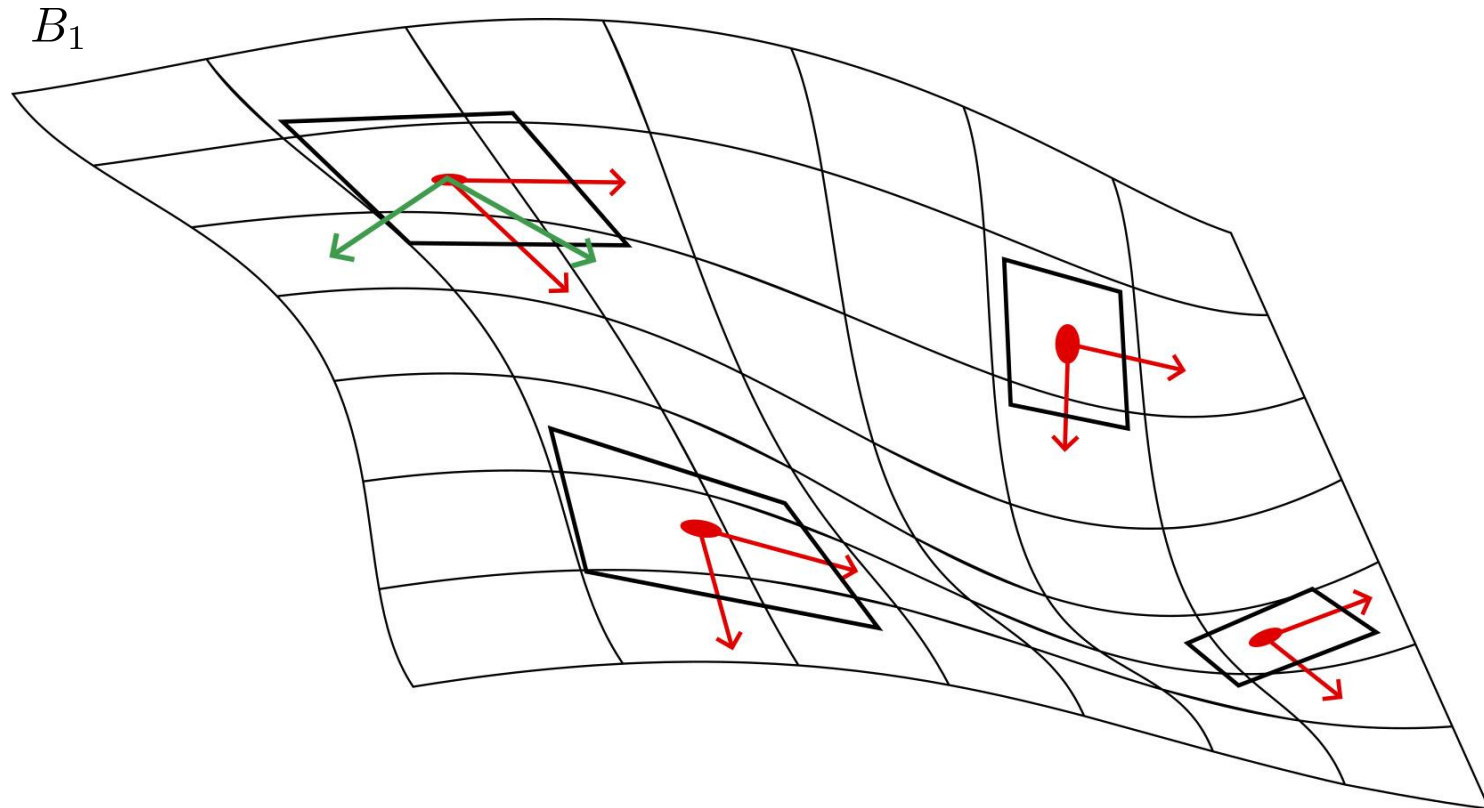
Geometry of General Relativity

B_0



$$B_0 \rightarrow B_1 : M_1 \in GL(4, \mathbb{R})$$

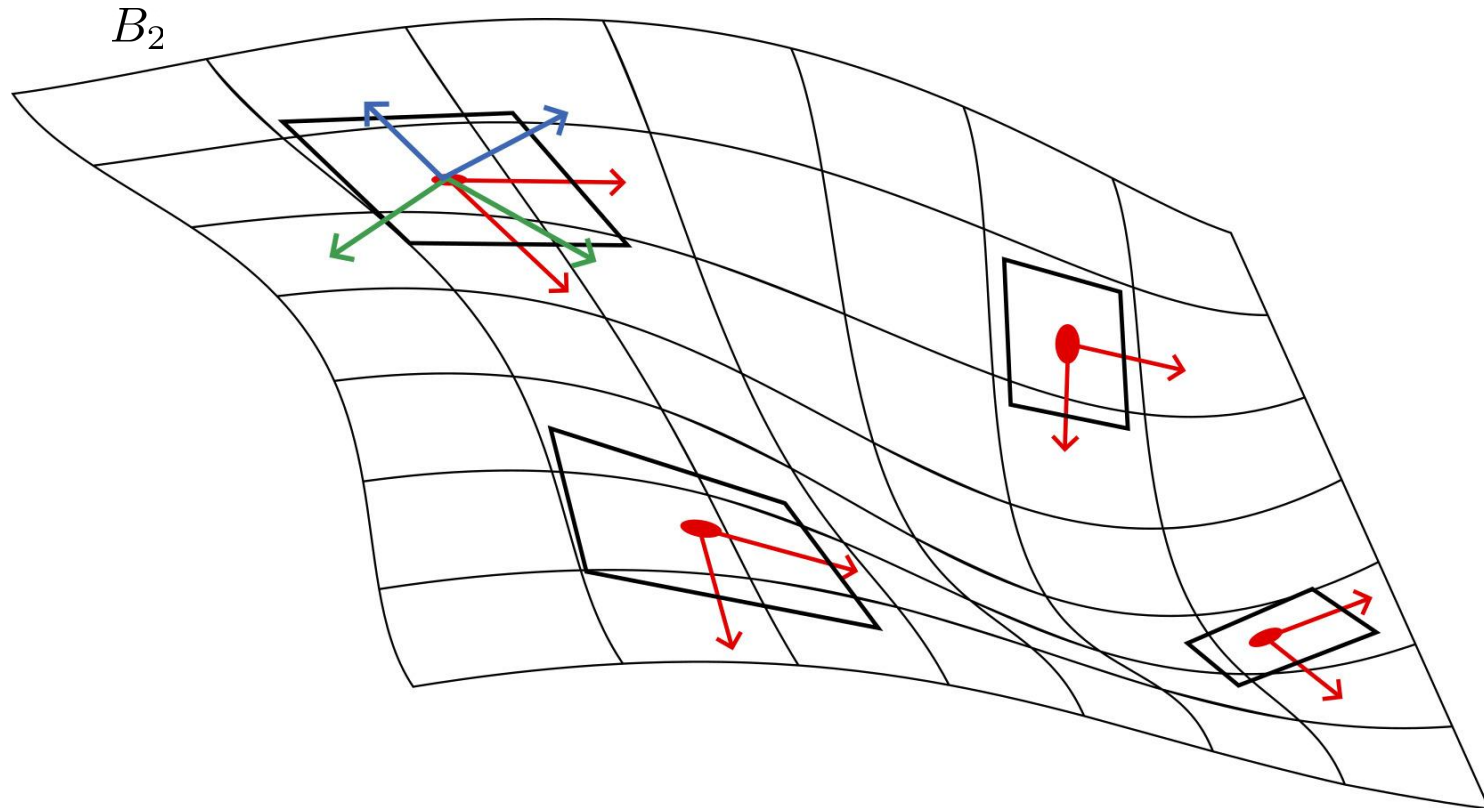
Geometry of General Relativity



$$B_0 \rightarrow B_1 : M_1 \in GL(4, \mathbb{R})$$

$$B_0 \rightarrow B_2$$

Geometry of General Relativity



$$B_0 \rightarrow B_1 : M_1 \in GL(4, \mathbb{R})$$

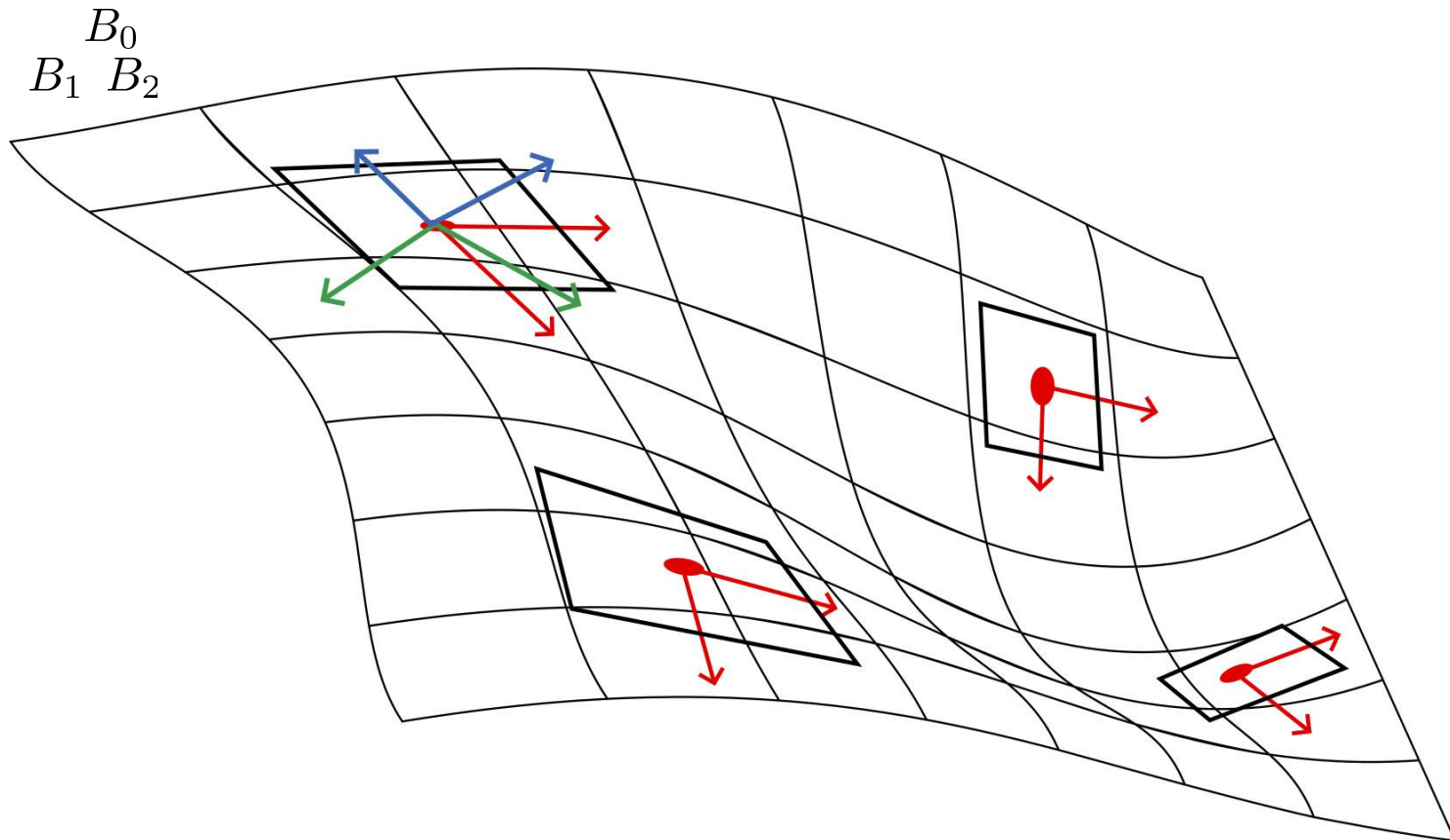
$$B_0 \rightarrow B_2 : M_2 \in GL(4, \mathbb{R})$$

$$B_0 \rightarrow B_1 : M_1 \in GL(4, \mathbb{R})$$

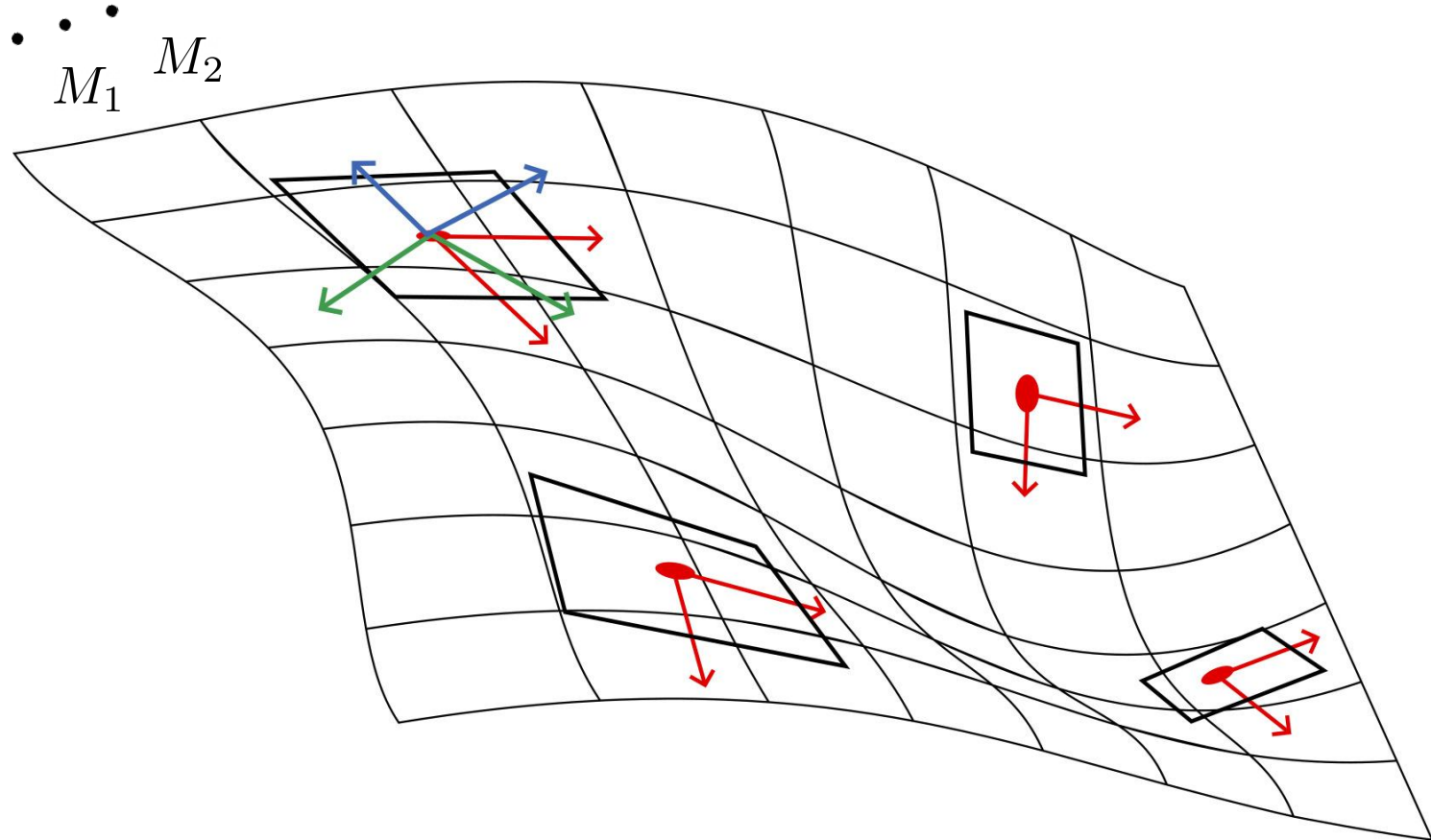
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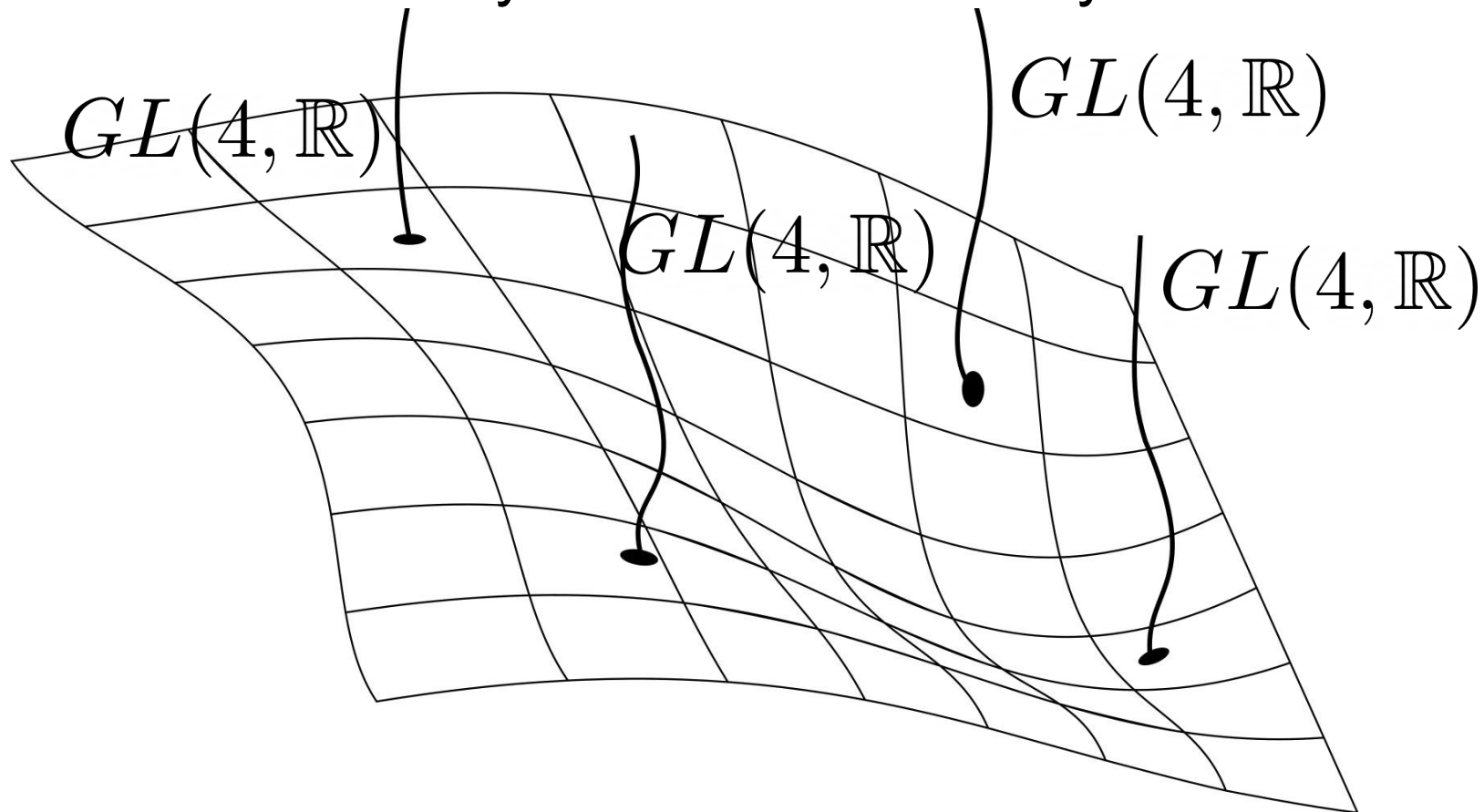
Geometry of General Relativity



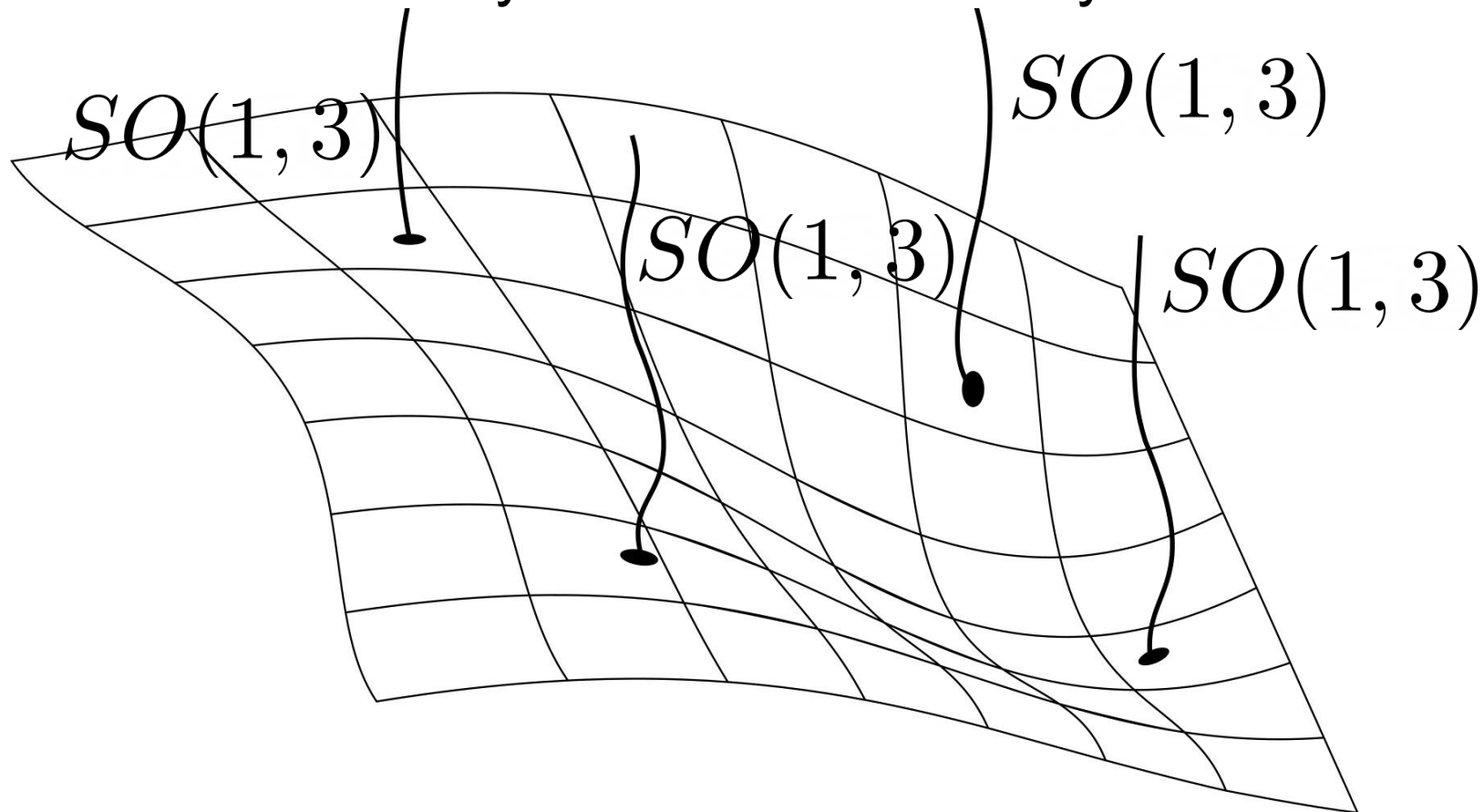
Geometry of General Relativity



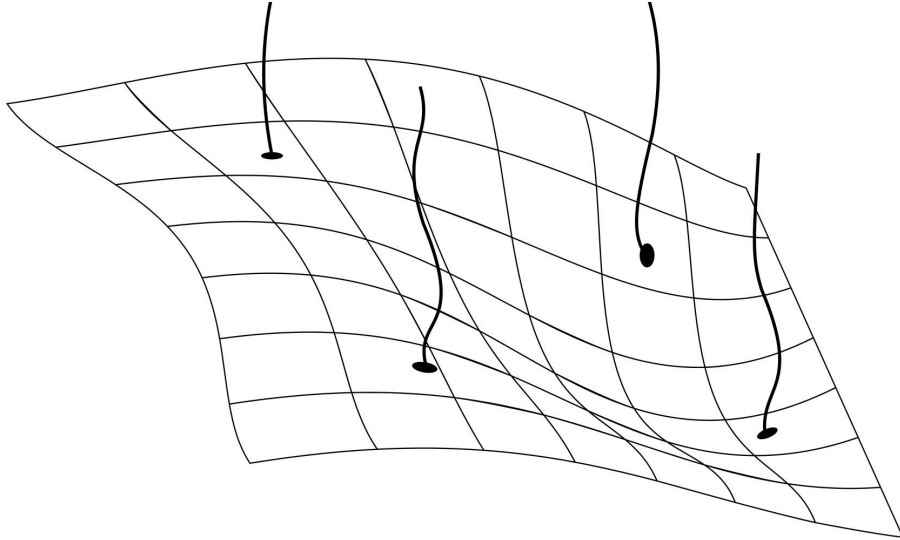
Geometry of General Relativity



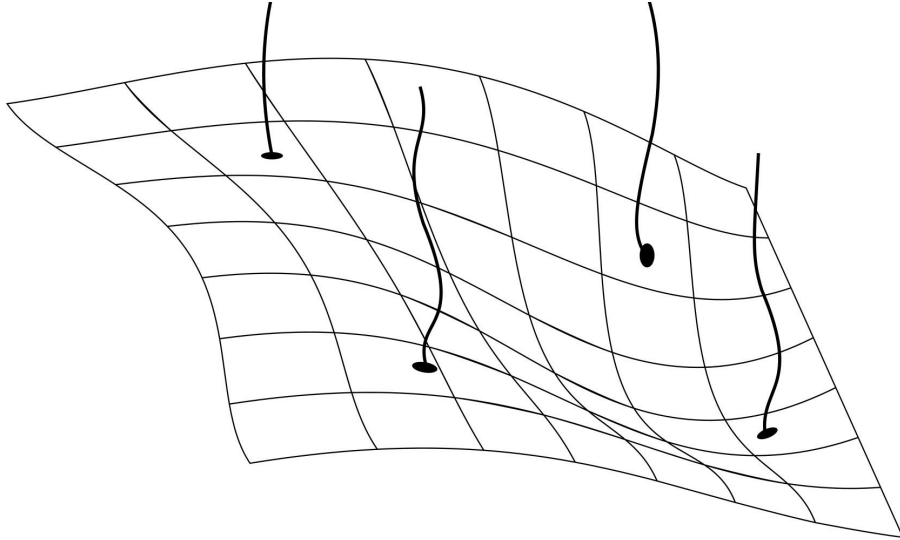
Geometry of General Relativity



Summary of the geometry of General Relativity

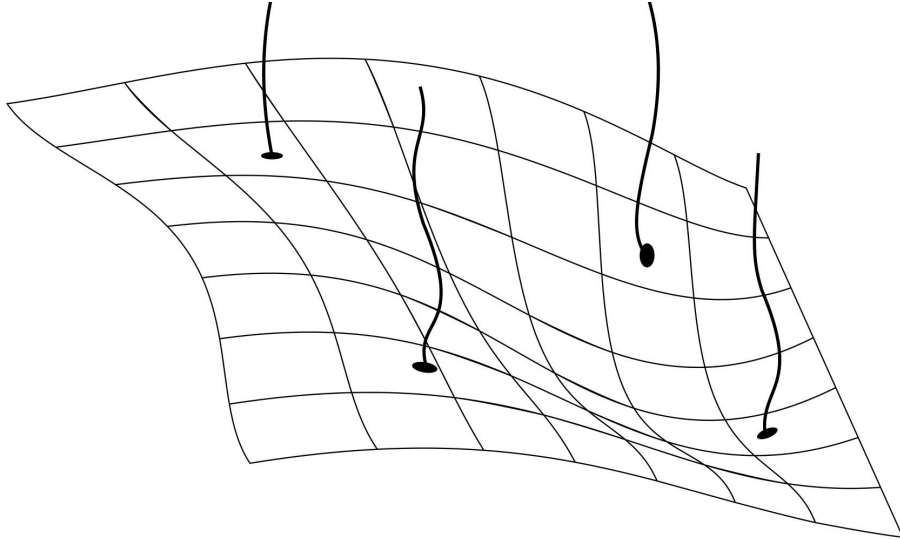


Summary of the geometry of General Relativity



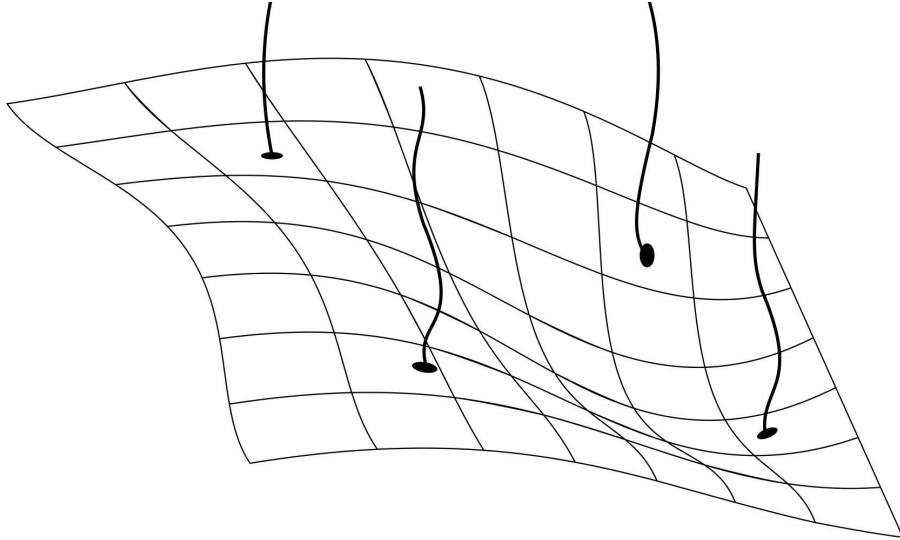
1. Principal $SO(1,3)$ -bundle

Summary of the geometry of General Relativity



1. Principal $SO(1,3)$ -bundle
2. Levi-Civita connection

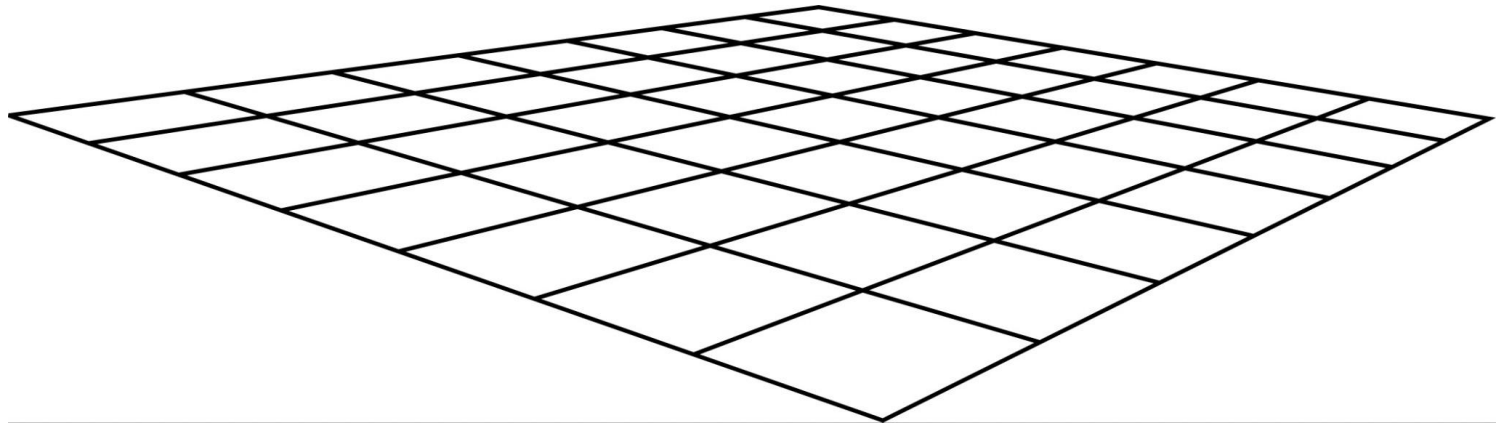
Summary of the geometry of General Relativity



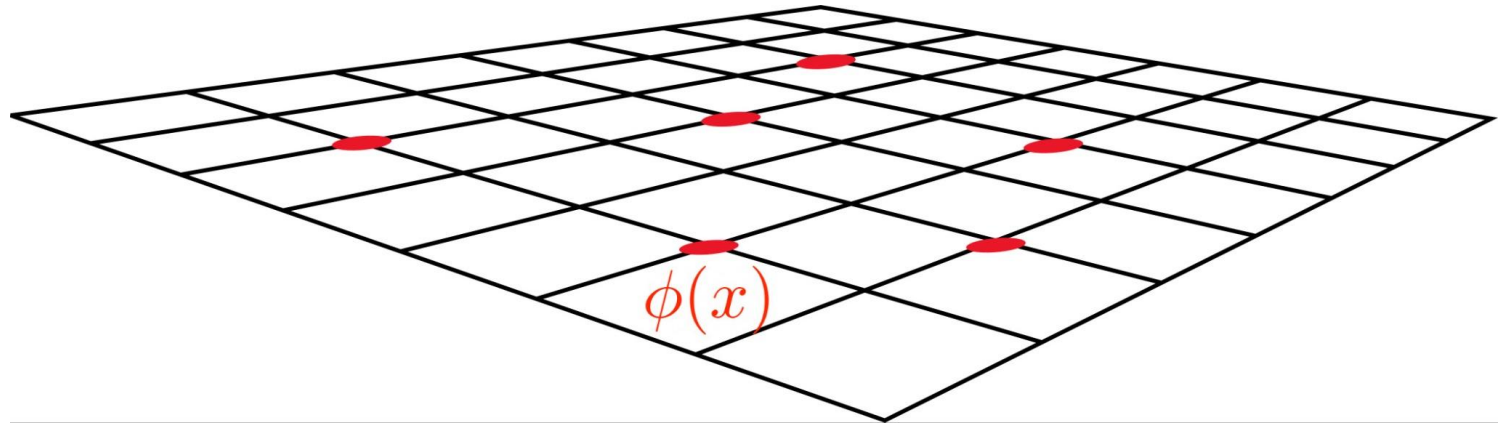
1. Principal $SO(1,3)$ -bundle
2. Levi-Civita connection
3. Covariant derivative

Gauge theories?

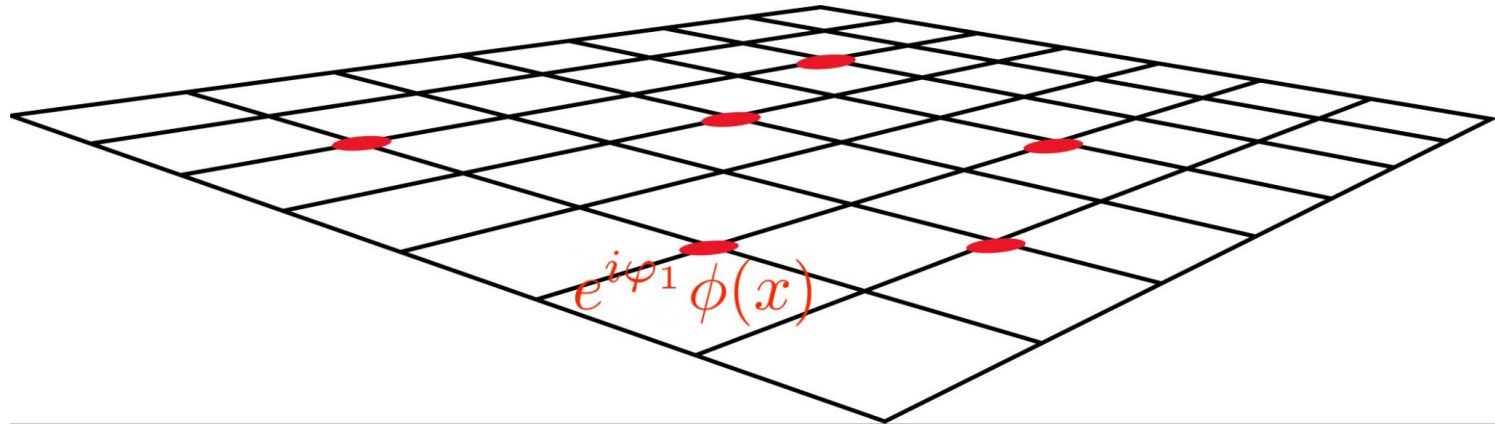
Geometry of Gauge theories



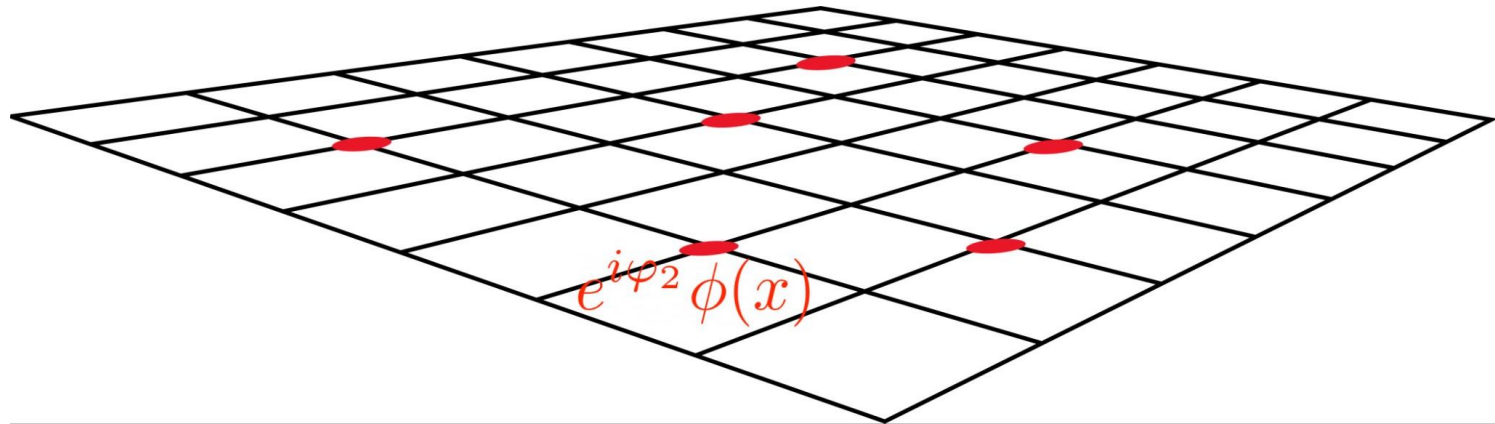
Geometry of Gauge theories



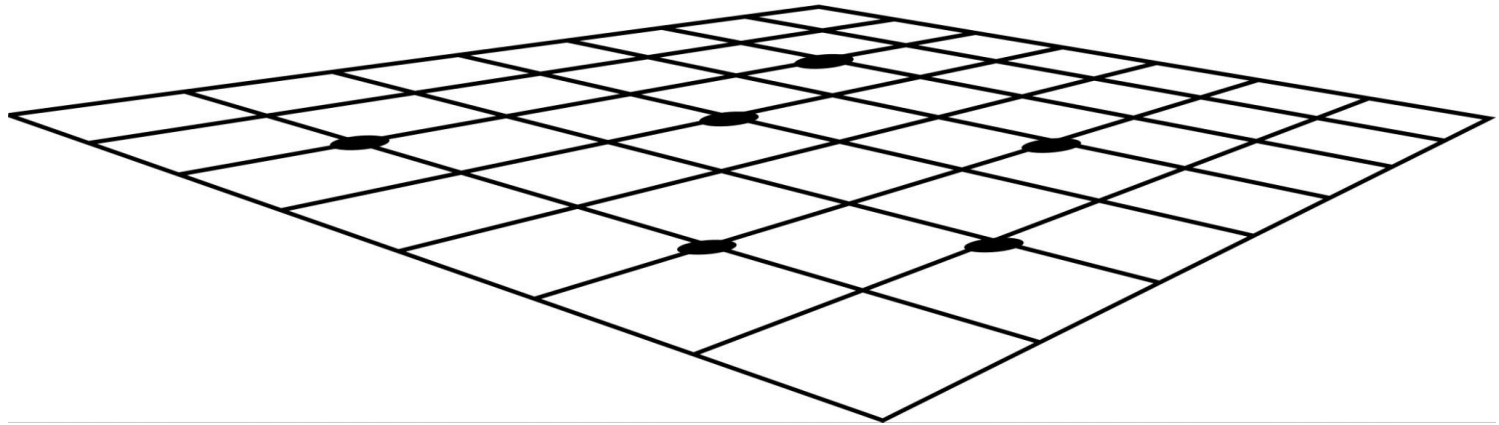
Geometry of Gauge theories



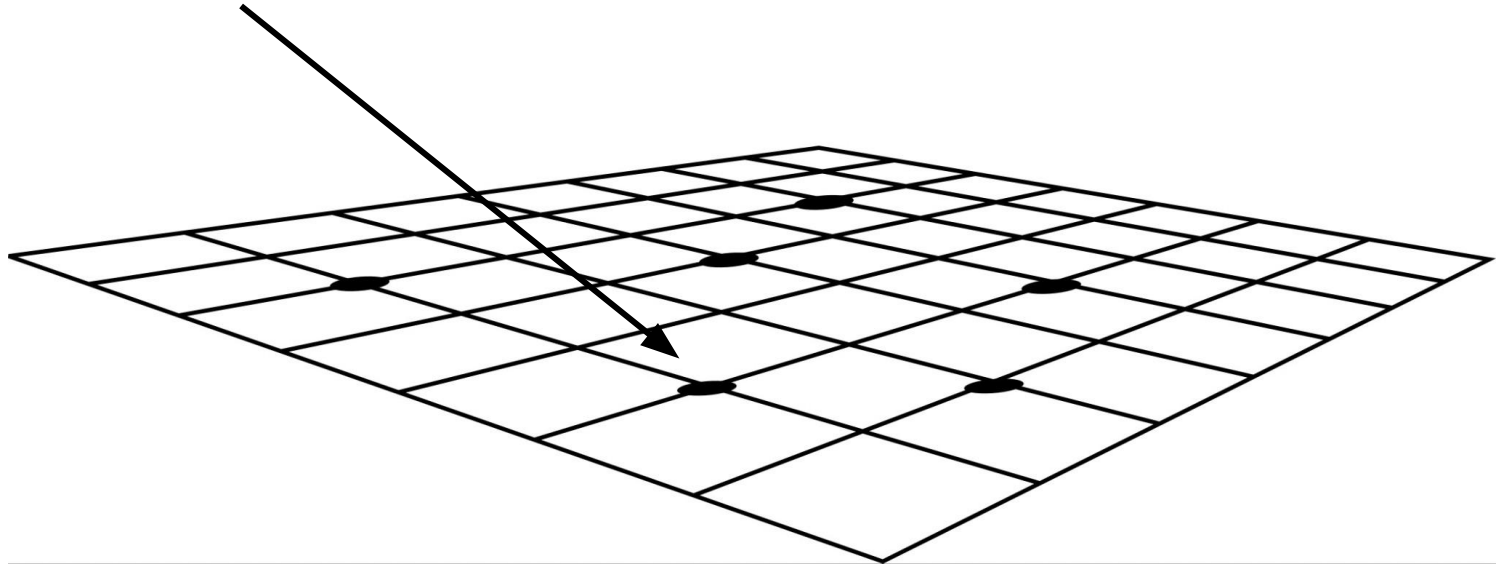
Geometry of Gauge theories



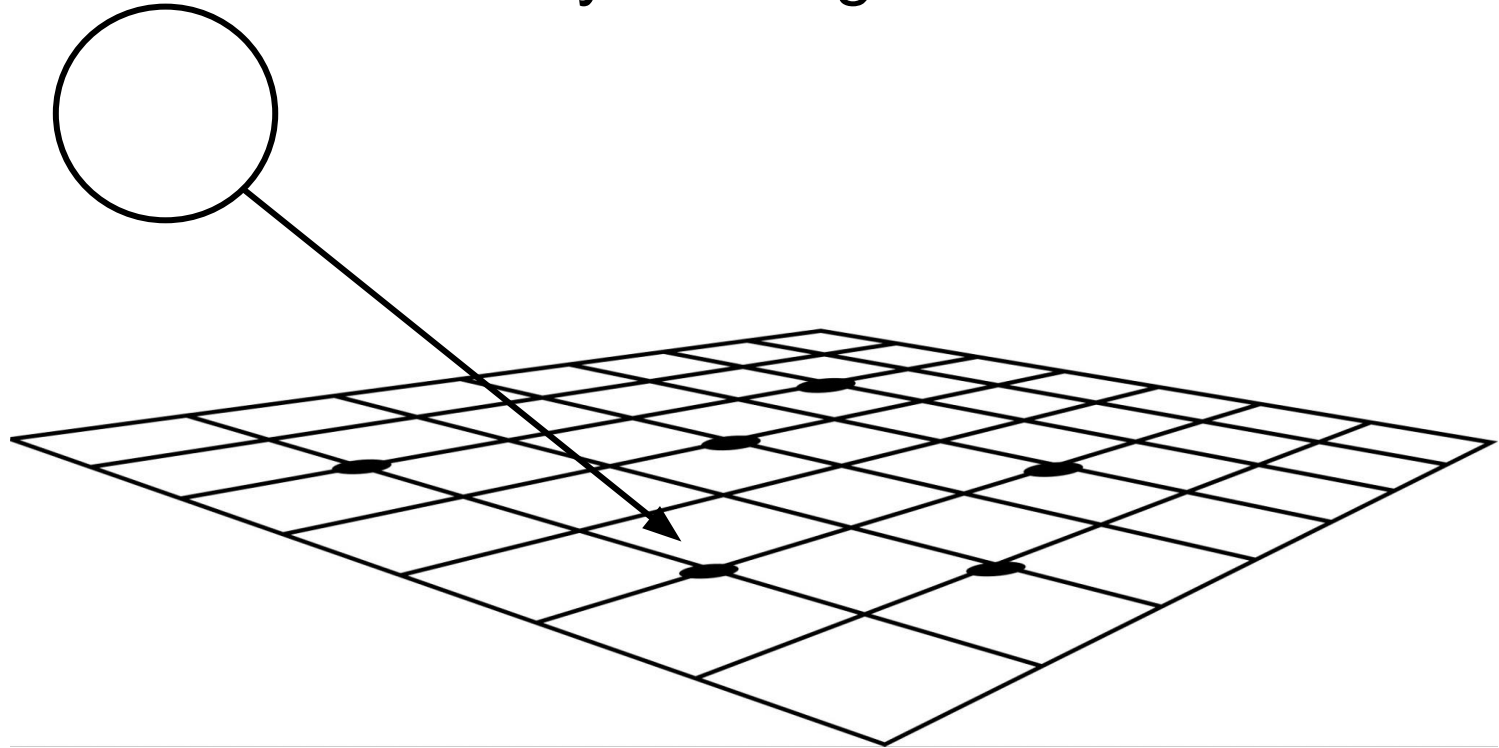
Geometry of Gauge theories



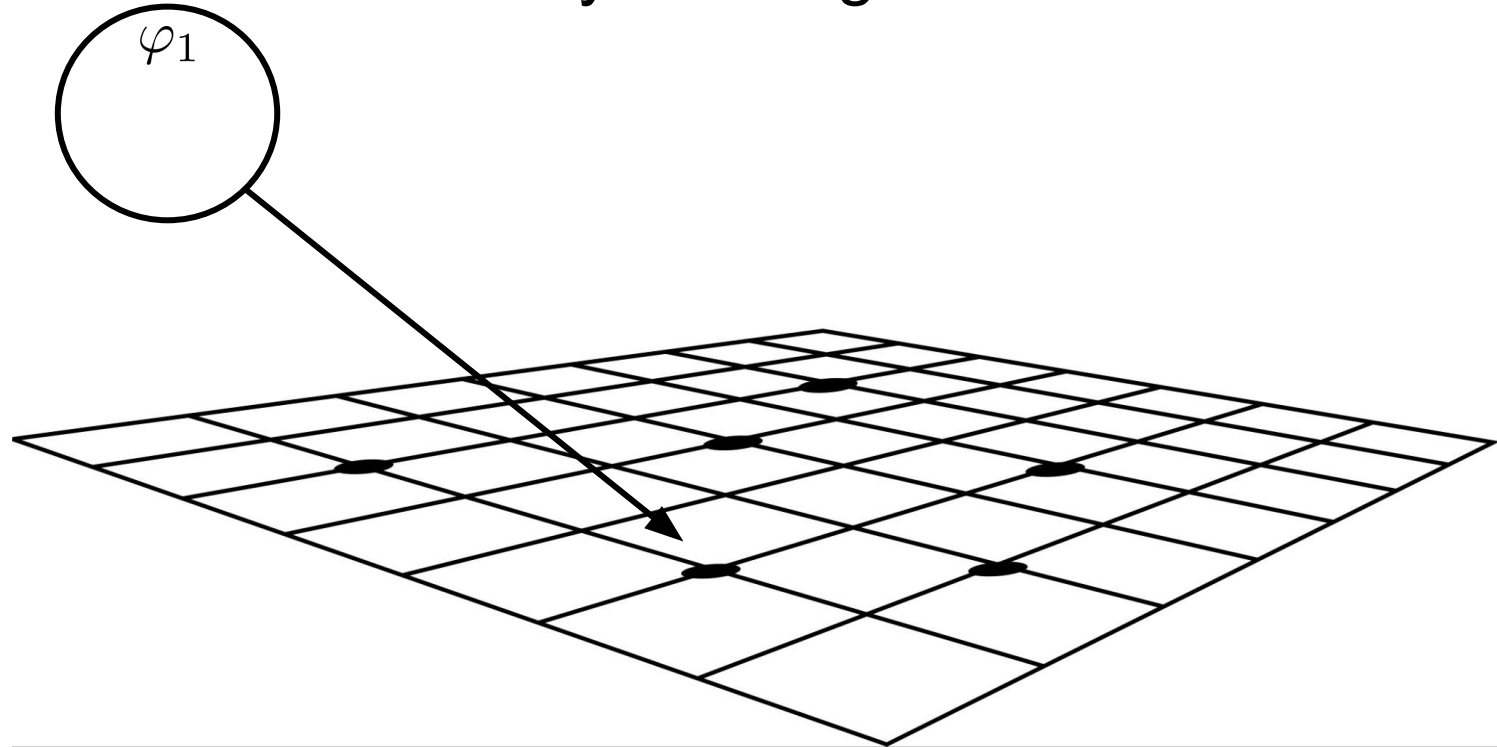
Geometry of Gauge theories



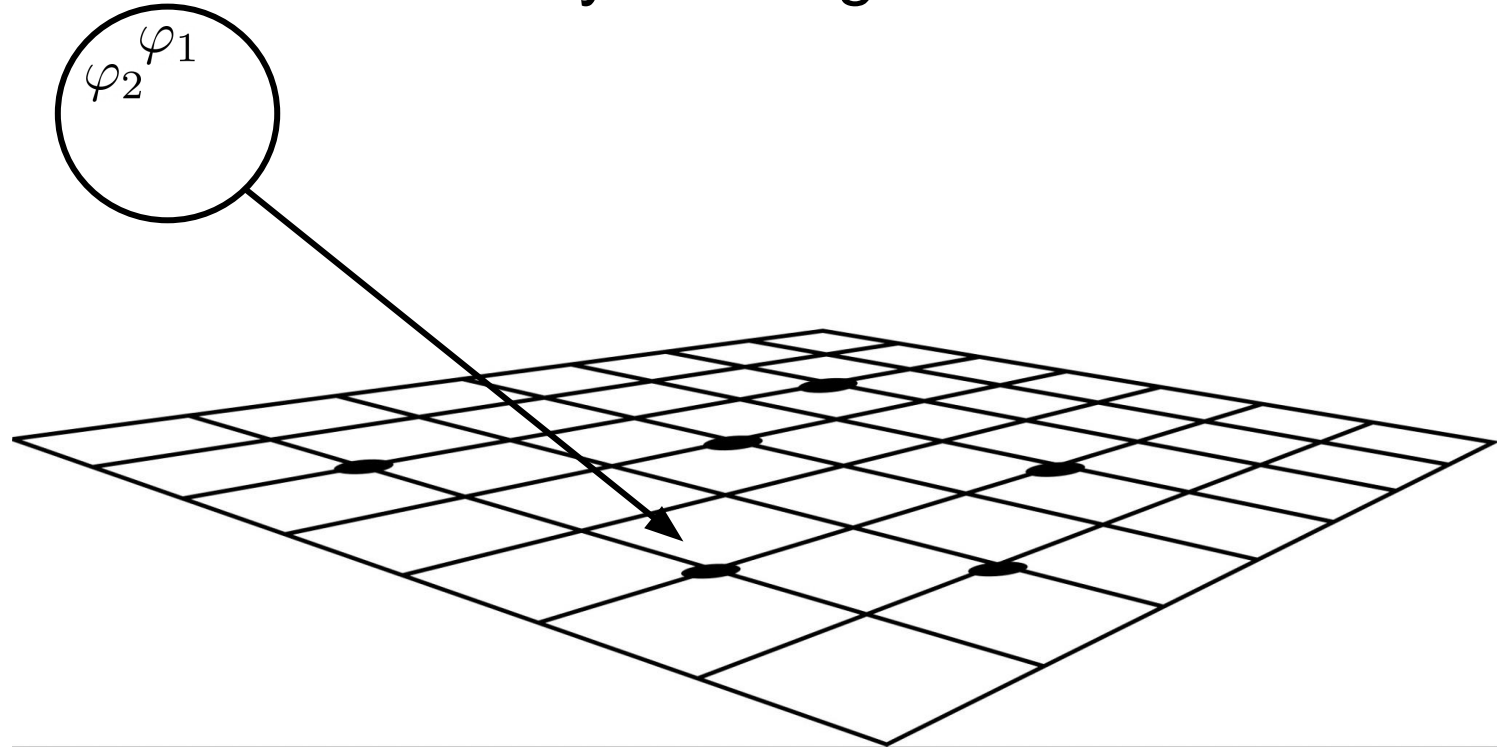
Geometry of Gauge theories



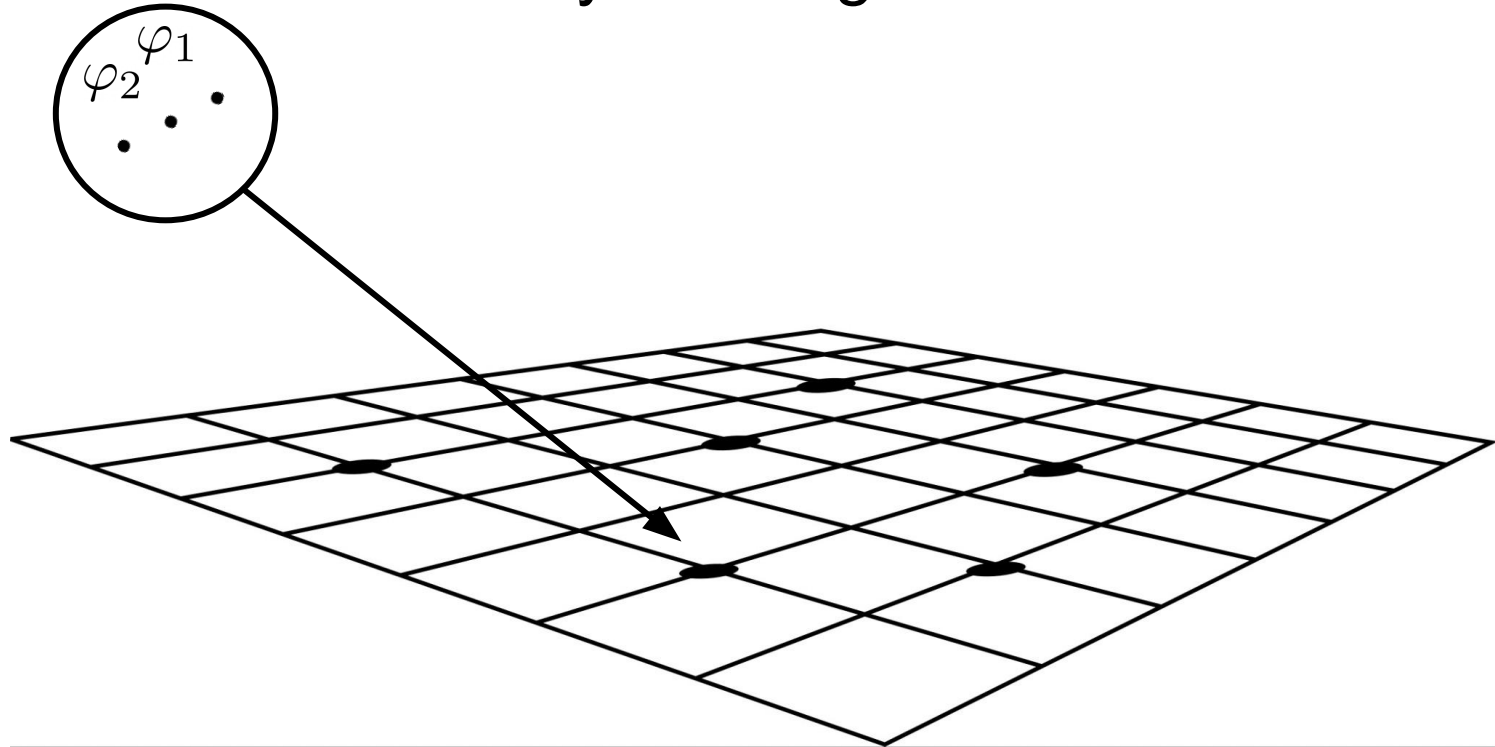
Geometry of Gauge theories



Geometry of Gauge theories

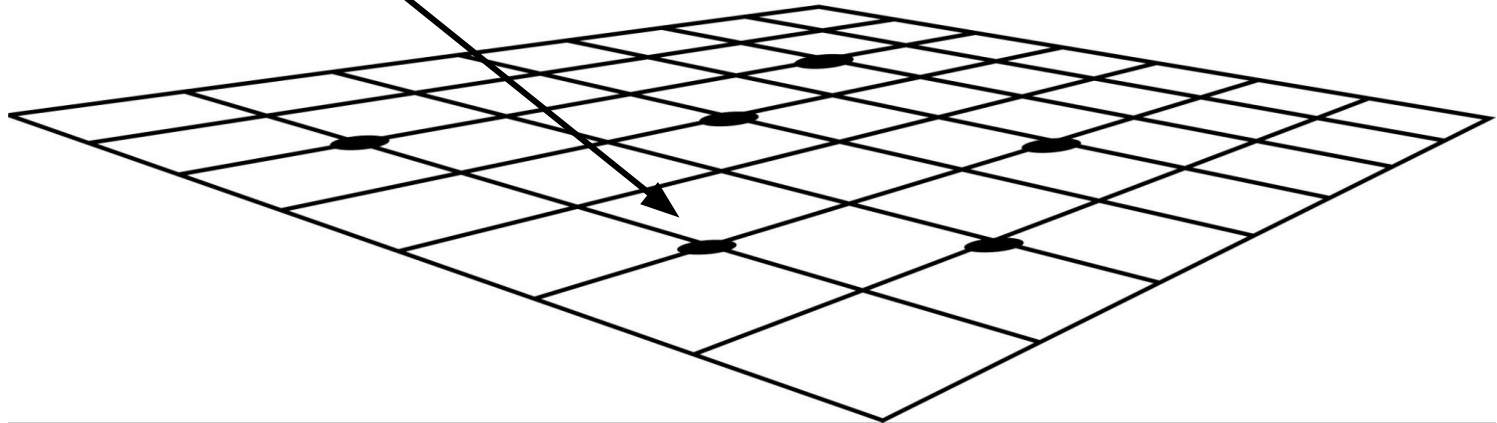


Geometry of Gauge theories

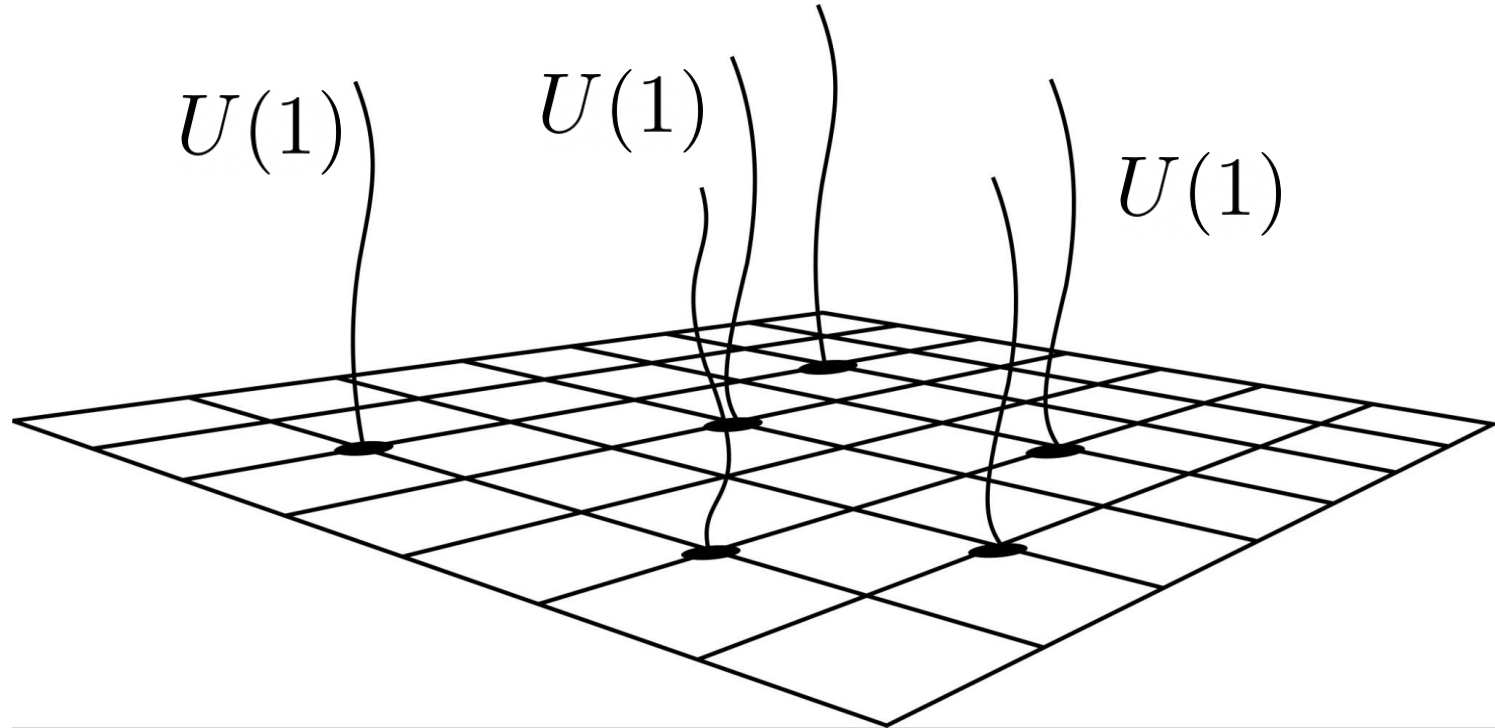


Geometry of Gauge theories

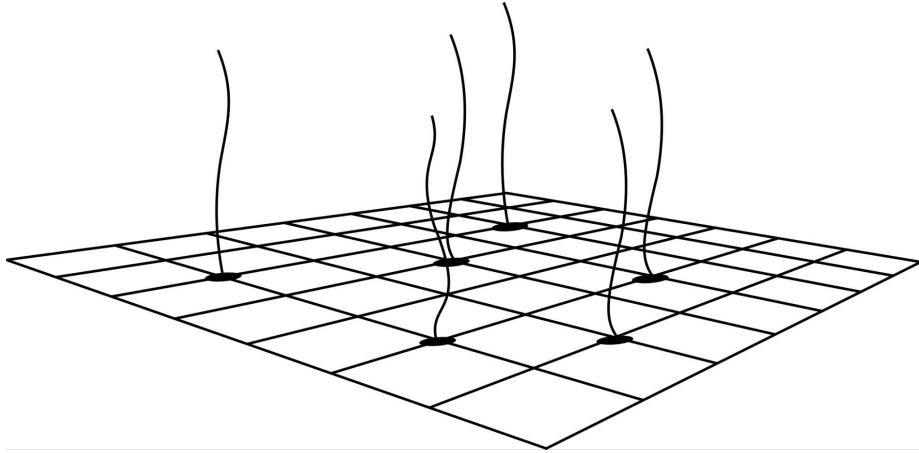
$U(1)$



Geometry of Gauge theories

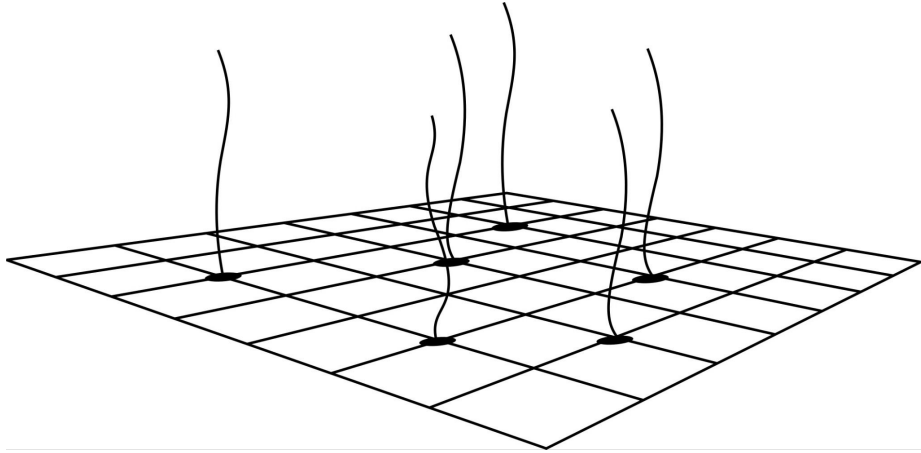


Summary of the geometry of Gauge theories



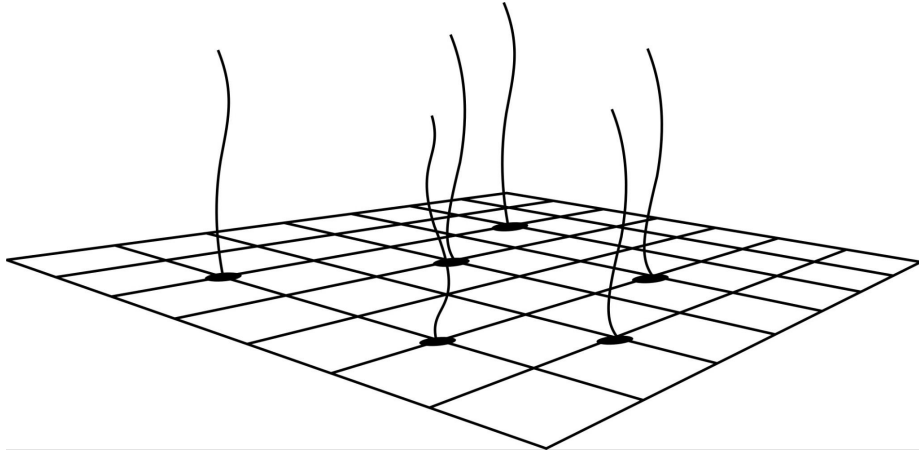
1. Principal $U(1)$ -bundle

Summary of the geometry of Gauge theories



1. Principal $U(1)$ -bundle
2. Connection $\equiv$ Gauge field

Summary of the geometry of Gauge theories



1. Principal $U(1)$ -bundle
2. Connection $\equiv$ Gauge field
3. Covariant derivative

1st comparison between General Relativity and Gauge theories

Theory	General Relativity	(Abelian) Gauge theory

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Table 1: Geometrical comparison between General Relativity and abelian Gauge theories

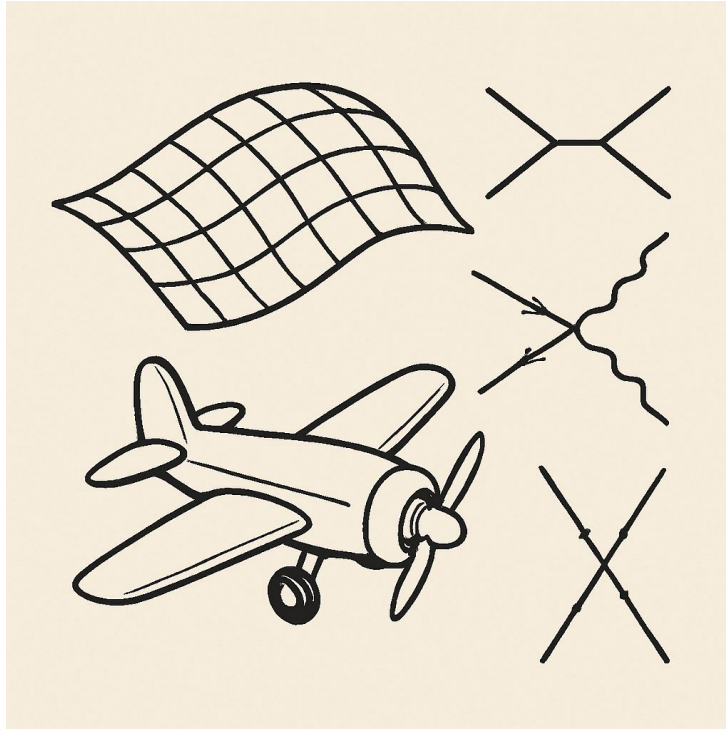
1st comparison between General Relativity and Gauge theories

Theory	General Relativity	Gauge theory
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	$(\omega, \sigma) \in GL(4, \mathbb{R})$	G
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Why is the Levi-Civita connection not a gauge field?

Table 1: Geometrical comparison between General Relativity and abelian Gauge theories

Our work



2nd comparison between General Relativity and Gauge theories

Theory	General Relativity	Gauge theories
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2nd comparison between General Relativity and Gauge theories

Theory	General Relativity	Gauge theories
Values of the connection	$\mathfrak{so}(1, 3)$	$\mathfrak{g}$

2nd comparison between General Relativity and Gauge theories

Theory	General Relativity	Gauge theories
Values of the connection	$\mathfrak{so}(1, 3)$	$\mathfrak{g}$
Noether currents	T^4	G

2nd comparison between General Relativity and Gauge theories

Theory	General Relativity	Gauge theories
Values of the connection	$\mathfrak{so}(1, 3)$	$\mathfrak{g}$
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Coupling	Energy of fields	Fields

Table 3: Group comparison between General Relativity, Gauge theories and our toy model

Toy model

Theory	General Relativity	Gauge theories	Toy model
Values of the connection	$\mathfrak{so}(1, 3)$	$\mathfrak{g}$	
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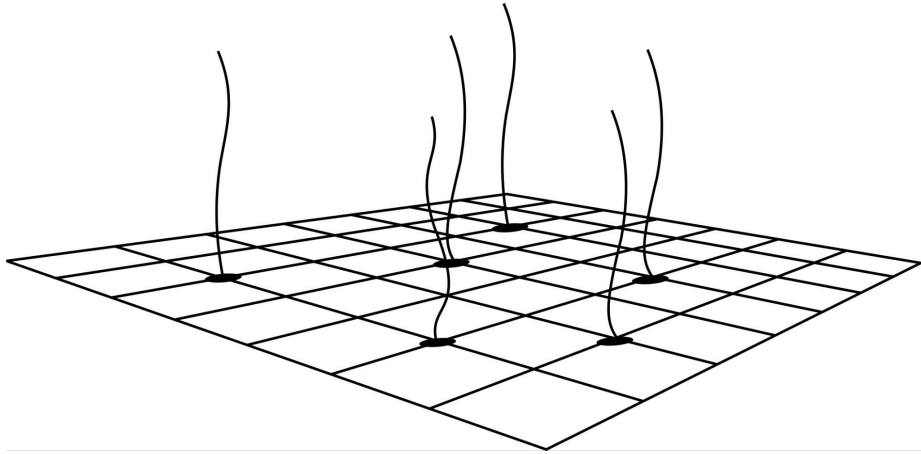
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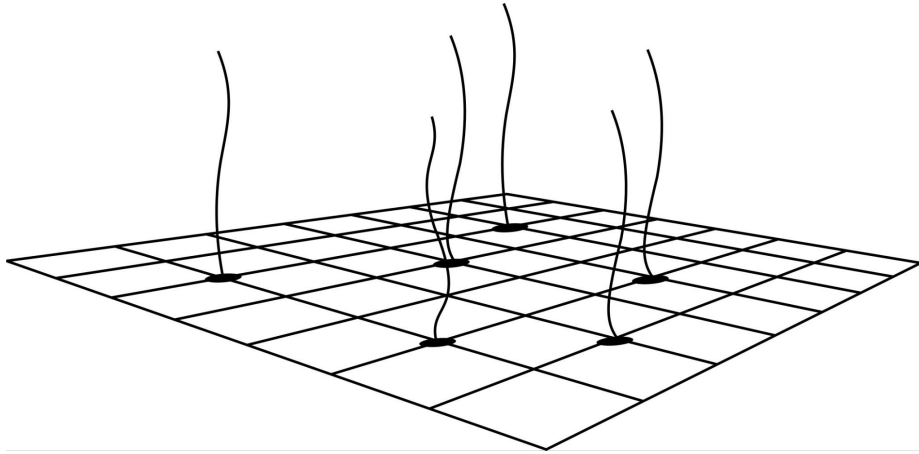
Table 3: Group comparison between General Relativity, Gauge theories and our toy model

Constructing the toy model



Constructing the toy model

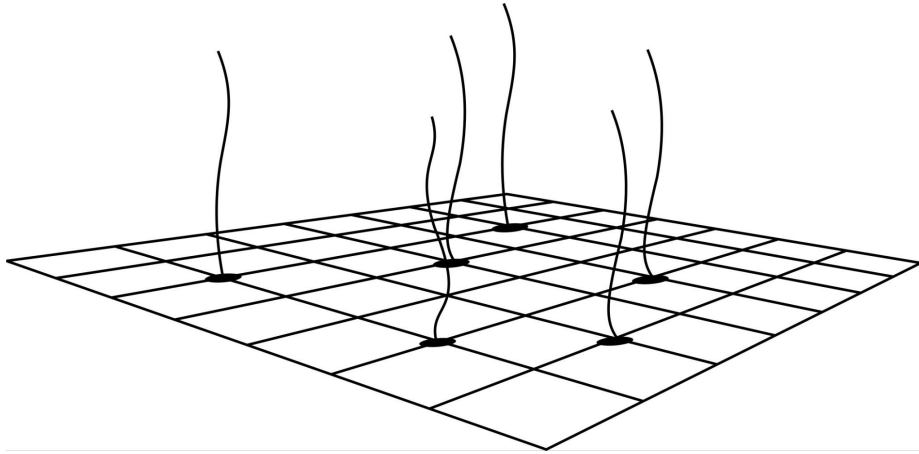
1. Principal G-bundle



Constructing the toy model

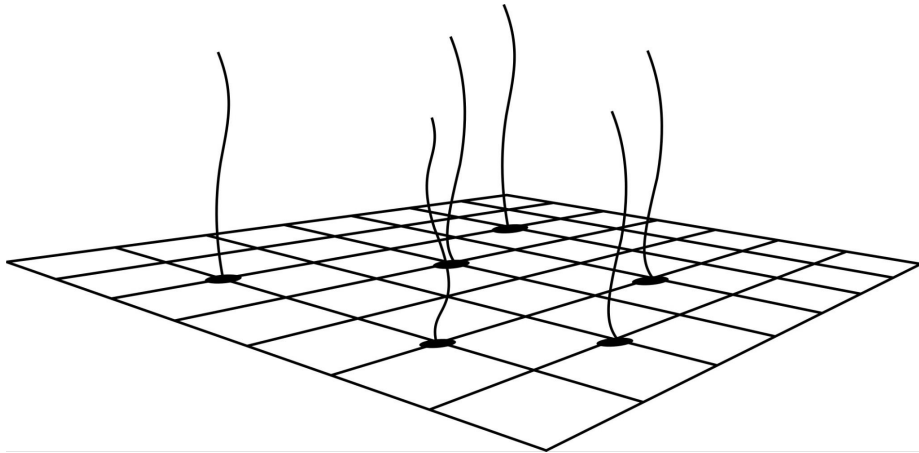
1. Principal G-bundle

1.1. Lie group G



Constructing the toy model

1. Principal G-bundle
 - 1.1. Lie group G
 - 1.2. Action of G (representation)



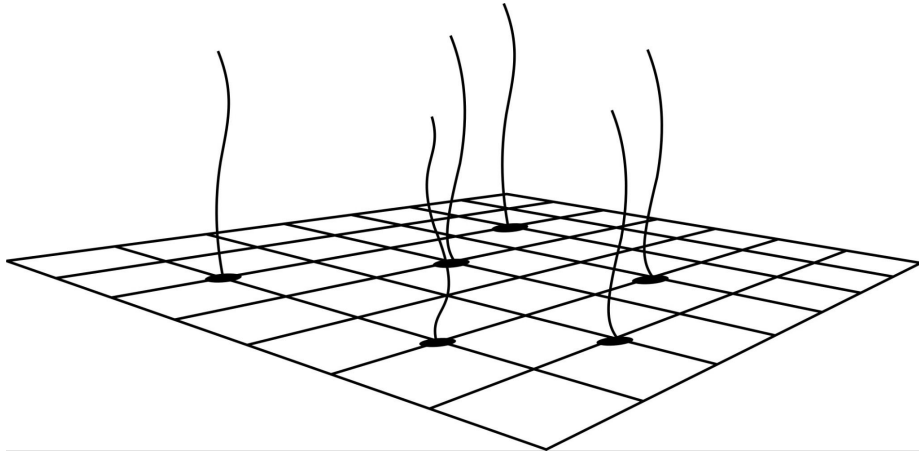
Constructing the toy model

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Constructing the toy model

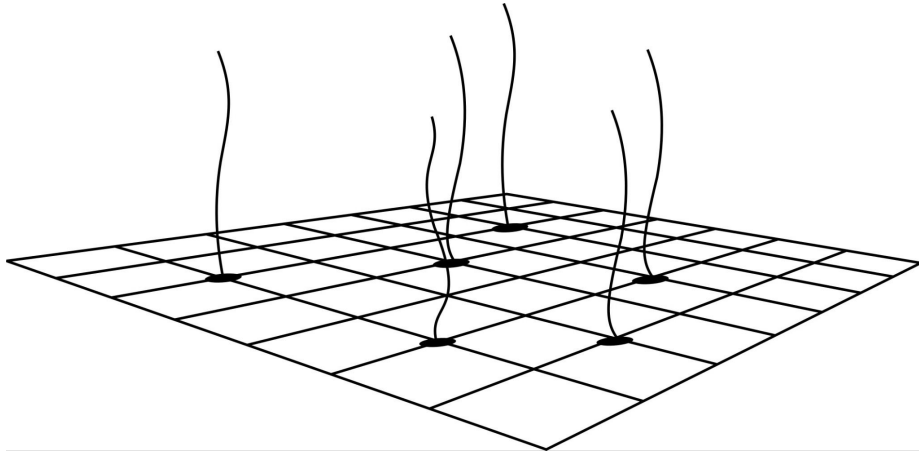
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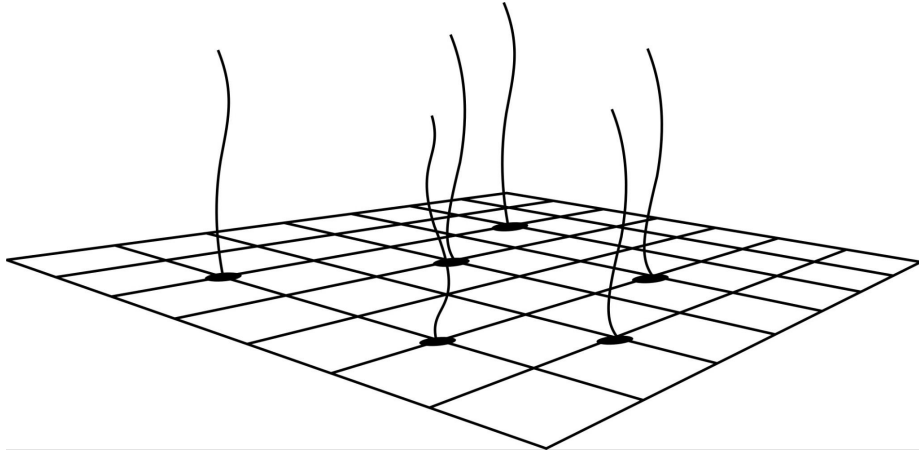
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1.4. Variation of action

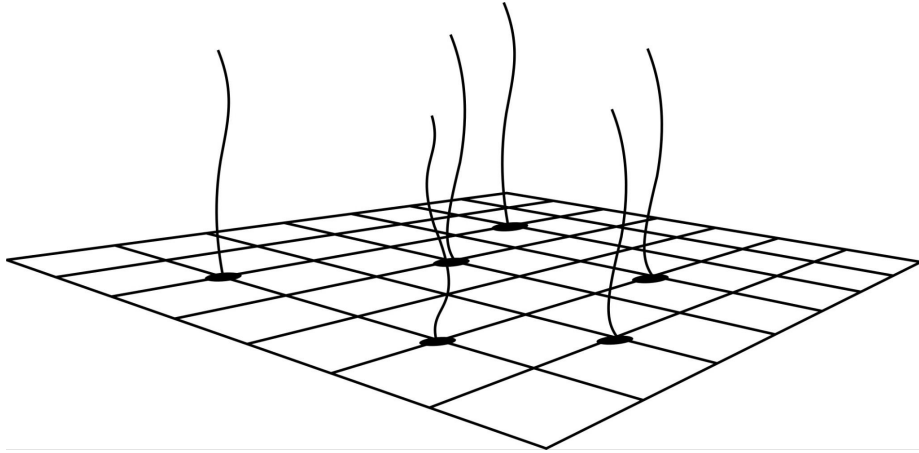


Constructing the toy model



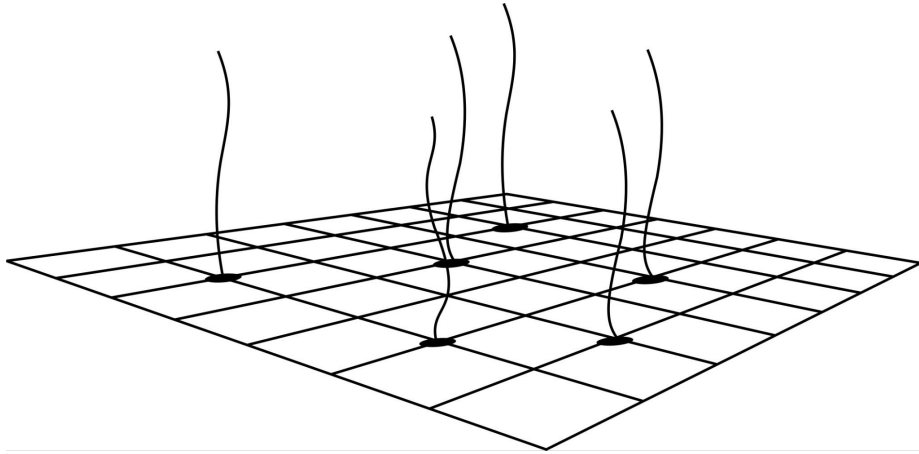
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Constructing the toy model



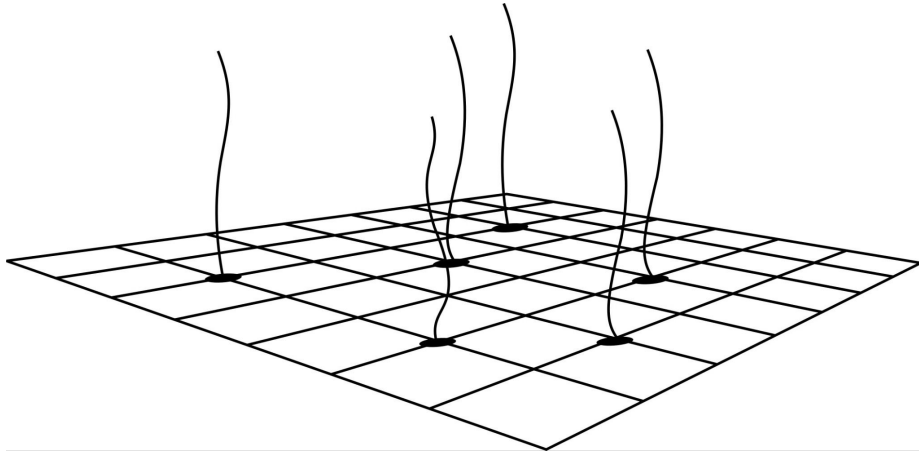
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Constructing the toy model



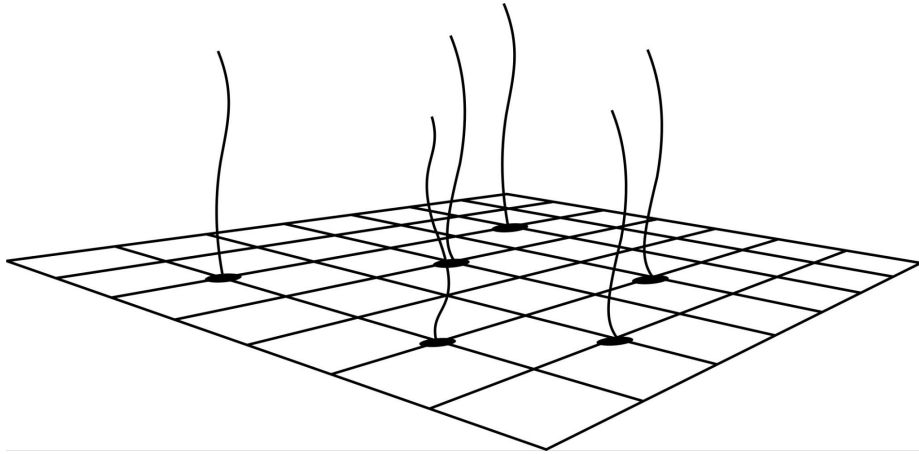
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Constructing the toy model



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3. Problems

Constructing the toy model



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 - 1.1. Lie group G
 - 1.2. Action of G (representation)
 - 1.3. Volume form
 - 1.4. Variation of action
2. Restoring local invariance
 - 2.1. Connection
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3. Problems
4. Case of $SO(1,3)$

General Lie group G and its action

General Lie group G :

General* Lie group G :

- Finite dimension

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- Connected (to identity)

General Lie group G :

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$$g(x) = e^{\varepsilon \alpha(x) X}$$

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- $\varepsilon \ll 1$

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Gauge theory

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~~$$g(x) = e^{\epsilon \alpha(x) X}$$~~

General Lie group G :

—————→ Representations

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General Lie group G :

————→ Representations

————→ Manifold

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————→ Representations

————→ Manifold: Γ

General Lie group G :

~~$$g(x) = e^{\epsilon \alpha(x) X}$$~~

————→ Representations

————→ Manifold: $\Gamma : G$

$(g$

General Lie group G :

~~$$g(x) = e^{\epsilon \alpha(x)} X$$~~

—————→ Representations

—————→ Manifold: $\Gamma : G \times \mathcal{M}$
 (g, x)

General Lie group G :

~~$$g(x) = e^{\epsilon \alpha(x) X}$$~~

————→ Representations

————→ Manifold: $\Gamma : G \times \mathcal{M} \rightarrow \mathcal{M}$
 $(g, x) \rightarrow x'$

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General Lie group G :

$$\rho(g(x)) = e^{\varepsilon\alpha(x)}\rho'(X)$$

————→ Representations

————→ Manifold: $\Gamma : G \times \mathcal{M} \rightarrow \mathcal{M}$

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2D rotations

2D rotations

$$\rho(g) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

2D rotations

$$\rho(g) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

$$\rho'(X) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

General Lie group G : $\rho(g(x)) = e^{\varepsilon\alpha(x)\rho'(X)}$

- $\varepsilon \ll 1 \longrightarrow \text{Infinitesimal} \longrightarrow g(x) \sim 1 + \varepsilon\alpha(x)X$
- $\alpha(x) \longrightarrow \text{Global / Local} \longrightarrow d\alpha = 0 \quad \text{or} \quad d\alpha \neq 0$
- $X \in \mathfrak{g} \longrightarrow \text{Abelian / Non-abelian}$

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- $\rho'(X)$

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- $X \in \mathfrak{g} \longrightarrow \text{Abelian / Non-abelian}$
- $\rho'(X) \longrightarrow \text{External / Internal}$

→ Internal

→ Internal

$$\rho(g)x = x$$

→ Internal

$$\rho(g)x = x$$

$$\iff \rho(g) \cong 1$$

→ Internal

$$\rho(g)x = x$$

$$\iff \rho(g) \cong 1$$

$$\iff \rho'(X) = 0$$

General Lie group G : $\rho(g(x)) = e^{\varepsilon\alpha(x)\rho'(X)}$

- $\varepsilon \ll 1 \longrightarrow$ Infinitesimal $\longrightarrow g(x) \sim 1 + \varepsilon\alpha(x)X$
- $\alpha(x) \longrightarrow$ Global / Local $\longrightarrow d\alpha = 0$ or $d\alpha \neq 0$
- $X \in \mathfrak{g} \longrightarrow$ Abelian / Non-abelian
- $\rho'(X) \longrightarrow$ External / Internal $\longrightarrow = 0$ or $\neq 0$

Variation of the action

Variation of the action

$$S[d\phi, d\phi^*, \omega] = \int_{\mathcal{M}} d\phi^* \wedge \star d\phi$$

Variation of the action

$$\begin{aligned} S[d\phi, d\phi^*, \omega] &= \int_{\mathcal{M}} d\phi^* \wedge \star d\phi \\ &= \int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega \end{aligned}$$

Variation of the action

$$S[d\phi, d\phi^*, \omega] = \int_{\mathcal{M}} d\phi^* \wedge \star d\phi$$

$$= \int_{\mathcal{M}} \langle \underline{d\phi^*}, \underline{d\phi} \rangle \omega$$

Variation of the action

$$S[d\phi, d\phi^*, \omega] = \int_{\mathcal{M}} d\phi^* \wedge \star d\phi$$

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Variation of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega$$

Variation of the action

$$\int_{\mathcal{M}} \langle \underline{d\phi^*}, \underline{d\phi} \rangle \omega$$

Variation of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega$$

$$\longrightarrow \int_{\mathcal{M}} \langle d(\sigma(g)\phi^*), d(\sigma(g)\phi) \rangle \omega$$

Variation of the action

$$\int_{\mathcal{M}} \langle d(\sigma(g)\phi^*), d(\sigma(g)\phi) \rangle \omega$$

Variation of the action

$$\int_{\mathcal{M}} \langle d(\sigma(g)\phi^*), d(\sigma(g)\phi) \rangle \omega$$

$$= \int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega$$

Variation of the action

$$\int_{\mathcal{M}} \langle d(\sigma(g)\phi^*), d(\sigma(g)\phi) \rangle \omega$$

$$= \int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$$

Variation of the action

$$\int_{\mathcal{M}} \langle d(\sigma(g)\phi^*), d(\sigma(g)\phi) \rangle \omega$$

$$= \int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

Variation of the action

$$\int_{\mathcal{M}} \langle d(\sigma(g)\phi^*), d(\sigma(g)\phi) \rangle \omega$$

$$= \int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \boxed{\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

Global U(1)

Variation of the action

$$\int_{\mathcal{M}} \langle d(\sigma(g)\phi^*), d(\sigma(g)\phi) \rangle \omega$$

$$= \int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

Local U(1)

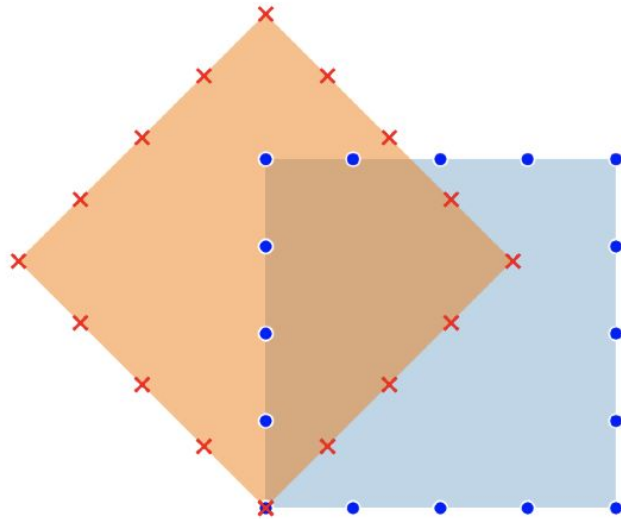
Variation of the action

$$\int_{\mathcal{M}} \langle d(\sigma(g)\phi^*), d(\sigma(g)\phi) \rangle_{\underline{\omega}}$$

Variation of the action

$$\int_{\mathcal{M}} \langle d(\sigma(g)\phi^*), d(\sigma(g)\phi) \rangle_{\underline{\omega}}$$

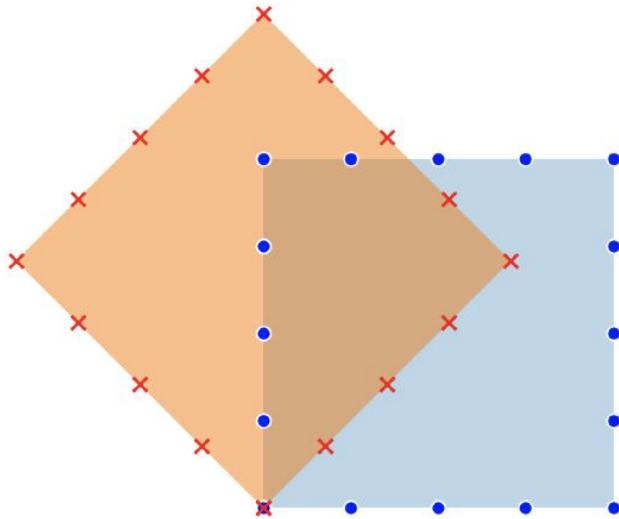
Global Rotation (45°)



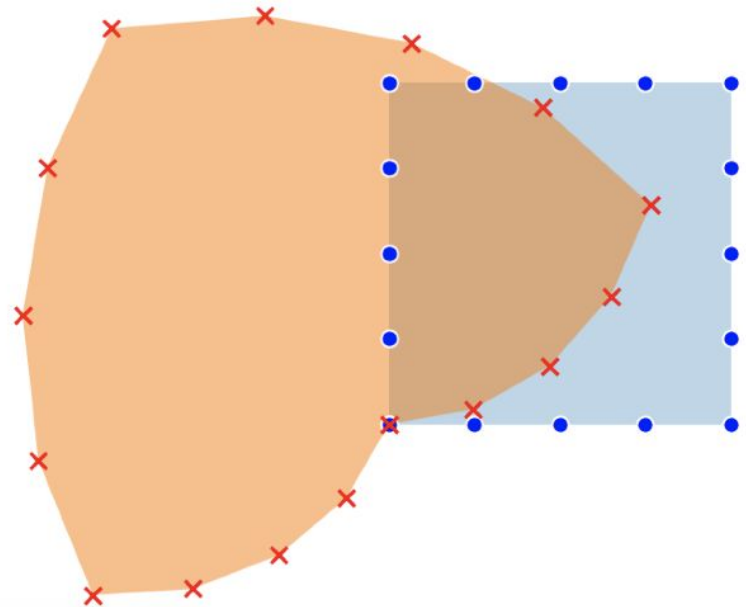
Variation of the action

$$\int_{\mathcal{M}} \langle d(\sigma(g)\phi^*), d(\sigma(g)\phi) \rangle_{\underline{\omega}}$$

Global Rotation (45°)



Local Rotation (10° × n)



Variation of the action

$$\int_{\mathcal{M}} \langle d(\sigma(g)\phi^*), d(\sigma(g)\phi) \rangle_{\underline{\omega}}$$

Variation of the action

$$\int_{\mathcal{M}} \langle d(\sigma(g)\phi^*), d(\sigma(g)\phi) \rangle \rho^*(g^{-1})\omega$$

Variation of the volume form

$$\rho^*(g^{-1})\omega$$

Variation of the volume form

$$\rho^*(g^{-1})\omega = [\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]\omega$$

Variation of the volume form

$$\rho^*(g^{-1})\omega = [\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]\omega + \varepsilon\sqrt{|\det(g)|}H^{\nu_1\nu_2\nu_3\nu_4} \wedge d\alpha$$

Variation of the volume form

$\rho'(X)$



$$\rho^*(g^{-1})\omega = [\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]\omega + \varepsilon\sqrt{|\det(g)|}H^{\nu_1\nu_2\nu_3\nu_4} \wedge d\alpha$$

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—————> External

Variation of the volume form

$$\rho^*(g^{-1})\omega = [\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(\underline{X}))]\omega + \varepsilon\sqrt{|\det(g)|}\underline{H}^{\nu_1\nu_2\nu_3\nu_4} \wedge d\alpha$$

External

Variation of the volume form

$$\rho^*(g^{-1})\omega = [\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(\underline{X}))]\omega + \varepsilon\sqrt{|\det(g)|}\underline{H^{\nu_1\nu_2\nu_3\nu_4}} \wedge d\alpha$$

$$d\alpha = 0$$

External

Variation of the volume form

$$\rho^*(g^{-1})\omega = [\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(\underline{X}))]\omega + \varepsilon\sqrt{|\det(g)|}\underline{H^{\nu_1\nu_2\nu_3\nu_4}} \wedge d\alpha$$

$$d\alpha = 0$$

External

Variation of the volume form

$$\rho^*(g^{-1})\omega = \underbrace{[\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]}_{\substack{\rho'(X) \text{ traceless} \\ \text{External}}} \omega + \varepsilon \sqrt{|\det(g)|} \underbrace{H^{\nu_1\nu_2\nu_3\nu_4}}_{\text{External}} \wedge d\alpha$$

$d\alpha = 0$

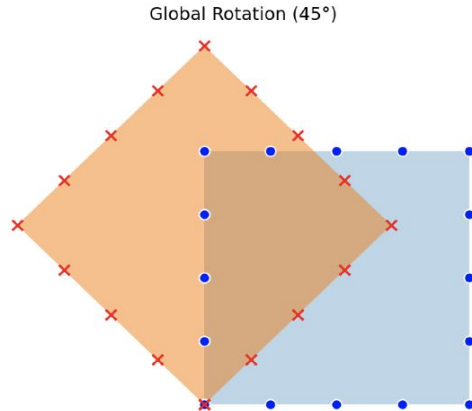
Variation of the volume form

$$\rho^*(g^{-1})\omega = \left[\mathbb{I} - \varepsilon \alpha \text{Tr}(\rho'(\underline{X})) \right] \omega + \varepsilon \sqrt{|\det(g)|} \underline{H^{\nu_1 \nu_2 \nu_3 \nu_4}} \wedge d\alpha$$

$\rho'(X)$ traceless

$d\alpha = 0$

External



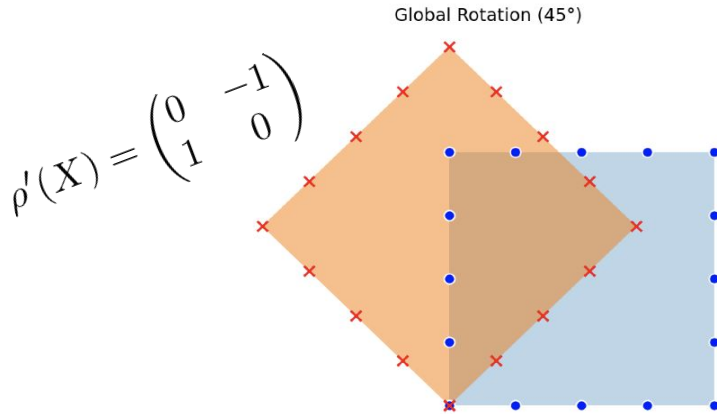
Variation of the volume form

$$\rho^*(g^{-1})\omega = \left[\mathbb{I} - \varepsilon \alpha \text{Tr}(\rho'(\underline{X})) \right] \omega + \varepsilon \sqrt{|\det(g)|} \underline{H^{\nu_1 \nu_2 \nu_3 \nu_4}} \wedge d\alpha$$

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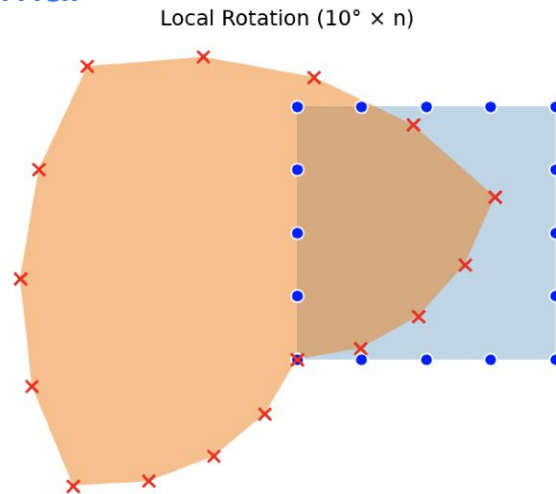
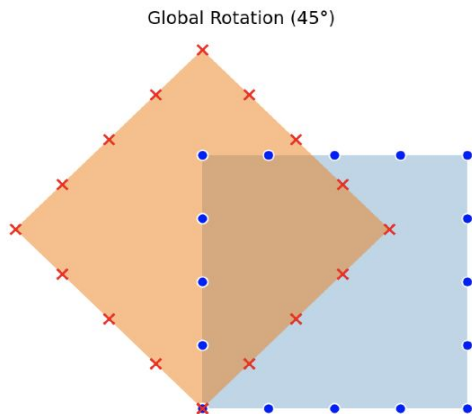
External



Variation of the volume form

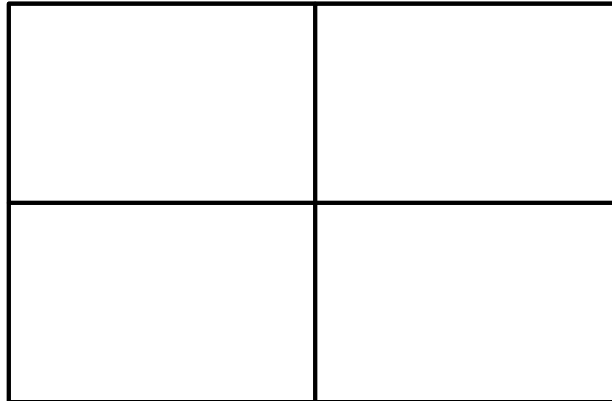
$$\rho^*(g^{-1})\omega = \underbrace{[\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]}_{\rho'(X) \text{ traceless}} \omega + \varepsilon \sqrt{|\det(g)|} \underbrace{H^{\nu_1\nu_2\nu_3\nu_4}}_{d\alpha=0} \wedge d\alpha$$

External



$$\rho^*(g^{-1})\omega = \underbrace{[\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]}_{\substack{\rho'(X) \text{ traceless} \\ \text{External}}} \omega + \varepsilon \sqrt{|\det(g)|} \underbrace{H^{\nu_1\nu_2\nu_3\nu_4}}_{d\alpha=0} \wedge d\alpha$$

$$\rho^*(g^{-1})\omega = \underbrace{[\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]}_{\substack{\rho'(X) \text{ traceless} \\ \text{External}}} \omega + \varepsilon \sqrt{|\det(g)|} \underbrace{H^{\nu_1\nu_2\nu_3\nu_4}}_{d\alpha=0} \wedge d\alpha$$

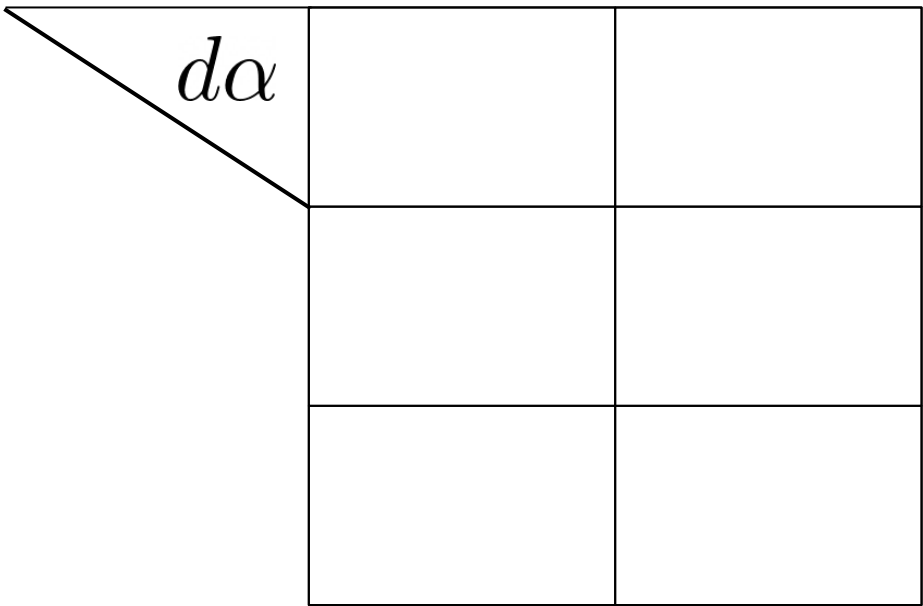


$$\rho^*(g^{-1})\omega = \boxed{[\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]\omega} + \boxed{\varepsilon\sqrt{|\det(g)|} \underline{H^{\nu_1\nu_2\nu_3\nu_4}} \wedge d\alpha}$$

$\rho'(X)$ traceless

External

$d\alpha = 0$



$$\rho^*(g^{-1})\omega = [\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]\omega + \varepsilon\sqrt{|\det(g)|} \underline{H^{\nu_1\nu_2\nu_3\nu_4}} \wedge d\alpha$$

$\rho'(X)$ traceless

External

$d\alpha = 0$

$d\alpha$	= 0	

$$\rho^*(g^{-1})\omega = \underbrace{[\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]}_{\substack{\rho'(X) \text{ traceless} \\ \text{External}}} \omega + \varepsilon \sqrt{|\det(g)|} \underbrace{H^{\nu_1\nu_2\nu_3\nu_4}}_{d\alpha = 0} \wedge d\alpha$$

$d\alpha$	$= 0$	$\neq 0$

$$\rho^*(g^{-1})\omega = \boxed{[\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]\omega} + \boxed{\varepsilon\sqrt{|\det(g)|} \underline{H^{\nu_1\nu_2\nu_3\nu_4}} \wedge d\alpha}$$

$\rho'(X)$ traceless

External

$d\alpha = 0$

$\rho'(X) \backslash d\alpha$	$= 0$	$\neq 0$

$$\rho^*(g^{-1})\omega = \underbrace{[\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]}_{\substack{\rho'(X) \text{ traceless} \\ \text{External}}} \omega + \varepsilon \sqrt{|\det(g)|} \underbrace{H^{\nu_1\nu_2\nu_3\nu_4}}_{d\alpha = 0} \wedge d\alpha$$

$\rho'(X) \backslash d\alpha$	$= 0$	$\neq 0$
traceless		

$$\rho^*(g^{-1})\omega = \underbrace{[\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]}_{\substack{\rho'(X) \text{ traceless} \\ \text{External}}} \omega + \varepsilon \sqrt{|\det(g)|} \underbrace{H^{\nu_1\nu_2\nu_3\nu_4}}_{d\alpha = 0} \wedge d\alpha$$

$\rho'(X) \backslash d\alpha$	$= 0$	$\neq 0$
traceless		
not traceless		

$$\rho^*(g^{-1})\omega = \underbrace{[\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]}_{\substack{\rho'(X) \text{ traceless} \\ \text{External}}} \omega + \varepsilon \sqrt{|\det(g)|} \underbrace{H^{\nu_1\nu_2\nu_3\nu_4}}_{d\alpha = 0} \wedge d\alpha$$

$\rho'(X) \backslash d\alpha$	$= 0$	$\neq 0$
traceless		not invariant
not traceless		not invariant

$$\rho^*(g^{-1})\omega = \underbrace{[\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]}_{\rho'(X) \text{ traceless}} \omega + \underbrace{\varepsilon \sqrt{|\det(g)|} \underline{H^{\nu_1\nu_2\nu_3\nu_4}} \wedge d\alpha}_{\text{External } d\alpha = 0}$$

$\rho'(X) \backslash d\alpha$	$= 0$	$\neq 0$
traceless	invariant	not invariant
not traceless	not invariant	not invariant

$$\rho^*(g^{-1})\omega = \underbrace{[\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]}_{\rho'(X) \text{ traceless}} \omega + \underbrace{\varepsilon \sqrt{|\det(g)|} \underline{H^{\nu_1\nu_2\nu_3\nu_4}}}_{\text{External}} \wedge d\alpha$$

$d\alpha = 0$

$\rho'(X) \backslash d\alpha$	$= 0$	$\neq 0$
traceless	invariant	not invariant
not traceless	not invariant	not invariant

$\omega?$

Variation of the action

$$\int_{\mathcal{M}} \langle d(\sigma(g)\phi^*), d(\sigma(g)\phi) \rangle \omega$$

$$= \int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

Variation of the action

$$\int_{\mathcal{M}} \langle d(\sigma(g)\phi^*), d(\sigma(g)\phi) \rangle \rho^*(g^{-1})\omega$$

$$= \int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

Variation of the action

$$\int_{\mathcal{M}} \langle d(\sigma(g)\phi^*), d(\sigma(g)\phi) \rangle \rho^*(g^{-1})\omega$$

$$= \int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

$$-d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\# dx^{\hat{\mu}}} (d\phi^* \wedge \star d\phi)$$

Variation of the action

$$-d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^{\mu}_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi)$$

Variation of the action

$$- \boxed{d\alpha} \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^{\mu}_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi)$$

Variation of the action

$$-d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^{\mu}_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi)$$

Local

Variation of the action

$$-d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^{\mu}_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi)$$

Local

Variation of the action

$$-d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^{\mu}_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi)$$

Local

External

Variation of the action

$$\rho^*(g^{-1})\omega = [\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]\omega + \varepsilon\sqrt{|\det(g)|}H^{\nu_1\nu_2\nu_3\nu_4} \wedge d\alpha$$

The diagram illustrates the variation of the action. The equation is split into two parts, each enclosed in a colored box. The first part, $[\mathbb{I} - \varepsilon\alpha \text{Tr}(\rho'(X))]\omega$, is enclosed in a green box. The second part, $\varepsilon\sqrt{|\det(g)|}H^{\nu_1\nu_2\nu_3\nu_4} \wedge d\alpha$, is enclosed in a red box. A red arrow points from the red box to the word 'Local' in red text. A green arrow points from the green box to the word 'External' in green text. Another green arrow points from the red box to the word 'External' in green text.

Local

External

Variation of the action

$$-d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^{\mu}_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi)$$

Local

External

Variation of the action

	Global	Local
Internal		
External		

Variation of the action

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \int_{\mathcal{M}} d(\star J_1)$	
External		

Variation of the action

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \int_{\mathcal{M}} d(\star J_1)$ 	
External		

Variation of the action

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \int_{\mathcal{M}} d(\star J_1)$	
External		

Variation of the action

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+ \varepsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+ \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External		

Variation of the action

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$ $+ \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External		

Variation of the action

Noether current

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$ $+\varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External		

Variation of the action

Noether current

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \epsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External		

Variation of the action

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \epsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External		

Variation of the action

Gauge field

Noether current

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \epsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External		

Variation of the action

Gauge field

Noether current

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \epsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \alpha \int_{\mathcal{M}} d(\star J_1)$	

Variation of the action

Gauge field

Noether current

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \epsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \alpha \int_{\mathcal{M}} d(\star J_1)$	

Variation of the action

Gauge field

Noether current

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \epsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \alpha \int_{\mathcal{M}} d(\star J_1)$	

Noether current

Variation of the action

Gauge field

Noether current

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \epsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \alpha \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \epsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$

Noether current

$$-d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^{\mu}_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \tilde{\star} d\phi)$$

Diagram illustrating the classification of terms in the equation:

- The term $d\alpha$ is highlighted in a red box and labeled **Local**.
- The term $\rho'(X)^{\mu}_{\tilde{\mu}}$ is highlighted in a green box and labeled **External**.

Variation of the action

Gauge field

Noether current

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+ \varepsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+ \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External	$S[d\phi, d\phi^*, \omega]$ $+ \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+ \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$

Noether current

Variation of the action

Gauge field

Noether current

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \epsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \alpha \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+ \epsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \epsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$ $+ d\alpha \wedge \epsilon \int_{\mathcal{M}} (\rho'(X)^\mu{}_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\sharp dx^{\hat{\mu}}} (d\phi^* \wedge \star d\phi)$

Variation of the action

Gauge field

Noether current

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$ $+\varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$ $+\varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$ $+d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu{}_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\sharp dx^{\hat{\mu}}} (d\phi^* \wedge \star d\phi)$

Noether current

Variation of the action

Gauge field

Noether current

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$ $+\varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$ $+\varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$ $+d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu{}_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\sharp dx^{\hat{\mu}}} (d\phi^* \wedge \star d\phi)$

Noether current

Noether current

Variation of the action

Gauge field

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$ $+\varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$ <u>$+\varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$</u> <u>$+d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu{}_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\sharp dx^{\hat{\mu}}} (d\phi^* \wedge \star d\phi)$</u>

Noether current

Noether current

Variation of the action

Gauge field

Noether current

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$ $+\varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$ <u>$+\varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$</u> <u>$+d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu{}_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\tilde{\mu}} \iota_{\sharp dx^{\tilde{\mu}}} (d\phi^* \wedge \star d\phi)$</u>

Gauge field?

Noether current

Noether current

Variation of the action

Gauge field

Noether current

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$ $+\varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$ <u>$+\varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$</u> <u>$+d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu{}_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\tilde{\mu}} \iota_{\sharp dx^{\tilde{\mu}}} (d\phi^* \wedge \star d\phi)$</u>

~~Gauge field?~~

Noether current

Noether current

Variation of the action

Gauge field

Noether current

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$ $+d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu{}_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\tilde{\mu}} \iota_{\sharp dx^{\tilde{\mu}}} (d\phi^* \wedge \star d\phi)$

Noether current

Noether current

Gauge field

Variation of the action

Gauge field

Noether current

Noether current

	Global	Local
Internal	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$
External	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1)$	$S[d\phi, d\phi^*, \omega]$ $+\varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$ $+d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu{}_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\tilde{\mu}} \iota_{\sharp dx^{\tilde{\mu}}} (d\phi^* \wedge \star d\phi)$

Noether current

Noether current

Gauge field

Restoring invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

Restoring invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

Restoring invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

→ Introduce field A

Restoring invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

→ Introduce field A $A \rightarrow A - d\alpha$

Restoring invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

→ Introduce field A $A \rightarrow A - d\alpha$

→ Define covariant derivative

Restoring invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

→ Introduce field A $A \rightarrow A - d\alpha$

→ Define covariant derivative $D\phi = d\phi + iqA\phi$

Restoring invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

→ Introduce field A $A \rightarrow A - d\alpha$

→ Define covariant derivative $D\phi = d\phi + iqA\phi$

→ A couples to fields in a specific way

Restoring invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \cancel{\int_{\mathcal{M}} d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

→ Introduce field A $A \rightarrow A - d\alpha$

→ Define covariant derivative $D\phi = d\phi + iqA\phi$

→ A couples to fields in a specific way $A \wedge \star J_2$

Restoring invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

→ Introduce field A $A \rightarrow A - d\alpha$

→ Define covariant derivative $D\phi = d\phi + iqA\phi$

→ A couples to fields in a specific way $A \wedge \star J_2$

$$d\phi^* \wedge \star d\phi$$

Restoring invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

→ Introduce field A $A \rightarrow A - d\alpha$

→ Define covariant derivative $D\phi = d\phi + iqA\phi$

→ A couples to fields in a specific way $A \wedge \star J_2$

$$d\phi^* \wedge \star d\phi$$

→ not invariant

Restoring invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

→ Introduce field A $A \rightarrow A - d\alpha$

→ Define covariant derivative $D\phi$ $= d\phi + iqA\phi$

→ A couples to fields in a specific way $A \wedge \star J_2$

$$d\phi^* \wedge \star d\phi$$

→ not invariant

Restoring invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

→ Introduce field A $A \rightarrow A - d\alpha$

→ Define covariant derivative $D\phi = d\phi + iqA\phi$

→ A couples to fields in a specific way $A \wedge \star J_2$

$$\underline{d\phi^* \wedge \star d\phi}$$

→ not invariant

Restoring invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

→ Introduce field A $A \rightarrow A - d\alpha$

→ Define covariant derivative $D\phi = d\phi + iqA\phi$

→ A couples to fields in a specific way $A \wedge \star J_2$

$$d\phi^* \wedge \star d\phi$$

$$D\phi^* \wedge \star D\phi$$

→ not invariant

Restoring invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

→ Introduce field A $A \rightarrow A - d\alpha$

→ Define covariant derivative $D\phi = d\phi + iqA\phi$

→ A couples to fields in a specific way $A \wedge \star J_2$

$$d\phi^* \wedge \star d\phi$$

$$D\phi^* \wedge \star D\phi$$

→ not invariant

Restoring invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

→ Introduce field A $A \rightarrow A - d\alpha$

→ Define covariant derivative $D\phi = d\phi + iqA\phi$

→ A couples to fields in a specific way $A \wedge \star J_2$

$$d\phi^* \wedge \star d\phi$$

$$D\phi^* \wedge \star D\phi$$

→ not invariant

Restoring invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

→ Introduce field A $A \rightarrow A - d\alpha$

→ Define covariant derivative $D\phi = d\phi + iqA\phi$

→ A couples to fields in a specific way $A \wedge \star J_2$

$$d\phi^* \wedge \star d\phi$$

→ not invariant

$$D\phi^* \wedge \star D\phi$$

→ invariant

Problems with the invariance of the action

$$\begin{aligned} \int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2 \\ - d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi) \end{aligned}$$

Problems with the invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

$$- d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi)$$

Problems with the invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

$$- d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi)$$

→ Introduce field A

Problems with the invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2 \\ - d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi)$$

→ Introduce field A $A \rightarrow A + \varepsilon \{ \alpha [X, A] - X d\alpha \}$

Problems with the invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

$$- d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi)$$

→ Introduce field A $A \rightarrow A + \varepsilon \{ \alpha [X, A] - X d\alpha \}$

→ Define covariant derivative

Problems with the invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

$$- d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi)$$

→ Introduce field A $A \rightarrow A + \varepsilon \{ \alpha[X, A] - X d\alpha \}$

→ Define covariant derivative $D\phi = d\phi - A[\hat{\rho}'(X)^b_a E_b - E_a(\xi_X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}} \partial_\mu] \phi^a$

Problems with the invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2 \\ - d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi)$$

→ Introduce field A $A \rightarrow A + \varepsilon \{ \alpha[X, A] - X d\alpha \}$

→ Define covariant derivative $D\phi = d\phi - A[\hat{\rho}'(X)^b_a E_b - E_a(\xi_X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}} \partial_\mu] \phi^a$

→ Coupling

Problems with the invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} \cancel{d(\star J_1)} + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2 \\ - d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi)$$

→ Introduce field A $A \rightarrow A + \varepsilon \{ \alpha [X, A] - X d\alpha \}$

→ Define covariant derivative $D\phi = d\phi - A[\hat{\rho}'(X)^b_a E_b - E_a(\xi_X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}} \partial_\mu] \phi^a$

→ Coupling $A \wedge \star J_2$

Problems with the invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

$$- d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi)$$

→ Introduce field A $A \rightarrow A + \varepsilon \{ \alpha [X, A] - X d\alpha \}$

→ Define covariant derivative $D\phi = d\phi - A[\hat{\rho}'(X)^b_a E_b - E_a(\xi_X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}} \partial_\mu] \phi^a$

→ Coupling $A \wedge \star J_2$

Problems with the invariance of the action

$$\int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2$$

$$-d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi)$$

→ Introduce field A $A \rightarrow A + \varepsilon \{ \alpha[X, A] - X d\alpha \}$

→ Define covariant derivative $D\phi = d\phi - A[\hat{\rho}'(X)^b_a E_b - E_a(\xi_X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}} \partial_\mu] \phi^a$

→ Coupling $A \wedge \star J_2$

Problems with the invariance of the action

$$d\phi^* \wedge \star d\phi$$

Problems with the invariance of the action

$$d\phi^* \wedge \star d\phi$$

→ not invariant

Problems with the invariance of the action

$$d\phi^* \wedge \star d\phi$$

→ not invariant

$$D\phi^* \wedge \star D\phi$$

Problems with the invariance of the action

$$d\phi^* \wedge \star d\phi$$

→ not invariant

$$D\phi^* \wedge \star D\phi$$

→ not invariant

Problems with the invariance of the action

$$d\phi^* \wedge \star d\phi$$

→ not invariant

$$D\phi^* \wedge \star D\phi$$

→ not invariant

$$D\phi^* \wedge \star D\phi + A \wedge \left((\rho'(X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\#} dx^{\hat{\mu}} (d\phi^* \wedge \star d\phi) \right)$$

Problems with the invariance of the action

Problems with the invariance of the action

Gauge symmetry

Problems with the invariance of the action

Gauge symmetry

internal

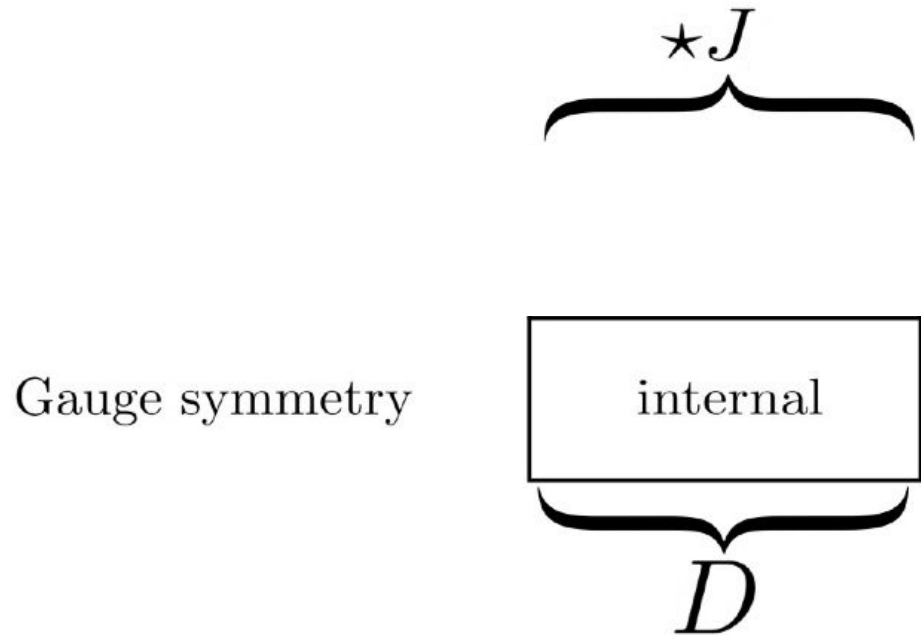
Problems with the invariance of the action

$$\overbrace{\star J}$$

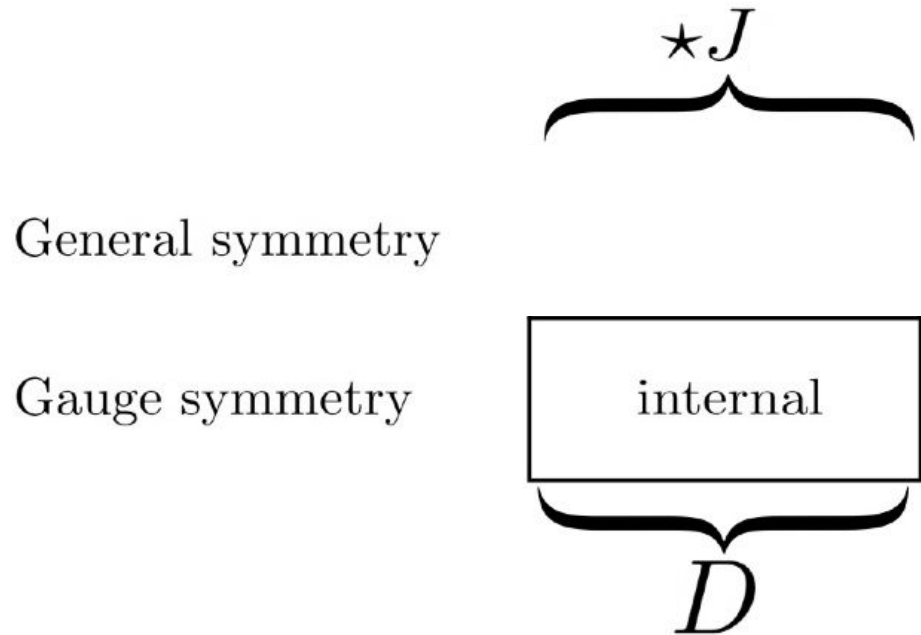
Gauge symmetry

internal

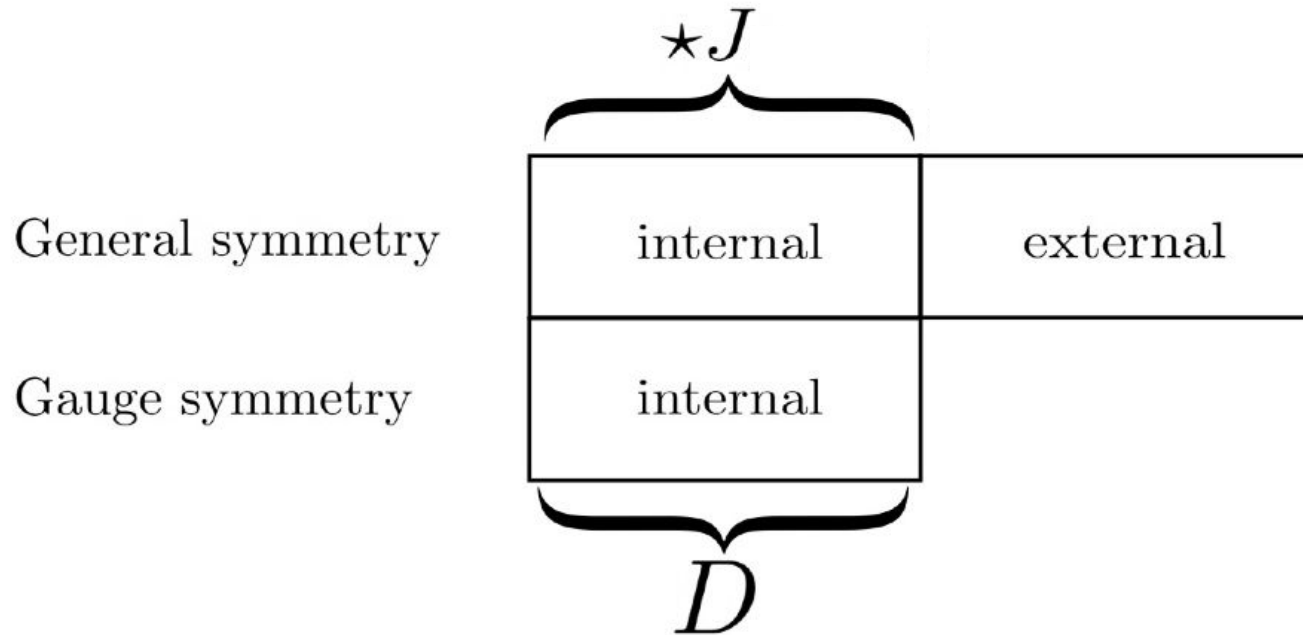
Problems with the invariance of the action



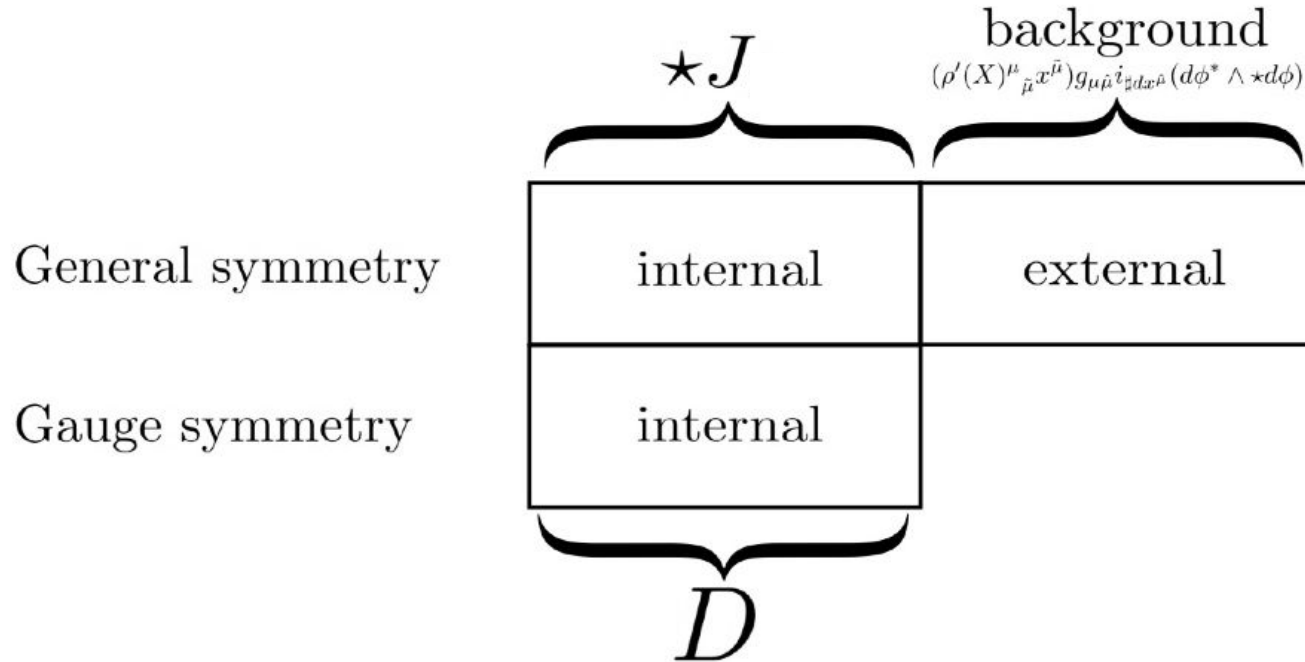
Problems with the invariance of the action



Problems with the invariance of the action



Problems with the invariance of the action



Summary

Summary

1. General Lie group

Summary

1. General Lie group
2. Local transformation of matter action

Summary

1. General Lie group
2. Local transformation of matter action
3. Restore invariance

Summary

1. General Lie group
2. Local transformation of matter action
3. Restore invariance: introduce field

Summary

1. General Lie group
2. Local transformation of matter action
3. Restore invariance: introduce field
4. Interpretation of field depending on group

Summary

4. Interpretation of field depending on group

Summary

4. Interpretation of field depending on group

$$\begin{aligned}
 = & \int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2 \\
 & - d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\# dx^{\hat{\mu}}} (d\phi^* \wedge \tilde{\star} d\phi)
 \end{aligned}$$

Summary

4. Interpretation of field depending on group

$$\begin{aligned}
 = & \int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2 \\
 & - d\alpha \wedge \varepsilon \int_{\mathcal{M}} (\rho'(X)^\mu_{\tilde{\mu}} x^{\tilde{\mu}}) g_{\mu\hat{\mu}} \iota_{\# dx^{\hat{\mu}}} (d\phi^* \wedge \tilde{\star} d\phi)
 \end{aligned}$$

Global / external

Summary

4. Interpretation of field depending on group

$$\begin{aligned}
 = & \int_{\mathcal{M}} \langle d\phi^*, d\phi \rangle \omega + \varepsilon \alpha \int_{\mathcal{M}} d(\star J_1) + \varepsilon \int_{\mathcal{M}} d\alpha \wedge \star J_2 \\
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Global / external (ex: Global Lorentz)

Summary

4. Interpretation of field depending on group

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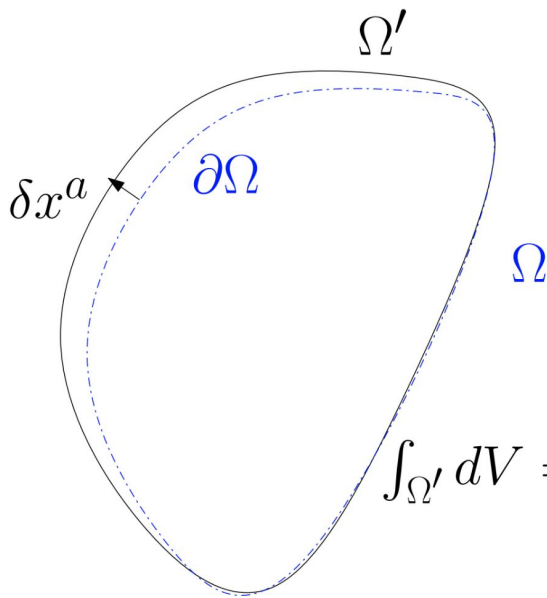
Further work

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Yi-Shi Duan, Jing-Ye Zhang, “Conservation law of energy-momentum in general relativity”, arXiv:1803.09651 [gr-qc]

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$$\int_{\Omega'} dV = \int_{\Omega} dV + \int_{\partial\Omega} \delta x^a dS_a$$

Thank you for your attention!

Appendix

G

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$\mathcal{M}$

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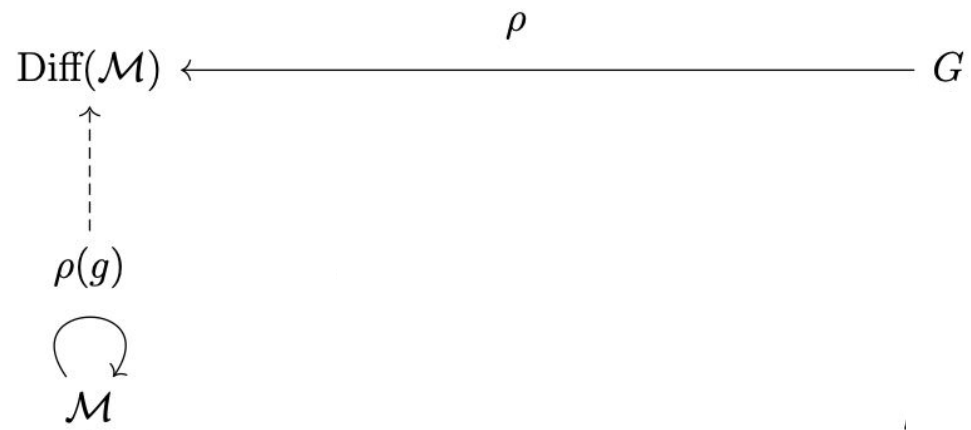
$\text{Diff}(\mathcal{M})$

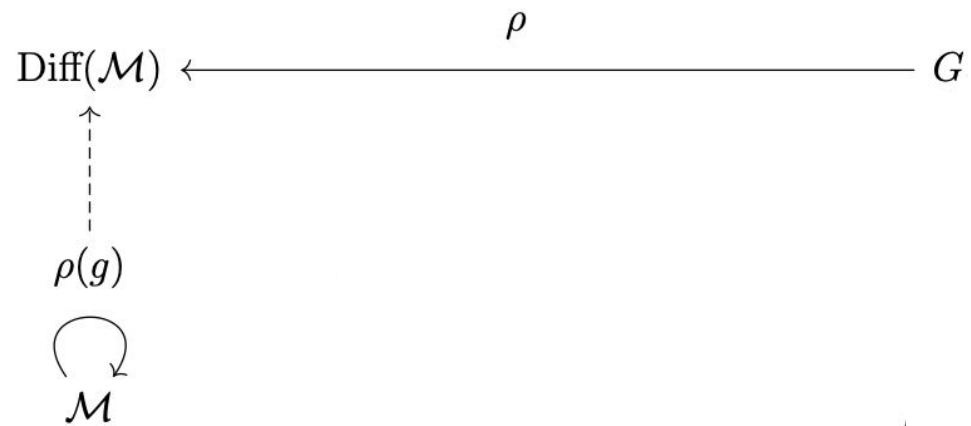
G



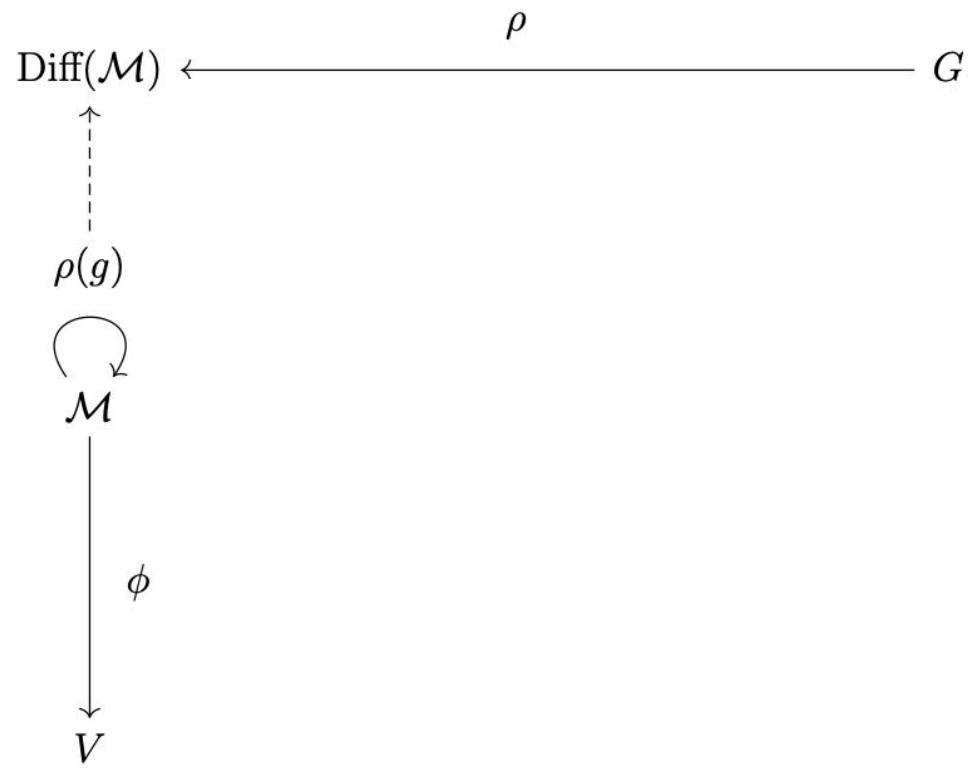
$$\mathrm{Diff}(\mathcal{M}) \xleftarrow{\rho} G$$



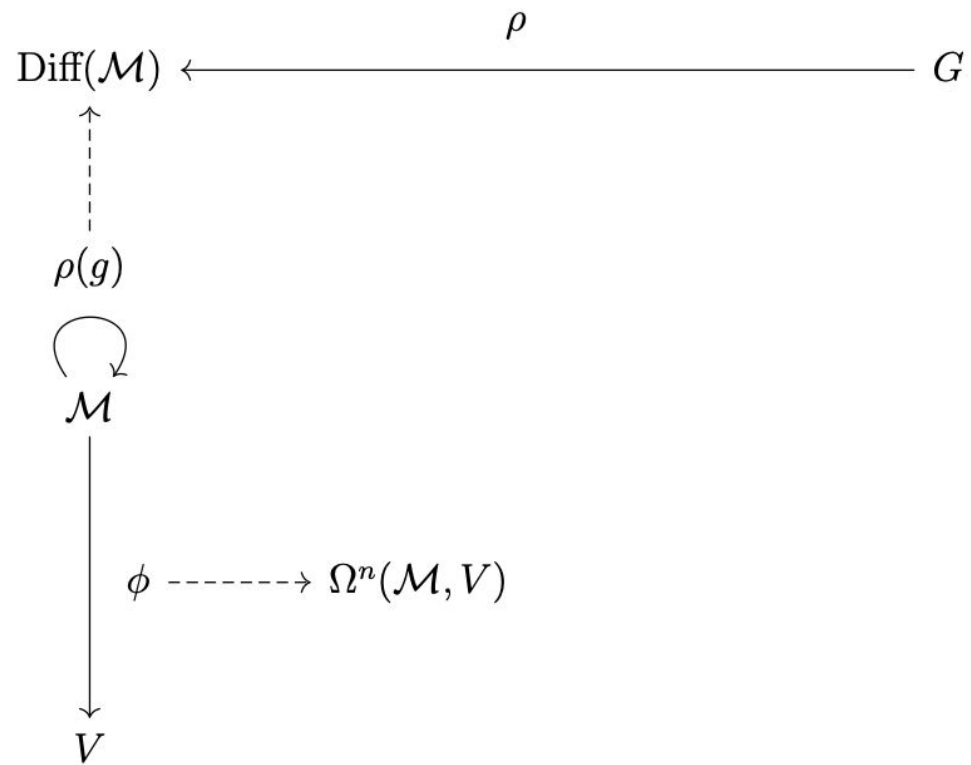




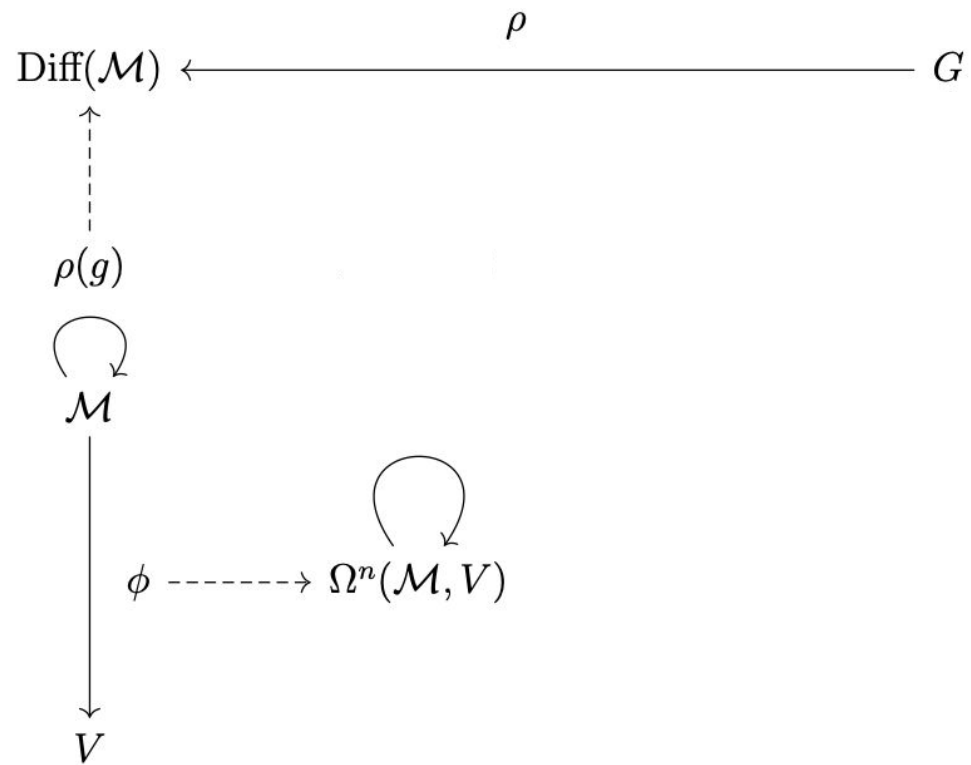
$$\begin{aligned} \Gamma : G \times \mathcal{M} &\rightarrow \mathcal{M} \\ (g, x) &\rightarrow L_g x =: \rho(g)x \end{aligned}$$



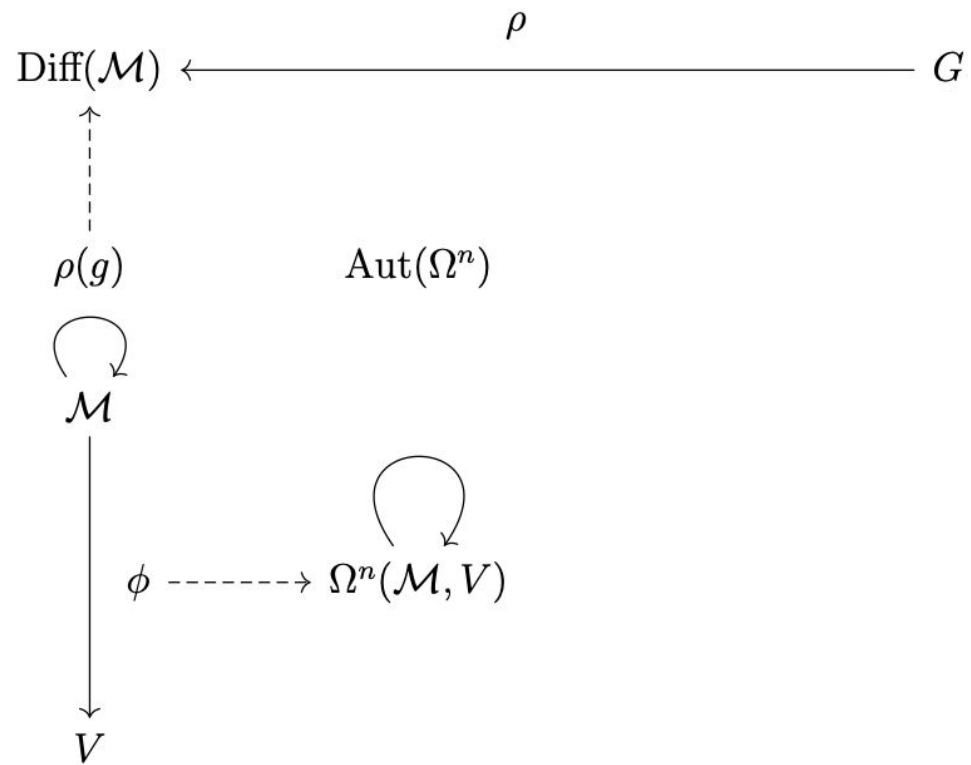
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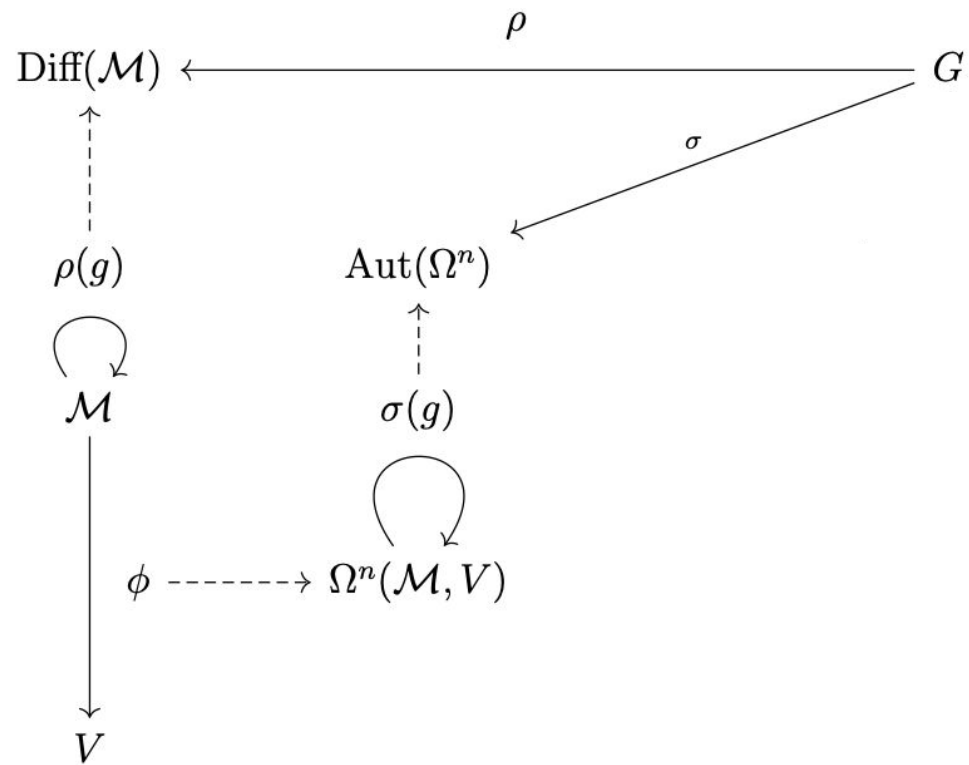
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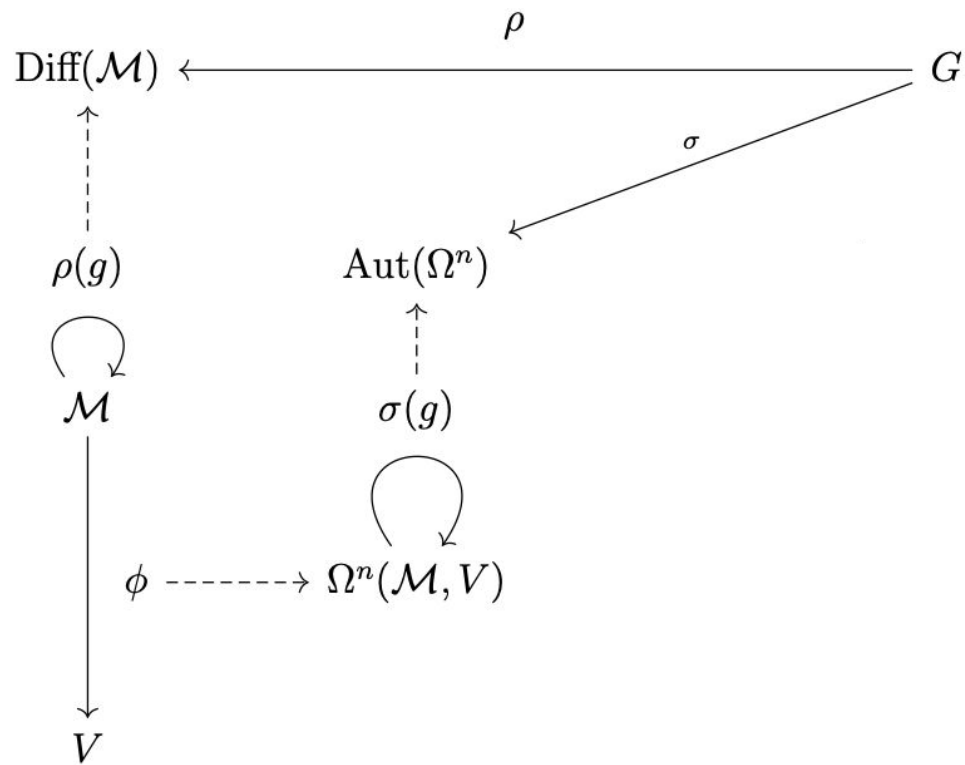
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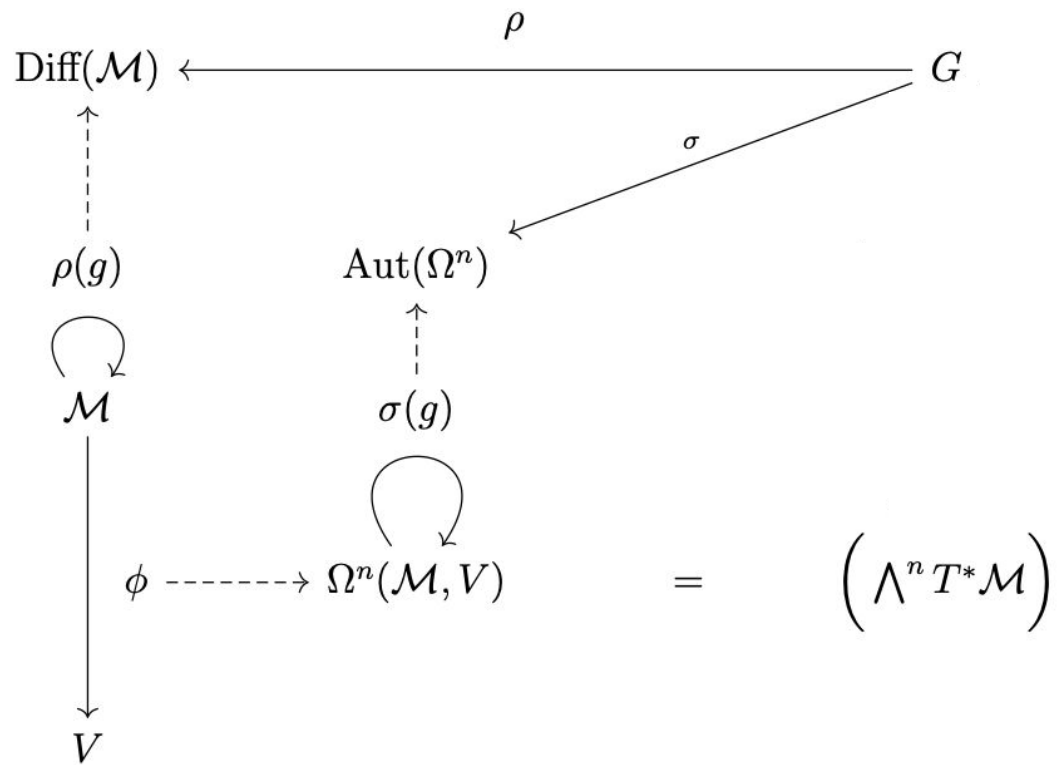


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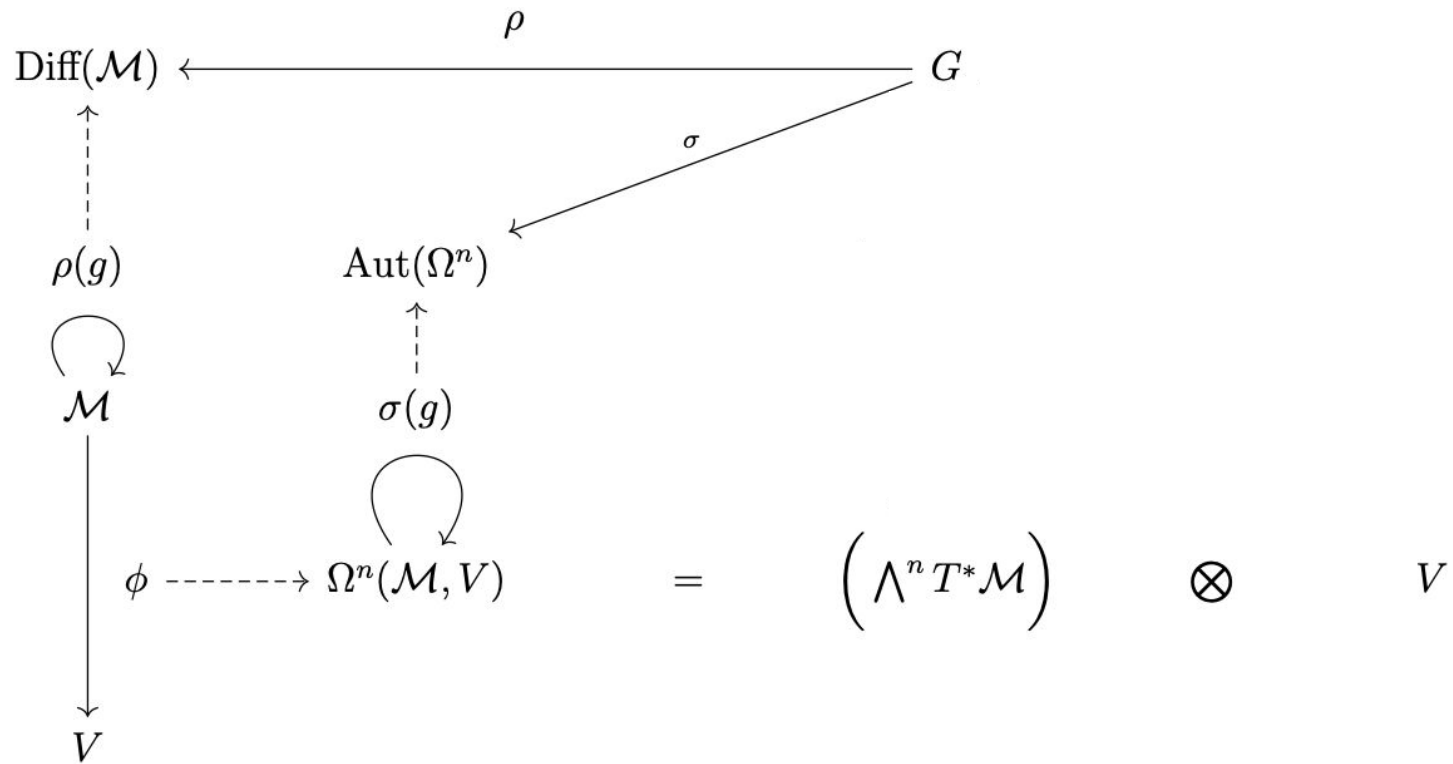
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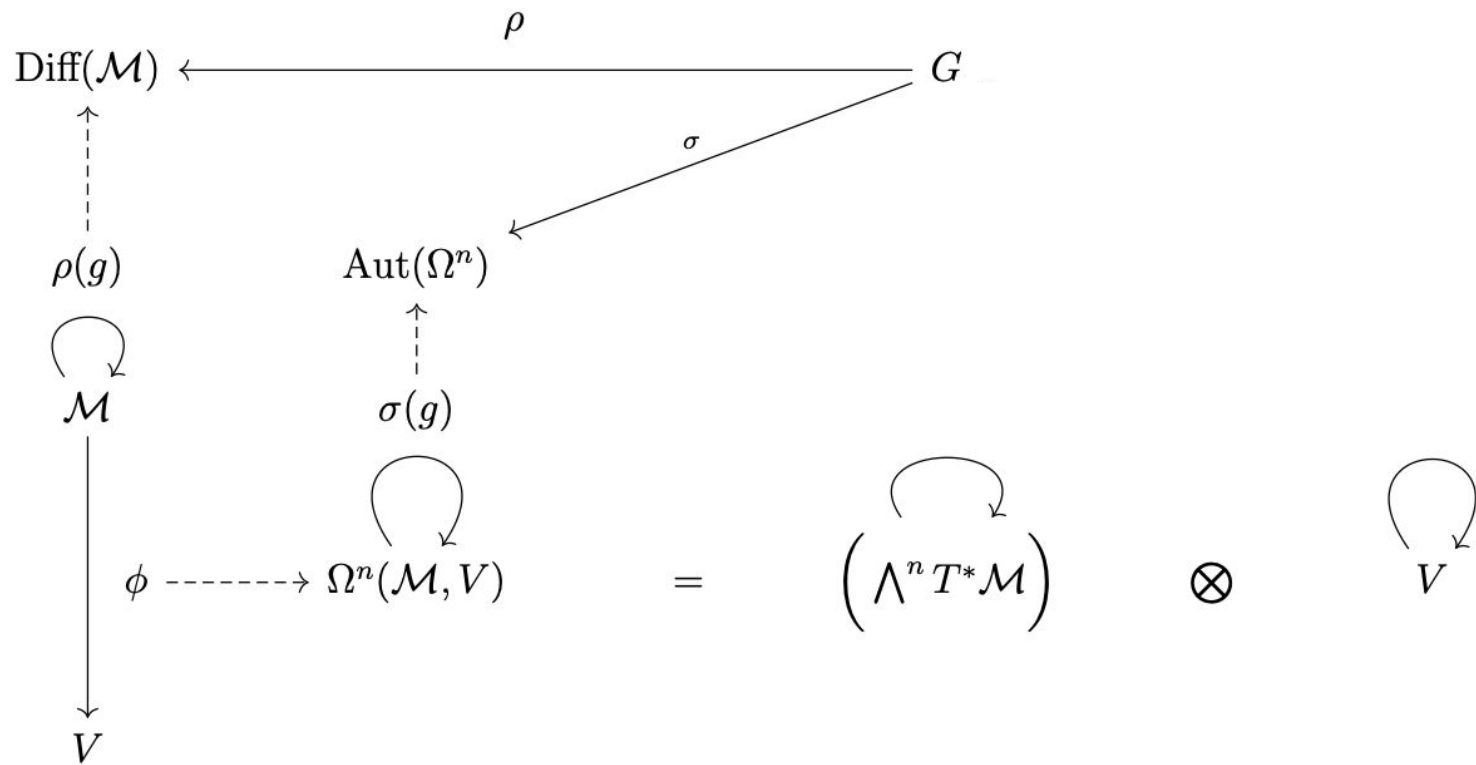
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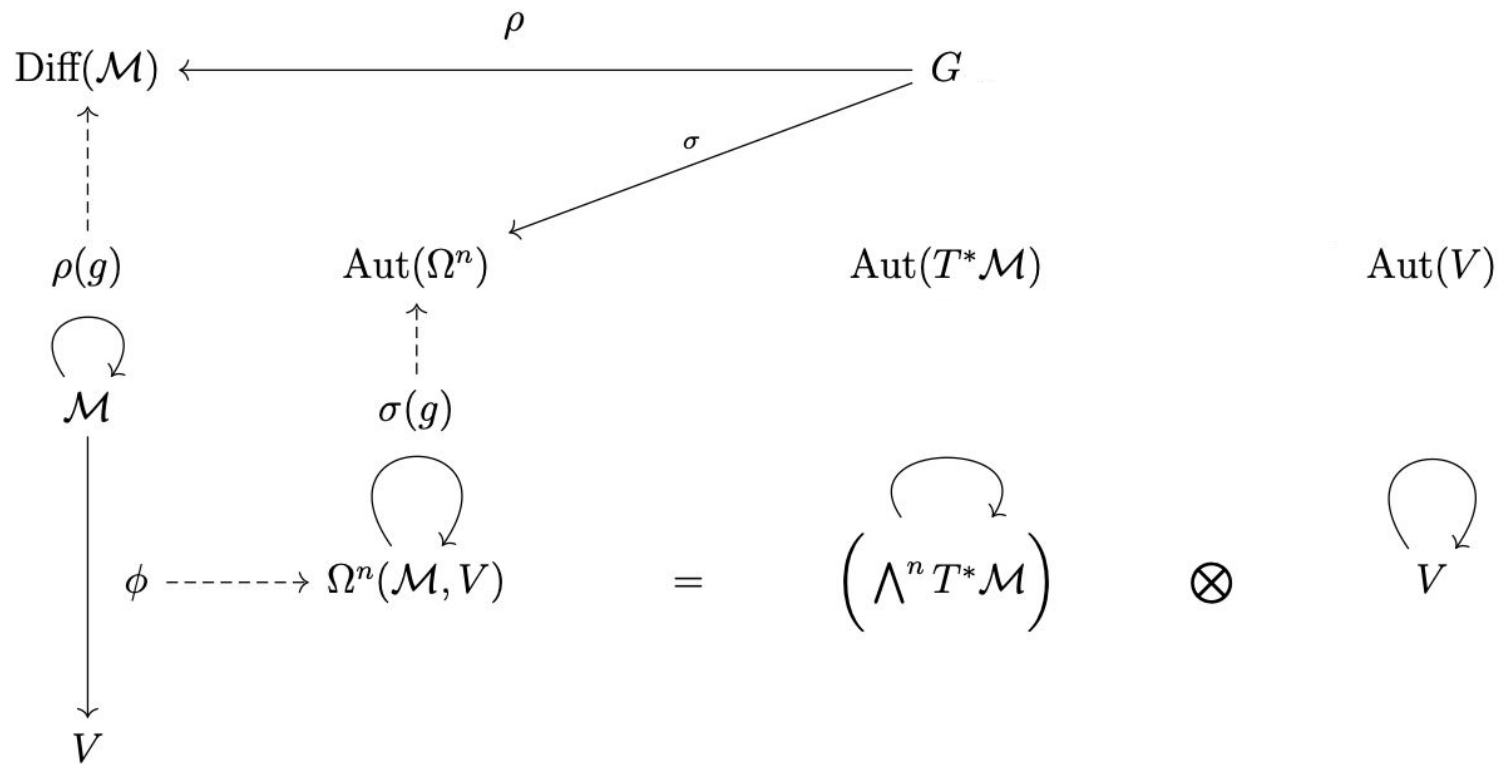
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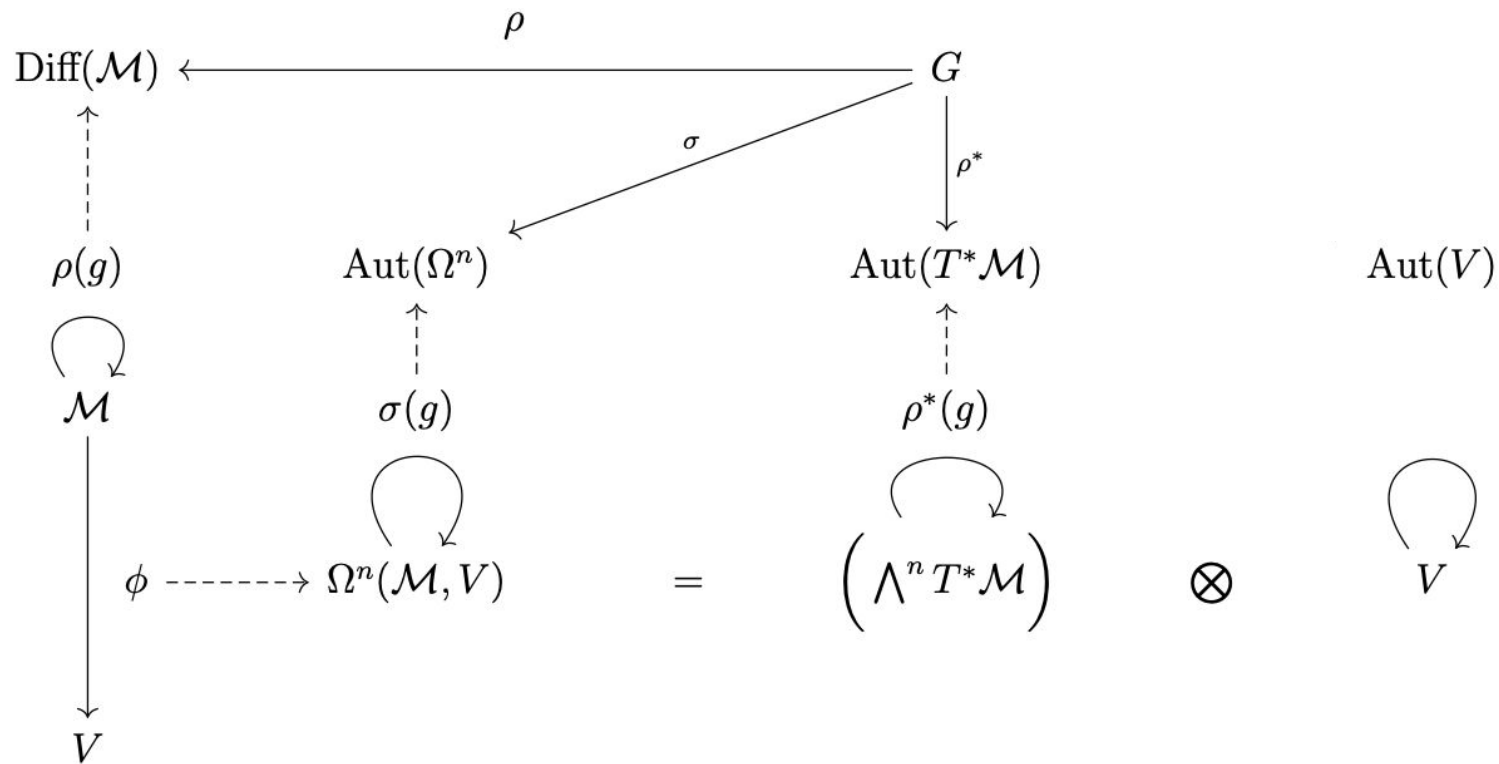
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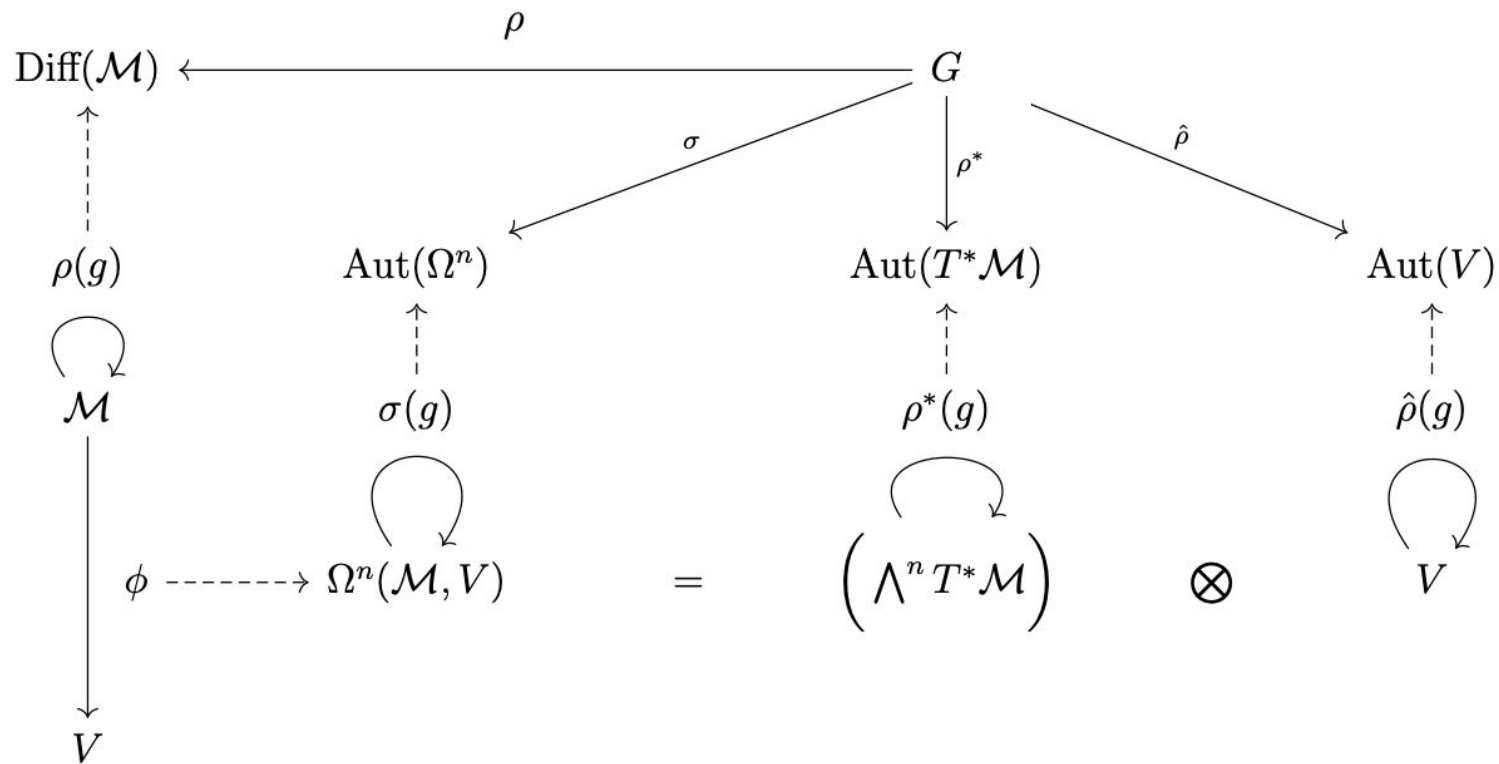
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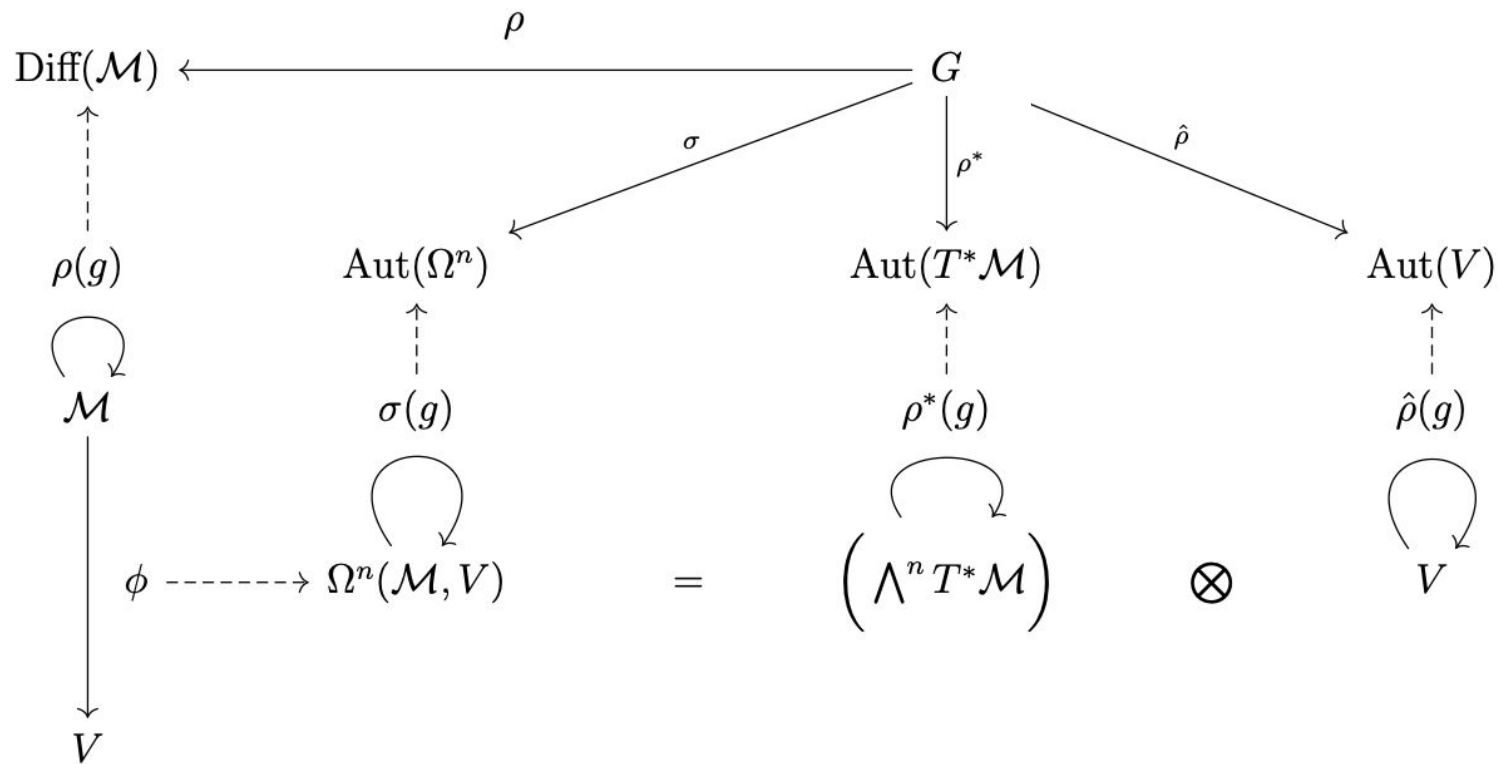
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